

CHAPTER 14



PROBLEM 14.1

An airline employee tosses two suitcases, of mass 15 kg and 20 kg, respectively, onto a 25-kg baggage carrier in rapid succession. Knowing that the carrier is initially at rest and that the employee imparts a 3-m/s horizontal velocity to the 15-kg suitcase and a 2-m/s horizontal velocity to the 20-kg suitcase, determine the final velocity of the baggage carrier if the first suitcase tossed onto the carrier is (a) the 15-kg suitcase, (b) the 20-kg suitcase.

SOLUTION

There are no horizontal external forces acting during the impacts. The baggage carrier is to coast between impacts.

(a) 15-kg suitcase tossed on carrier first:

Let v_1 be the common velocity of suitcase A and the carrier after the first impact and v_2 be the common velocity of the two suitcases and the carrier of the second impact.

Conservation of momentum:

$$(15)(3) = (40)v_1 \quad v_1 = 1.125 \text{ m/s} \rightarrow$$

20-kg suitcase tossed next:

$$(20)(2) + (40)(1.125) = 60v_2 \quad v_2 = 1.417 \text{ m/s} \rightarrow \blacktriangleleft$$

(b) 20-kg suitcase tossed on carrier first:

Let v'_1 be the common velocity of suitcase B and the carrier after the first impact and v'_2 be the common velocity of all after the second impact.

Conservation of momentum:

$$(20)(2) = (45)v'_1 \quad v'_1 = 0.8889 \text{ m/s} \rightarrow$$

15-kg suitcase tossed next:

$$(15)(3) + (45)(0.8889) = 60v'_2$$

$$v'_2 = 1.417 \text{ m/s} \rightarrow \blacktriangleleft$$



PROBLEM 14.2

An airline employee tosses two suitcases in rapid succession, with a horizontal velocity of 2.4 m/s, onto a 25-kg baggage carrier, which is initially at rest. (a) Knowing that the final velocity of the baggage carrier is 1.2 m/s and that the first suitcase the employee tosses onto the carrier has a mass of 15 kg, determine the mass of the other suitcase. (b) What would be the final velocity of the carrier if the employee reversed the order in which he tosses the suitcases?

SOLUTION

There are no horizontal external forces acting during the impacts. The baggage carrier is to coast between impacts.

(a) Conservation of momentum:

$$\begin{array}{c} m_B (2.4 \text{ m/s}) \\ \boxed{B} \end{array} + \begin{array}{c} (15 \text{ kg}) (2.4 \text{ m/s}) \\ \boxed{A} \end{array} = \begin{array}{c} (m_B + 15 + 25) v_{\text{FINAL}} \\ \boxed{B} \boxed{A} \end{array}$$

$$2.4(m_B + 15) = (m_B + 40)v_{\text{final}} \quad (1)$$

Let $v_{\text{final}} = 1.2 \text{ m/s}$:

$$2.4m_B + 36 = 1.2m_B + 48 \quad m_B = 10.00 \text{ kg} \quad \blacktriangleleft$$

(b) Conservation of momentum equation is still valid.

Let $m_B = 10.00 \text{ kg}$ in Eq. (1):

$$2.4(10 + 15) = (10 + 40)v_{\text{final}} \quad v_{\text{final}} = 1.200 \text{ m/s} \quad \rightarrow \blacktriangleleft$$



PROBLEM 14.3

A 90-kg man and a 60 kg woman stand side by side at the same end of a 150-kg boat ready to dive, each with a 5-m/s velocity relative to the boat. Determine the velocity of the boat after they have both dived, if (a) the woman dives first, (b) the man dives first.

SOLUTION

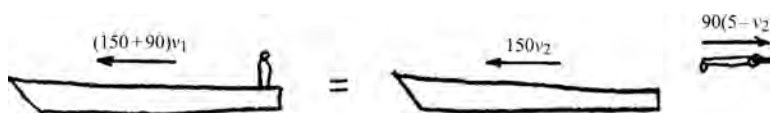
(a) Woman dives first.

Conservation of momentum:

$$60(5 - v_1) - (150 + 90)v_1 = 0$$

$$v_1 = \frac{300}{300} = 1 \text{ m/s} \leftarrow$$

Man dives next. Conservation of momentum:



$$-240v_1 = -150v_2 + 90(5 - v_2)$$

$$v_2 = \frac{240v_1 + 450}{240} = 2.875 \text{ m/s}$$

$$v_2 = 2.88 \text{ m/s} \leftarrow \blacktriangleleft$$

(b) Man dives first.

Conservation of momentum:

$$90(5 - v'_1) - 210v'_1 = 0$$

$$v'_1 = \frac{450}{300} = 1.5 \text{ m/s}$$

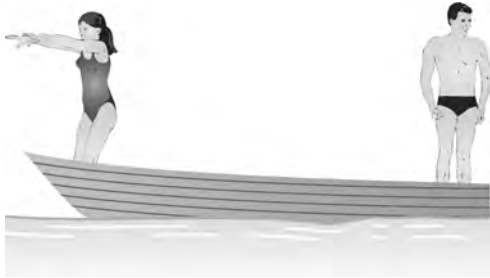
Woman dives next. Conservation of momentum:

$$-210v'_1 = -150v'_2 + 60(5 - v'_2)$$

$$v'_2 = \frac{210v'_1 + 300}{210} = 2.9286 \text{ m/s}$$

$$v'_2 = 2.93 \text{ m/s} \leftarrow \blacktriangleleft$$





PROBLEM 14.4

A 90-kg man and a 60-kg woman stand at opposite ends of a 150-kg boat, ready to dive, each with a 5-m/s velocity relative to the boat. Determine the velocity of the boat after they have both dived, if (a) the woman dives first, (b) the man dives first.

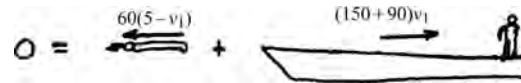
SOLUTION

(a) Woman dives first.

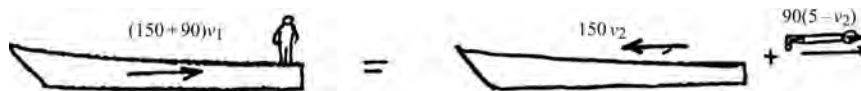
Conservation of momentum:

$$-60(5 - v_1) + 240v_1 = 0$$

$$v_1 = \frac{300}{300} = 1 \text{ m/s} \rightarrow$$



Man dives next. Conservation of momentum:



$$240v_1 = -150v_2 + 90(5 - v_2)$$

$$v_2 = \frac{450 + 240v_1}{240} = 0.875 \text{ m/s} \quad v_2 = 0.875 \text{ m/s} \leftarrow \blacktriangleleft$$

(b) Man dives first.

Conservation of momentum:

$$90(5 - v'_1) - 210v'_1 = 0$$

$$v'_1 = \frac{450}{300} = 1.5 \text{ m/s} \leftarrow$$

Woman dives next. Conservation of momentum:

$$-210v'_1 = 150v'_2 - 60(5 - v'_2)$$

$$v'_2 = -210v'_1 = 210v'_2 - 300$$

$$v'_2 = \frac{-210v'_1 + 300}{210} = \frac{-210 \times 1.5 + 300}{210} = -0.0714 \text{ m/s}$$

$$v'_2 = -0.0714 \text{ m/s} \leftarrow \blacktriangleleft$$



PROBLEM 14.5

A bullet is fired with a horizontal velocity of 450 m/s through a 3-kg block A and becomes embedded in a 2.5-kg block B. Knowing that blocks A and B start moving with velocities of 1.5 m/s and 2.7 m/s, respectively, determine (a) the weight of the bullet, (b) its velocity as it travels from block A to block B.

SOLUTION

The masses are m for the bullet and m_A and m_B for the blocks.

- (a) The bullet passes through block A and embeds in block B. Momentum is conserved.

$$\text{Initial momentum:} \quad mv_0 + m_A(0) + m_B(0) = mv_0$$

$$\text{Final momentum:} \quad mv_B + m_A v_A + m_B v_B$$

$$\text{Equating,} \quad mv_0 = mv_B + m_A v_A + m_B v_B$$

$$m = \frac{m_A v_A + m_B v_B}{v_0 - v_B} = \frac{(3)(1.5) + (2.5)(2.7)}{(450 - 2.7)} = 0.0252 \text{ kg}$$

$$m = 25.2 \text{ g} \quad \blacktriangleleft$$

- (b) The bullet passes through block A. Momentum is conserved.

$$\text{Initial momentum:} \quad mv_0 + m_A(0) = mv_0$$

$$\text{Final momentum:} \quad mv_1 + m_A v_A$$

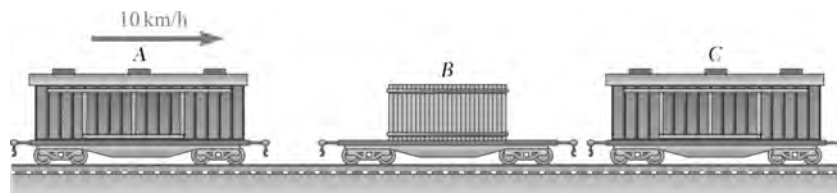
$$\text{Equating,} \quad mv_0 = mv_1 + m_A v_A$$

$$v_1 = \frac{mv_0 - m_A v_A}{m} = v_0 - \frac{m_A v_A}{m} = 450 - \frac{3(1.5)}{0.02515} = 271.07 \text{ m/s}$$

$$v_1 = 271 \text{ m/s} \rightarrow \quad \blacktriangleleft$$

PROBLEM 14.6

A 45,000 kg boxcar A is moving in a railroad switchyard with a velocity of 10 km/h toward cars B and C, which are both at rest with their brakes off at a short distance from each other. Car B is a 25000 kg flatcar supporting a 30,000 kg container, and car C is a 40,000 kg boxcar. As the cars hit each other, they get automatically and tightly coupled. Determine the velocity of car A immediately after each of the two couplings, assuming that the container (a) does not slide on the flatcar, (b) slides after the first coupling but hits a stop before the second coupling occurs, (c) slides and hits the stop only after the second coupling has occurred.



SOLUTION

Each term of the conservation of momentum equation is mass times velocity. As long as the same units are used in all terms, any unit may be used for mass and for velocity. We use kg for mass and km/h for velocity.

Conservation of momentum:

(a) Container does not slide

$$\begin{array}{c}
 (45,000)(10) \\
 \boxed{45,000 \text{ kg}} \\
 \longrightarrow
 \end{array}
 =
 \begin{array}{c}
 (100,000)v_1 \\
 \boxed{45,000 \text{ kg}} \quad \boxed{55,000 \text{ kg}} \\
 \longrightarrow
 \end{array}
 =
 \begin{array}{c}
 (140,000)v_2 \\
 \boxed{45,000 \text{ kg}} \quad \boxed{55,000 \text{ kg}} \quad \boxed{40,000 \text{ kg}} \\
 \longrightarrow
 \end{array}$$

$$450,000 = 100,000v_1 = 140,000v_2$$

$$v_1 = 4.50 \text{ km/h} \longrightarrow \blacktriangleleft$$

$$v_2 = 3.21 \text{ km/h} \longrightarrow \blacktriangleleft$$

(b) Container slides after 1st coupling, stops before 2nd

$$\begin{array}{c}
 (45,000)(10) \\
 \boxed{45,000 \text{ kg}} \\
 \longrightarrow
 \end{array}
 =
 \begin{array}{c}
 (70,000)v_1 \\
 \boxed{45,000 \text{ kg}} \quad \boxed{25,000 \text{ kg}} \\
 \longrightarrow
 \end{array}
 =
 \begin{array}{c}
 (140,000)v_2 \\
 \boxed{45,000 \text{ kg}} \quad \boxed{35,000 \text{ kg}} \quad \boxed{40,000 \text{ kg}} \\
 \longrightarrow
 \end{array}$$

$$(45,000)(10) = (70,000)v_1 = (140,000)v_2$$

$$v_1 = 6.43 \text{ km/h} \longrightarrow \blacktriangleleft$$

$$v_2 = 3.21 \text{ km/h} \longrightarrow \blacktriangleleft$$

(c) Container slides and stops only after 2nd coupling

$$\begin{array}{c}
 (45,000)(10) \\
 \boxed{45,000 \text{ kg}} \\
 \longrightarrow
 \end{array}
 =
 \begin{array}{c}
 (70,000)v_1 \\
 \boxed{45,000 \text{ kg}} \quad \boxed{25,000 \text{ kg}} \\
 \longrightarrow
 \end{array}
 =
 \begin{array}{c}
 (110,000)v_2 \\
 \boxed{45,000 \text{ kg}} \quad \boxed{25,000 \text{ kg}} \quad \boxed{40,000 \text{ kg}} \\
 \longrightarrow
 \end{array}$$

$$(45,000)(10) = (70,000)v_1 = (110,000)v_2$$

$$v_1 = 6.43 \text{ km/h} \longrightarrow \blacktriangleleft$$

$$v_2 = 4.09 \text{ km/h} \longrightarrow \blacktriangleleft$$

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PROBLEM 14.7

At an amusement park, there are 200-kg bumper cars A, B, and C that have riders with masses of 40 kg, 60 kg, and 35 kg, respectively. Car A is moving to the right with a velocity $\mathbf{v}_A = 2 \text{ m/s}$ and car C has a velocity $\mathbf{v}_B = 1.5 \text{ m/s}$ to the left, but car B is initially at rest. The coefficient of restitution between each car is 0.8. Determine the final velocity of each car, after all impacts, assuming (a) cars A and C hit car B at the same time, (b) car A hits car B before car C does.



SOLUTION

Assume that each car with its rider may be treated as a particle. The masses are:

$$m_A = 200 + 40 = 240 \text{ kg},$$

$$m_B = 200 + 60 = 260 \text{ kg},$$

$$m_C = 200 + 35 = 235 \text{ kg}.$$

Assume velocities are positive to the right. The initial velocities are:

$$v_A = 2 \text{ m/s} \quad v_B = 0 \quad v_C = -1.5 \text{ m/s}$$

Let v'_A , v'_B , and v'_C be the final velocities.

(a) Cars A and C hit B at the same time. Conservation of momentum for all three cars.

$$\begin{aligned} m_A v_A + m_B v_B + m_C v_C &= m_A v'_A + m_B v'_B + m_C v'_C \\ (240)(2) + 0 + (235)(-1.5) &= 240v'_A + 260v'_B + 235v'_C \end{aligned} \quad (1)$$

Coefficient of restitution for cars A and B.

$$v'_B - v'_A = e(v_A - v_B) = (0.8)(2 - 0) = 1.6 \quad (2)$$

Coefficient of restitution for cars B and C.

$$v'_C - v'_B = e(v_B - v_C) = (0.8)[0 - (-1.5)] = 1.2 \quad (3)$$

Solving Eqs. (1), (2), and (3) simultaneously,

$$v'_A = -1.288 \text{ m/s} \quad v'_B = 0.312 \text{ m/s} \quad v'_C = 1.512 \text{ m/s}$$

$$\mathbf{v}'_A = 1.288 \text{ m/s} \leftarrow \blacktriangleleft$$

$$\mathbf{v}'_B = 0.312 \text{ m/s} \rightarrow \blacktriangleleft$$

$$\mathbf{v}'_C = 1.512 \text{ m/s} \rightarrow \blacktriangleleft$$

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PROBLEM 14.7 (Continued)

(b) Car A hits Car B before C does.

First impact. Car A hits car B. Let v'_A and v'_B be the velocities after this impact. Conservation of momentum for cars A and B.

$$\begin{aligned} m_A v_A + m_B v_B &= m_A v'_A + m_B v'_B \\ (240)(2) + 0 &= 240v'_A + 260v'_B \end{aligned} \quad (4)$$

Coefficient of restitution for cars A and B.

$$v'_B - v'_A = e(v_A - v_B) = (0.8)(2 - 0) = 1.6 \quad (5)$$

Solving Eqs. (4) and (5) simultaneously,

$$v'_A = 0.128 \text{ m/s}, \quad v'_B = 1.728 \text{ m/s}$$

$$\mathbf{v}'_A = 0.128 \text{ m/s} \rightarrow$$

$$\mathbf{v}'_B = 1.728 \text{ m/s} \rightarrow$$

Second impact. Cars B and C hit. Let v''_B and v''_C be the velocities after this impact. Conservation of momentum for cars B and C.

$$\begin{aligned} m_B v'_B + m_C v_C &= m_B v''_B + m_C v''_C \\ (260)(1.728) + (235)(-1.5) &= 260v''_B + 235v''_C \end{aligned} \quad (6)$$

Coefficient of restitution for cars B and C.

$$v''_C - v''_B = e(v'_B - v_C) = (0.8)[1.728 - (-1.5)] = 2.5824 \quad (7)$$

Solving Eqs. (6) and (7) simultaneously,

$$v''_B = -1.03047 \text{ m/s} \quad v''_C = 1.55193 \text{ m/s}$$

$$\mathbf{v}''_B = 1.03047 \text{ m/s} \leftarrow$$

$$\mathbf{v}''_C = 1.55193 \text{ m/s} \rightarrow$$

Third impact. Cars A and B hit again. Let v'''_A and v'''_B be the velocities after this impact. Conservation of momentum for cars A and B.

$$\begin{aligned} m_A v'_A + m_B v''_B &= m_A v'''_A + m_B v'''_B \\ (240)(0.128) + (260)(-1.03047) &= 240v'''_A + 260v'''_B \end{aligned} \quad (8)$$

Coefficient of restitution for cars A and B.

$$v'''_B - v'''_A = e(v'_A - v''_B) = (0.8)[0.128 - (-1.03047)] = 0.926776 \quad (9)$$

Solving Eqs. (8) and (9) simultaneously,

$$v'''_A = -0.95633 \text{ m/s}$$

$$v'''_B = -0.02955 \text{ m/s}$$

$$\mathbf{v}'''_A = 0.95633 \text{ m/s} \leftarrow$$

$$\mathbf{v}'''_B = 0.02955 \text{ m/s} \leftarrow$$

PROBLEM 14.7 (Continued)

There are no more impacts. The final velocities are:

$$\mathbf{v}_A''' = 0.956 \text{ m/s} \leftarrow \blacktriangleleft$$

$$\mathbf{v}_B''' = 0.0296 \text{ m/s} \leftarrow \blacktriangleleft$$

$$\mathbf{v}_C'' = 1.552 \text{ m/s} \rightarrow \blacktriangleleft$$

We may check our results by considering conservation of momentum of all three cars over all three impacts.

$$\begin{aligned} m_A v_A + m_B v_B + m_C v_C &= (240)(2) + 0 + (235)(-1.5) \\ &= 127.5 \text{ kg} \cdot \text{m/s} \end{aligned}$$

$$\begin{aligned} m_A v_A''' + m_B v_B''' + m_C v_C'' &= (240)(-0.95633) + (260)(-0.02955) + (235)(1.55193) \\ &= 127.50 \text{ kg} \cdot \text{m/s}. \end{aligned}$$

PROBLEM 14.8

At an amusement park, there are 200-kg bumper cars A , B , and C that have riders with masses of 40 kg, 60 kg, and 35 kg, respectively. Car A is moving to the right with a velocity $\mathbf{v}_A = 2$ m/s when it hits stationary car B . The coefficient of restitution between each car is 0.8. Determine the velocity of car C so that after car B collides with car C , the velocity of car B is zero.



SOLUTION

Assume that each car with its rider may be treated as a particle. The masses are:

$$m_A = 200 + 40 = 240 \text{ kg}$$

$$m_B = 200 + 60 = 260 \text{ kg}$$

$$m_C = 200 + 35 = 235 \text{ kg}$$

Assume velocities are positive to the right. The initial velocities are:

$$v_A = 2 \text{ m/s}, \quad v_B = 0, \quad v_C = ?$$

First impact. Car A hits car B . Let v'_A and v'_B be the velocities after this impact. Conservation of momentum for cars A and B .

$$\begin{aligned} m_A v_A + m_B v_B &= m_A v'_A + m_B v'_B \\ (240)(2) + 0 &= 240 v'_A + 260 v'_B \end{aligned} \quad (1)$$

Coefficient of restitution for cars A and B .

$$v'_B - v'_A = e(v_A - v_B) = (0.8)(2 - 0) = 1.6 \quad (2)$$

Solving Eqs. (1) and (2) simultaneously,

$$v'_A = 0.128 \text{ m/s}$$

$$v'_B = 1.728 \text{ m/s}$$

$$\mathbf{v}'_A = 0.128 \text{ m/s} \rightarrow$$

$$\mathbf{v}'_B = 1.728 \text{ m/s} \rightarrow$$

Second impact. Cars B and C hit. Let v''_B and v''_C be the velocities after this impact. $v''_B = 0$. Coefficient of restitution for cars B and C .

$$\begin{aligned} v''_C - v''_B &= e(v'_B - v'_C) = (0.8)(1.728 - v'_C) \\ v''_C &= 1.3824 - 0.8v'_C \end{aligned}$$

PROBLEM 14.8 (Continued)

Conservation of momentum for cars B and C .

$$m_B v_B' + m_C v_C = m_B v_B'' + m_C v_C''$$

$$(260)(1.728) + 235v_C = (260)(0) + (235)(1.3824 - 0.8v_C)$$

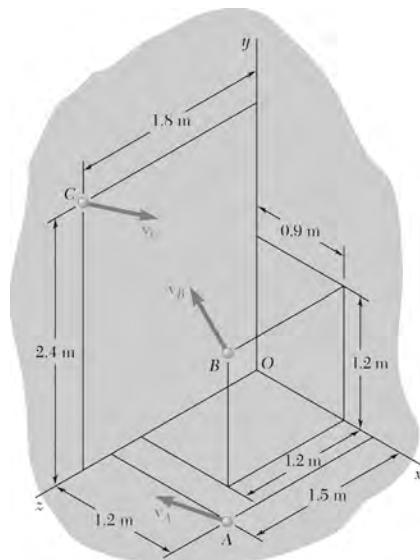
$$(235)(1.8)v_C = (235)(1.3824) - (260)(1.728)$$

$$v_C = -0.294 \text{ m/s}$$

$$\mathbf{v}_C = 0.294 \text{ m/s} \leftarrow \blacktriangleleft$$

Note: There will be another impact between cars A and B .

PROBLEM 14.9



A system consists of three particles A, B, and C. We know that $m_A = 3$ kg, $m_B = 4$ kg, and $m_C = 5$ kg, and that the velocities of the particles expressed in m/s are, respectively, $\mathbf{v}_A = -4\mathbf{i} + 4\mathbf{j} + 6\mathbf{k}$, $\mathbf{v}_B = -6\mathbf{i} + 8\mathbf{j} + 4\mathbf{k}$, and $\mathbf{v}_C = 2\mathbf{i} - 6\mathbf{j} - 4\mathbf{k}$. Determine the angular momentum $\mathbf{H}_O$ of the system about O.

SOLUTION

Linear momentum of each particle (kg·m/s):

$$m_A \mathbf{v}_A = -12\mathbf{i} + 12\mathbf{j} + 18\mathbf{k}$$

$$m_B \mathbf{v}_B = -24\mathbf{i} + 32\mathbf{j} + 16\mathbf{k}$$

$$m_C \mathbf{v}_C = 10\mathbf{i} - 30\mathbf{j} - 20\mathbf{k}$$

Position vectors, (meters):

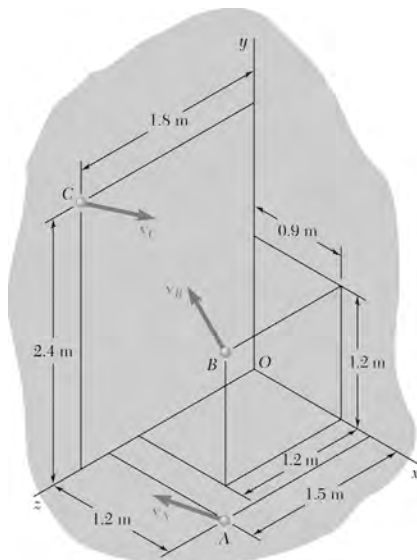
$$\mathbf{r}_A = 1.2\mathbf{i} + 1.5\mathbf{k}, \quad \mathbf{r}_B = 0.9\mathbf{i} + 1.2\mathbf{j} + 1.2\mathbf{k}, \quad \mathbf{r}_C = 2.4\mathbf{j} + 1.8\mathbf{k}$$

Angular momentum about O (kg·m²/s):

$$\mathbf{H}_O = \mathbf{r}_A \times m_A \mathbf{v}_A + \mathbf{r}_B \times m_B \mathbf{v}_B + \mathbf{r}_C \times m_C \mathbf{v}_C$$

$$\begin{aligned} &= \begin{vmatrix} \mathbf{i} & \mathbf{j} & \mathbf{k} \\ 1.2 & 0 & 1.5 \\ -12 & 12 & 18 \end{vmatrix} + \begin{vmatrix} \mathbf{i} & \mathbf{j} & \mathbf{k} \\ 0.9 & 1.2 & 1.2 \\ -24 & 32 & 16 \end{vmatrix} + \begin{vmatrix} \mathbf{i} & \mathbf{j} & \mathbf{k} \\ 0 & 2.4 & 1.8 \\ 10 & -30 & -20 \end{vmatrix} \\ &= (-18\mathbf{i} - 39.6\mathbf{j} + 14.4\mathbf{k}) + (-19.2\mathbf{i} - 43.2\mathbf{j} + 57.6\mathbf{k}) + (6\mathbf{i} + 18\mathbf{j} - 24\mathbf{k}) \\ &= -31.2\mathbf{i} - 64.8\mathbf{j} + 48.0\mathbf{k} \end{aligned}$$

$$\mathbf{H}_O = -(31.2 \text{ kg} \cdot \text{m}^2/\text{s})\mathbf{i} - (64.8 \text{ kg} \cdot \text{m}^2/\text{s})\mathbf{j} + (48.0 \text{ kg} \cdot \text{m}^2/\text{s})\mathbf{k} \quad \blacktriangleleft$$



PROBLEM 14.10

For the system of particles of Problem 14.9, determine (a) the position vector $\bar{\mathbf{r}}$ of the mass center G of the system, (b) the linear momentum $m\bar{\mathbf{v}}$ of the system, (c) the angular momentum $\mathbf{H}_G$ of the system about G . Also verify that the answers to this problem and to Problem 14.9 satisfy the equation given in Problem 14.27.

PROBLEM 14.9 A system consists of three particles A , B , and C . We know that $m_A = 3$ kg, $m_B = 4$ kg, and $m_C = 5$ kg and that the velocities of the particles expressed in m/s are, respectively, $\mathbf{v}_A = -4\mathbf{i} + 4\mathbf{j} + 6\mathbf{k}$, $\mathbf{v}_B = -6\mathbf{i} + 8\mathbf{j} + 4\mathbf{k}$, and $\mathbf{v}_C = 2\mathbf{i} - 6\mathbf{j} - 4\mathbf{k}$. Determine the angular momentum $\mathbf{H}_O$ of the system about O .

SOLUTION

Position vectors, (meters):

$$\mathbf{r}_A = 1.2\mathbf{i} + 1.5\mathbf{k}, \quad \mathbf{r}_B = 0.9\mathbf{i} + 1.2\mathbf{j} + 1.2\mathbf{k}, \quad \mathbf{r}_C = 2.4\mathbf{j} + 1.8\mathbf{k}$$

(a) Mass center:

$$(m_A + m_B + m_C)\bar{\mathbf{r}} = m_A\mathbf{r}_A + m_B\mathbf{r}_B + m_C\mathbf{r}_C$$

$$12\bar{\mathbf{r}} = (3)(1.2\mathbf{i} + 1.5\mathbf{k}) + (4)(0.9\mathbf{i} + 1.2\mathbf{j} + 1.2\mathbf{k}) + (5)(2.4\mathbf{j} + 1.8\mathbf{k})$$

$$\bar{\mathbf{r}} = 0.6\mathbf{i} + 1.4\mathbf{j} + 1.525\mathbf{k}$$

$$\bar{\mathbf{r}} = (0.600 \text{ m})\mathbf{i} + (1.400 \text{ m})\mathbf{j} + (1.525 \text{ m})\mathbf{k} \quad \blacktriangleleft$$

Linear momentum of each particle (kg · m/s):

$$m_A\mathbf{v}_A = -12\mathbf{i} + 12\mathbf{j} + 18\mathbf{k}$$

$$m_B\mathbf{v}_B = -24\mathbf{i} + 32\mathbf{j} + 16\mathbf{k}$$

$$m_C\mathbf{v}_C = 10\mathbf{i} - 30\mathbf{j} - 20\mathbf{k}$$

(b) Linear momentum of the system (kg · m/s):

$$m\bar{\mathbf{v}} = m_A\mathbf{v}_A + m_B\mathbf{v}_B + m_C\mathbf{v}_C = -26\mathbf{i} + 14\mathbf{j} + 14\mathbf{k}$$

$$m\bar{\mathbf{v}} = -(26.0 \text{ kg} \cdot \text{m/s})\mathbf{i} + (14.00 \text{ kg} \cdot \text{m/s})\mathbf{j} + (14.00 \text{ kg} \cdot \text{m/s})\mathbf{k} \quad \blacktriangleleft$$

Position vectors relative to the mass center (meters).

$$\mathbf{r}'_A = \mathbf{r}_A - \bar{\mathbf{r}} = 0.6\mathbf{i} - 1.4\mathbf{j} - 0.025\mathbf{k}$$

$$\mathbf{r}'_B = \mathbf{r}_B - \bar{\mathbf{r}} = 0.3\mathbf{i} - 0.2\mathbf{j} - 0.325\mathbf{k}$$

$$\mathbf{r}'_C = \mathbf{r}_C - \bar{\mathbf{r}} = -0.6\mathbf{i} + 1.0\mathbf{j} + 0.275\mathbf{k}$$

PROBLEM 14.10 (Continued)

(c) Angular momentum about G ($\text{kg} \cdot \text{m}^2/\text{s}$):

$$\begin{aligned}\mathbf{H}_G &= \mathbf{r}'_A \times m_A \mathbf{v}_A + \mathbf{r}'_B \times m_B \mathbf{v}_B + \mathbf{r}'_C \times m_C \mathbf{v}_C \\ &= \begin{vmatrix} \mathbf{i} & \mathbf{j} & \mathbf{k} \\ 0.6 & -1.4 & -0.025 \\ -12 & 12 & 18 \end{vmatrix} + \begin{vmatrix} \mathbf{i} & \mathbf{j} & \mathbf{k} \\ 0.3 & -0.2 & -0.325 \\ -24 & 32 & 16 \end{vmatrix} + \begin{vmatrix} \mathbf{i} & \mathbf{j} & \mathbf{k} \\ -0.6 & 1.0 & 0.275 \\ 10 & -30 & -20 \end{vmatrix} \\ &= (-24.9\mathbf{i} - 10.5\mathbf{j} - 9.6\mathbf{k}) + (7.2\mathbf{i} + 3.0\mathbf{j} + 4.8\mathbf{k}) + (-11.75\mathbf{i} - 9.25\mathbf{j} + 8.0\mathbf{k}) \\ &= -29.45\mathbf{i} - 16.75\mathbf{j} + 3.2\mathbf{k}\end{aligned}$$

$$\mathbf{H}_G = -(29.5 \text{ kg} \cdot \text{m}^2/\text{s})\mathbf{i} - (16.75 \text{ kg} \cdot \text{m}^2/\text{s})\mathbf{j} + (3.20 \text{ kg} \cdot \text{m}^2/\text{s})\mathbf{k} \quad \blacktriangleleft$$

$$\bar{\mathbf{r}} \times m\bar{\mathbf{v}} = \begin{vmatrix} \mathbf{i} & \mathbf{j} & \mathbf{k} \\ 0.6 & 1.4 & 1.525 \\ -26 & 14 & 14 \end{vmatrix} = -1.75\mathbf{i} - 48.05\mathbf{j} + 44.8\mathbf{k}$$

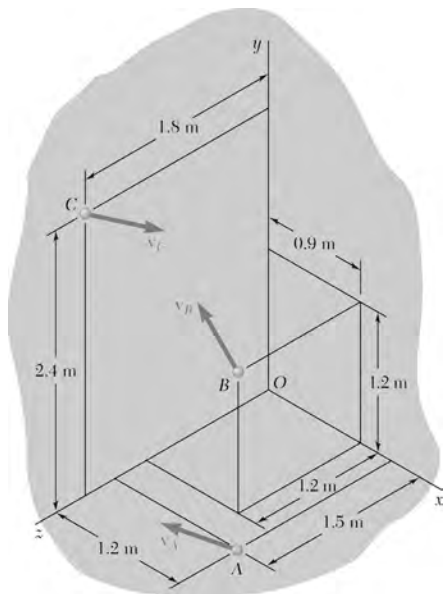
$$\mathbf{H}_G + \bar{\mathbf{r}} \times m\bar{\mathbf{v}} = -(31.2 \text{ kg} \cdot \text{m}^2/\text{s})\mathbf{i} - (64.8 \text{ kg} \cdot \text{m}^2/\text{s})\mathbf{j} + (48.0 \text{ kg} \cdot \text{m}^2/\text{s})\mathbf{k}$$

Angular momentum about O ($\text{kg} \cdot \text{m}^2/\text{s}$):

$$\begin{aligned}\mathbf{H}_O &= \mathbf{r}_A \times m_A \mathbf{v}_A + \mathbf{r}_B \times m_B \mathbf{v}_B + \mathbf{r}_C \times m_C \mathbf{v}_C \\ &= \begin{vmatrix} \mathbf{i} & \mathbf{j} & \mathbf{k} \\ 1.2 & 0 & 1.5 \\ -12 & 12 & 18 \end{vmatrix} + \begin{vmatrix} \mathbf{i} & \mathbf{j} & \mathbf{k} \\ 0.9 & 1.2 & 1.2 \\ -24 & 32 & 16 \end{vmatrix} + \begin{vmatrix} \mathbf{i} & \mathbf{j} & \mathbf{k} \\ 0 & 2.4 & 1.8 \\ 10 & -30 & -20 \end{vmatrix} \\ &= (-18\mathbf{i} - 39.6\mathbf{j} + 14.4\mathbf{k}) + (-19.2\mathbf{i} - 43.2\mathbf{j} + 57.6\mathbf{k}) + (6\mathbf{i} + 18\mathbf{j} - 24\mathbf{k}) \\ &= -(31.2 \text{ kg} \cdot \text{m}^2/\text{s})\mathbf{i} - (64.8 \text{ kg} \cdot \text{m}^2/\text{s})\mathbf{j} + (48.0 \text{ kg} \cdot \text{m}^2/\text{s})\mathbf{k}\end{aligned}$$

Note that

$$\mathbf{H}_O = \mathbf{H}_G + \bar{\mathbf{r}} \times m\bar{\mathbf{v}}.$$



PROBLEM 14.11

A system consists of three particles A, B, and C. We know that $m_A = 3$ kg, $m_B = 4$ kg, and $m_C = 5$ kg and that the velocities of the particles expressed in m/s are, respectively, $\mathbf{v}_A = -4\mathbf{i} + 4\mathbf{j} + 6\mathbf{k}$, $\mathbf{v}_B = v_x\mathbf{i} + v_y\mathbf{j} + 4\mathbf{k}$, and $\mathbf{v}_C = 2\mathbf{i} - 6\mathbf{j} - 4\mathbf{k}$. Determine (a) the components v_x and v_y of the velocity of particle B for which the angular momentum $\mathbf{H}_O$ of the system about O is parallel to the z axis, (b) the corresponding value of $\mathbf{H}_O$.

SOLUTION

$$\mathbf{H}_O = \mathbf{r}_A \times m_A \mathbf{v}_A + \mathbf{r}_B \times m_B \mathbf{v}_B + \mathbf{r}_C \times m_C \mathbf{v}_C$$

$$= (3 \text{ kg}) \begin{vmatrix} \mathbf{i} & \mathbf{j} & \mathbf{k} \\ 1.2 \text{ m} & 0 & 1.5 \text{ m} \\ -4 \text{ m/s} & 4 \text{ m/s} & 6 \text{ m/s} \end{vmatrix} + 4 \begin{vmatrix} \mathbf{i} & \mathbf{j} & \mathbf{k} \\ 0.9 & 1.2 & 1.2 \\ v_x & v_y & 4 \end{vmatrix} + 5 \begin{vmatrix} \mathbf{i} & \mathbf{j} & \mathbf{k} \\ 0 & 2.4 & 1.8 \\ 2 & -6 & -4 \end{vmatrix}$$

$$= -18\mathbf{i} - 39.6\mathbf{j} + 14.4\mathbf{k} + (19.2 - 4.8v_y)\mathbf{i} + (4.8v_x - 14.4)\mathbf{j} + (3.6v_y - 4.8v_x)\mathbf{k} + 6\mathbf{i} + 18\mathbf{j} - 24\mathbf{k}$$

$$\mathbf{H}_O = (7.2 - 4.8v_y)\mathbf{i} + (-36 + 4.8v_x)\mathbf{j} + (-9.6 + 3.6v_y - 4.8v_x)\mathbf{k}$$

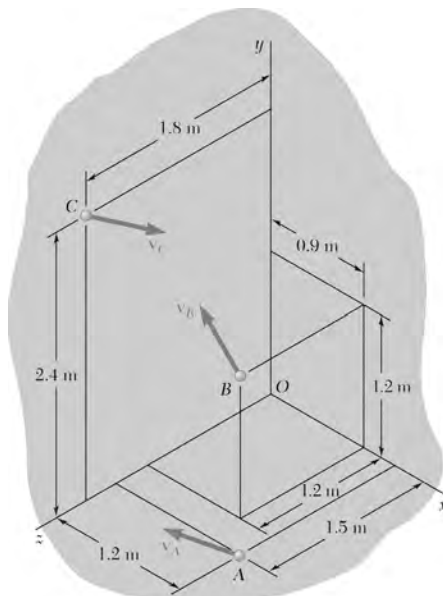
(a) For $\mathbf{H}_O$ to be parallel to the z axis:

$$H_x = 7.2 - 4.8v_y = 0 \quad H_y = -36 + 4.8v_x = 0$$

$$v_x = 7.50 \text{ m/s}, \quad v_y = 1.500 \text{ m/s} \quad \blacktriangleleft$$

(b) $\mathbf{H}_O = H_z \mathbf{k} = (-9.6 + 3.6 \times 1.500 - 4.8 \times 7.50)\mathbf{k}$

$$\mathbf{H}_O = -(40.2 \text{ kg} \cdot \text{m}^2/\text{s})\mathbf{k} \quad \blacktriangleleft$$



PROBLEM 14.12

For the system of particles of Problem 14.11, determine (a) the components v_x and v_y of the velocity of particle B for which the angular momentum $\mathbf{H}_O$ of the system about O is parallel to the y-axis, (b) the corresponding value of $\mathbf{H}_O$.

PROBLEM 14.11 A system consists of three particles A, B, and C. We know that $m_A = 3$ kg, $m_B = 4$ kg, and $m_C = 5$ kg and that the velocities of the particles expressed in m/s are, respectively, $\mathbf{v}_A = -4\mathbf{i} + 4\mathbf{j} + 6\mathbf{k}$, $\mathbf{v}_B = v_x\mathbf{i} + v_y\mathbf{j} + 4\mathbf{k}$, and $\mathbf{v}_C = -2\mathbf{i} - 6\mathbf{j} - 4\mathbf{k}$. Determine (a) the components v_x and v_y of the velocity of particle B for which the angular momentum $\mathbf{H}_O$ of the system about O is parallel to the z axis, (b) the corresponding value of $\mathbf{H}_O$.

SOLUTION

$$\mathbf{H}_O = \mathbf{r}_A \times m_A \mathbf{v}_A + \mathbf{r}_B \times m_B \mathbf{v}_B + \mathbf{r}_C \times m_C \mathbf{v}_C$$

$$= (3 \text{ kg}) \begin{vmatrix} \mathbf{i} & \mathbf{j} & \mathbf{k} \\ 1.2 \text{ m} & 0 & 1.5 \text{ m} \\ -4 \text{ m/s} & 4 \text{ m/s} & 6 \text{ m/s} \end{vmatrix} + 4 \begin{vmatrix} \mathbf{i} & \mathbf{j} & \mathbf{k} \\ 0.9 & 1.2 & 1.2 \\ v_x & v_y & 4 \end{vmatrix} + 5 \begin{vmatrix} \mathbf{i} & \mathbf{j} & \mathbf{k} \\ 0 & 2.4 & 1.8 \\ 2 & -6 & -4 \end{vmatrix}$$

$$= -18\mathbf{i} - 39.6\mathbf{j} + 14.4\mathbf{k} + (19.2 - 4.8v_y)\mathbf{i} + (4.8v_x - 14.4)\mathbf{j} + (3.6v_y - 4.8v_x)\mathbf{k} + 6\mathbf{i} + 18\mathbf{j} - 24\mathbf{k}$$

$$\mathbf{H}_O = (7.2 - 4.8v_y)\mathbf{i} + (-36 + 4.8v_x)\mathbf{j} + (-9.6 + 3.6v_y - 4.8v_x)\mathbf{k}$$

(a) For $\mathbf{H}_O$ to be parallel to the y axis:

$$H_x = 7.2 - 4.8v_y = 0 \quad H_z = -9.6 + 3.6v_y - 4.8v_x = 0$$

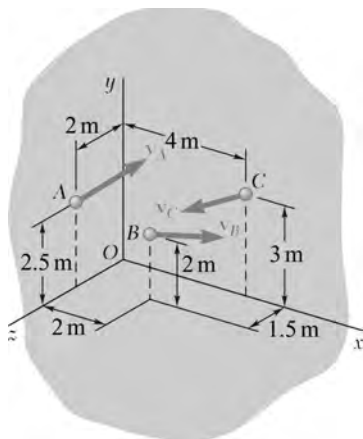
$$v_y = 1.500 \text{ m/s} \quad -9.6 + 3.6(1.500) - 4.8v_x = 0$$

$$v_x = -0.875 \text{ m/s}$$

$$v_x = -0.875 \text{ m/s}, \quad v_y = 1.500 \text{ m/s} \quad \blacktriangleleft$$

(b) $\mathbf{H}_O = H_y \mathbf{j} = [-36 + 4.8 \times (-0.875)]\mathbf{j}$

$$\mathbf{H}_O = -(40.2 \text{ kg} \cdot \text{m}^2/\text{s})\mathbf{j} \quad \blacktriangleleft$$



PROBLEM 14.13

A system consists of three particles A, B, and C. We know that $m_A = 2.5$ kg, $m_B = 2$ kg, and $m_C = 1.5$ kg, and that the velocities of the particles expressed in m/s are, respectively, $\mathbf{v}_A = \mathbf{i} + 1.5\mathbf{j} - \mathbf{k}$, $\mathbf{v}_B = v_x\mathbf{i} + v_y\mathbf{j} + v_z\mathbf{k}$, and $\mathbf{v}_C = -1.5\mathbf{i} - \mathbf{j} + 0.5\mathbf{k}$. Determine (a) the components v_x and v_y of the velocity of particle B for which the angular momentum $\mathbf{H}_O$ of the system about O is parallel to the x axis, (b) the value of $\mathbf{H}_O$.

SOLUTION

$$\begin{aligned}\mathbf{H}_O &= \sum \mathbf{r}_i \times m\mathbf{v}_i = \sum m_i \begin{vmatrix} \mathbf{i} & \mathbf{j} & \mathbf{k} \\ x_i & y_i & z_i \\ (v_i)_x & (v_i)_y & (v_i)_z \end{vmatrix} \\ &= 2.5 \begin{vmatrix} \mathbf{i} & \mathbf{j} & \mathbf{k} \\ 0 & 2.5 & 2 \\ 1 & 1.5 & -1 \end{vmatrix} + 2 \begin{vmatrix} \mathbf{i} & \mathbf{j} & \mathbf{k} \\ 2 & 2 & 1.5 \\ v_x & v_y & v_z \end{vmatrix} + 1.5 \begin{vmatrix} \mathbf{i} & \mathbf{j} & \mathbf{k} \\ 4 & 3 & 0 \\ -1.5 & -1 & 0.5 \end{vmatrix} \\ &= [2.5(-2.5 - 3) + 2(2v_z - 1.5v_y) + 1.5(1.5 - 0)]\mathbf{i} \\ &\quad + [2.5(2 - 0) + 2(1.5v_x - 2v_z + 1.5(0.2))]\mathbf{j} \\ &\quad + [2.5(0 - 2.5) + 2(2v_y - 2v_x) + 1.5(-4 + 4.5)]\mathbf{k} \\ \mathbf{H}_O &= [(4v_z - 3v_y - 11.5)\mathbf{i} + (3v_x - 4v_z + 2)\mathbf{j} + (-4v_x + 4v_y - 5.5)\mathbf{k}] \quad (1)\end{aligned}$$

(a) For $\mathbf{H}_O$ to be parallel to the x axis, we must have $H_y = H_z = 0$:

$$H_z = 0: \quad -4v_x + 4v_y - 5.5 = 0 \quad v_x = \frac{1}{4}(4v_y - 5.5) \text{ m/s} \quad \blacktriangleleft$$

$$H_y = 0: \quad (3v_x - 4v_z + 2) = 0 \quad v_z = \frac{1}{4}(3v_y - 2.125) \text{ m/s} \quad \blacktriangleleft$$

(b) Substitute into Eq. (1):

$$v_z = \frac{1}{4}(3v_x + 2) = \frac{1}{4} \left\{ \frac{3}{4}(4v_y - 5.5) + 2 \right\}$$

$$\mathbf{H}_O = (4v_z - 3v_y - 11.5)\mathbf{i} = (3v_y - 2.125 - 3v_y - 11.5)\mathbf{i} = 13.625 \mathbf{i} \text{ N} \cdot \text{m} \cdot \text{s}$$

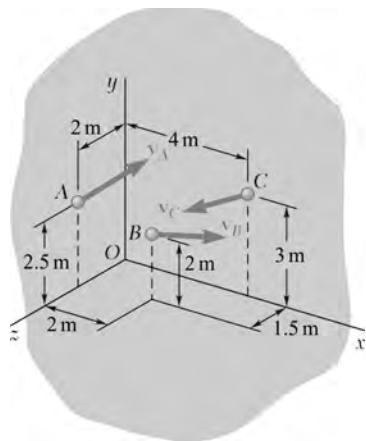
$$\mathbf{H}_O = (-13.63 \text{ N} \cdot \text{m} \cdot \text{s})\mathbf{i} \quad \blacktriangleleft$$

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PROBLEM 14.14

For the system of particles of Problem 14.13, determine (a) the components v_x and v_z of the velocity of particle B for which the angular momentum $\mathbf{H}_O$ of the system about O is parallel to the z axis, (b) the value of $\mathbf{H}_O$.

PROBLEM 14.13 A system consists of three particles A , B , and C . We know that $m_A = 2.5 \text{ kg}$, $m_B = 2 \text{ kg}$, and $m_C = 1.5 \text{ kg}$ and that the velocities of the particles expressed in m/s are, respectively, $\mathbf{v}_A = \mathbf{i} + 1.5\mathbf{j} - \mathbf{k}$, $\mathbf{v}_B = v_x\mathbf{i} + v_y\mathbf{j} + v_z\mathbf{k}$, and $\mathbf{v}_C = -1.5\mathbf{i} - \mathbf{j} + 0.5\mathbf{k}$. Determine (a) the components v_x and v_z of the velocity of particle B for which the angular momentum $\mathbf{H}_O$ of the system about O is parallel to the x axis, (b) the value of $\mathbf{H}_O$.



SOLUTION

$$\begin{aligned}\mathbf{H}_O &= \sum \mathbf{r}_i \times m\mathbf{v}_i = \sum m_i \begin{vmatrix} \mathbf{i} & \mathbf{j} & \mathbf{k} \\ x_i & y_i & z_i \\ (v_i)_x & (v_i)_y & (v_i)_z \end{vmatrix} \\ &= 2.5 \begin{vmatrix} \mathbf{i} & \mathbf{j} & \mathbf{k} \\ 0 & 2.5 & 2 \\ 1 & 1.5 & -1 \end{vmatrix} + 2 \begin{vmatrix} \mathbf{i} & \mathbf{j} & \mathbf{k} \\ 2 & 2 & 1.5 \\ v_x & v_y & v_z \end{vmatrix} + 1.5 \begin{vmatrix} \mathbf{i} & \mathbf{j} & \mathbf{k} \\ 4 & 3 & 0 \\ -1.5 & -1 & 0.5 \end{vmatrix} \\ &= [2.5(-2.5 - 3) + 2(2v_z - 1.5v_y) + 1.5(1.5 - 0)]\mathbf{i} \\ &\quad + [2.5(2 - 0) + 2(1.5v_x - 2v_z + 1.5(0.2))]\mathbf{j} \\ &\quad + [2.5(0 - 2.5) + 2(2v_y - 2v_x) + 1.5(-4 + 4.5)]\mathbf{k} \\ \mathbf{H}_O &= [(4v_z - 3v_y - 11.5)\mathbf{i} + (3v_x - 4v_z + 2)\mathbf{j} + (-4v_x + 4v_y - 5.5)\mathbf{k}] \quad (1)\end{aligned}$$

(a) For $\mathbf{H}_O$ to be parallel to the z axis, we must have $H_x = H_y = 0$:

$$H_x = 0: \quad 4v_z - 3v_y - 11.5 = 0 \quad v_z = \frac{1}{4}(3v_y + 11.5) \text{ m/s} \quad \blacktriangleleft$$

$$H_y = 0: \quad 3v_x - 4v_z + 2 = 0$$

$$3v_x - (3v_y + 11.5) = 0$$

$$v_x = \frac{1}{3}(3v_y + 11.5) \quad v_x = \frac{1}{3}(3v_y + 11.5) \text{ m/s} \quad \blacktriangleleft$$

(b) Substituting into Eq. (1):

$$\mathbf{H}_O = \frac{-4}{3}(3v_y + 11.5) + (3v_y + 11.5) - 5.5 \left(-v_y - \frac{28}{3} \right) \mathbf{k} \quad \mathbf{H}_O = - \left(v_y + \frac{28}{3} \right) \text{ N} \cdot \text{m} \cdot \text{s} \quad \blacktriangleleft$$

PROBLEM 14.15

A 450-kg space vehicle traveling with a velocity $\mathbf{v}_0 = (360 \text{ m/s})\mathbf{i}$ passes through the origin O at $t = 0$. Explosive charges then separate the vehicle into three Parts A, B, and C, weighing, respectively, 225 kg, 150 kg, and 75 kg. Knowing that at $t = 4 \text{ s}$, the positions of Parts A and B are observed to be A (1200 m, -300 m , -600 m) and B (1300 m, 3500 m, 800 m), determine the corresponding position of Part C. Neglect the effect of gravity.

SOLUTION

Motion of mass center:

Since there is no external force,

$$\bar{\mathbf{r}} = \mathbf{v}_0 t = (360 \text{ m/s})(4 \text{ s}) = (1440 \text{ m})\mathbf{i}$$

Equation (14.12):

$$m\bar{\mathbf{r}} = \sum m_i \mathbf{r}_i:$$

$$(450)(1440\mathbf{i}) = 225(1200\mathbf{i} - 300\mathbf{j} - 600\mathbf{k})$$

$$+ 150(1300\mathbf{i} + 350\mathbf{j} + 800\mathbf{k})$$

$$+ 75\mathbf{r}_C$$

$$75\mathbf{r}_C = (648000 - 270000 - 195000)\mathbf{i}$$

$$+ (67500 - 52500)\mathbf{j}$$

$$+ (135000 - 120000)\mathbf{k}$$

$$75\mathbf{r}_C = (183000\mathbf{i} + 15000\mathbf{j} + 15000\mathbf{k})$$

$$\mathbf{r}_C = (2440 \text{ m})\mathbf{i} + (200 \text{ m})\mathbf{j} + (200 \text{ m})\mathbf{k} \quad \blacktriangleleft$$

PROBLEM 14.16

A 15-kg projectile is passing through the origin O with a velocity $\mathbf{v}_0 = (40 \text{ m/s})\mathbf{i}$ when it explodes into two fragments A and B , of weight 6 kg and 9 kg, respectively. Knowing that 3 s later the position of fragment A is (100 m, 10 m, -20 m), determine the position of fragment B at the same instant. Assume $a_y = -g = -9.81 \text{ m/s}^2$ and neglect air resistance.

SOLUTION

Motion of mass center:

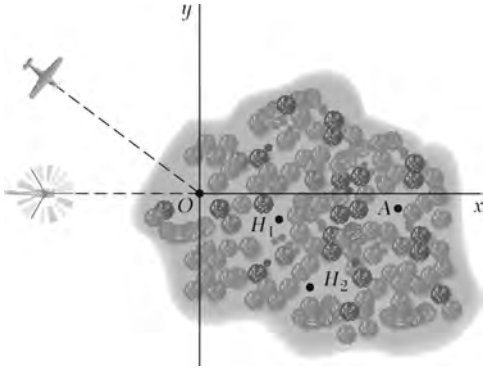
It moves as if projectile had not exploded.

$$\begin{aligned}\bar{\mathbf{r}} &= v_0 t \mathbf{i} - \frac{1}{2} g t^2 \mathbf{j} \\ &= (40 \text{ m/s})(3 \text{ s})\mathbf{i} - \frac{1}{2}(9.81 \text{ m/s}^2)(3 \text{ s})^2 \mathbf{j} \\ &= (120 \text{ m})\mathbf{i} - (44.145 \text{ m})\mathbf{j}\end{aligned}$$

Equation (14.12):

$$\begin{aligned}m\bar{\mathbf{r}} &= \sum m_i \mathbf{r}_i: \\ m\bar{\mathbf{r}} &= m_A \mathbf{r}_A + m_B \mathbf{r}_B \\ 15(120\mathbf{i} - 44.145\mathbf{j}) &= 6(100\mathbf{i} + 10\mathbf{j} - 20\mathbf{k}) + 9\mathbf{r}_B \\ 9\mathbf{r}_B &= (15 \times 120 - 6 \times 100)\mathbf{i} - (15 \times 44.145 + 6 \times 10)\mathbf{j} + (6 \times 20)\mathbf{k} \\ &= 1200\mathbf{i} - 722.175\mathbf{j} + 120\mathbf{k} \\ \mathbf{r}_B &= (133.3 \text{ m})\mathbf{i} - (80.2 \text{ m})\mathbf{j} + (13.33 \text{ m})\mathbf{k} \quad \blacktriangleleft\end{aligned}$$

PROBLEM 14.17



A small airplane of mass 1500 kg and a helicopter of mass 3000 kg flying at an altitude of 1200 m are observed to collide directly above a tower located at O in a wooded area. Four minutes earlier, the helicopter had been sighted 8.4 km due west of the tower and the airplane 16 km west and 12 km north of the tower. As a result of the collision, the helicopter was split into two pieces, H_1 and H_2 , of mass $m_1 = 1000$ kg and $m_2 = 2000$ kg, respectively; the airplane remained in one piece as it fell to the ground. Knowing that the two fragments of the helicopter were located at Points H_1 (500 m, -100 m) and H_2 (600 m, -500 m), respectively, and assuming that all pieces hit the ground at the same time, determine the coordinates of the Point A where the wreckage of the airplane will be found.

SOLUTION

Motion of mass center G .

At collision:

$$\mathbf{v}_H = \frac{(8400 \text{ m})}{4(60 \text{ s})} \mathbf{i} = (35.00 \text{ m/s}) \mathbf{i}$$

$$\begin{aligned} \mathbf{v}_A &= \frac{(16,000 \text{ m}) \mathbf{i} - (12,000 \text{ m}) \mathbf{j}}{4(60 \text{ s})} \\ &= (66.67 \text{ m/s}) \mathbf{i} - (50 \text{ m/s}) \mathbf{j} \end{aligned}$$

Velocity of mass center:

$$\begin{aligned} (m_H + m_A) \bar{\mathbf{v}} &= m_H \mathbf{v}_H + m_A \mathbf{v}_A \\ 4500 \bar{\mathbf{v}} &= 3000(35.00 \mathbf{i}) + 1500(66.67 \mathbf{i} - 50 \mathbf{j}) \\ \bar{\mathbf{v}} &= (45.556 \text{ m/s}) \mathbf{i} - (16.667 \text{ m/s}) \mathbf{j} \end{aligned}$$

Vertical motion of G :

$$h = \frac{1}{2} g t^2 \quad t = \sqrt{\frac{2h}{g}} = \sqrt{\frac{2(1200 \text{ m})}{9.81 \text{ m/s}^2}} = 15.641 \text{ s}$$

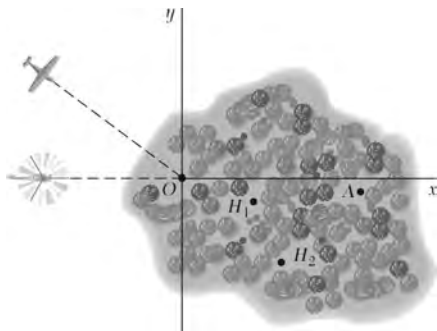
Position of G at time of ground impact:

$$\begin{aligned} \bar{\mathbf{r}} &= \bar{\mathbf{v}} t = (45.556 \mathbf{i} - 16.667 \mathbf{j})(15.641) \\ \bar{\mathbf{r}} &= (712.55 \text{ m}) \mathbf{i} - (260.69 \text{ m}) \mathbf{j} \end{aligned} \quad (1)$$

From Eq. (14.12):

$$\begin{aligned} (m_H + m_A) \bar{\mathbf{r}} &= m_{H_1} \mathbf{r}_{H_1} + m_{H_2} \mathbf{r}_{H_2} + m_A \mathbf{r}_A \\ 4500(712.55 \mathbf{i} - 260.69 \mathbf{j}) &= 1000(500 \mathbf{i} - 100 \mathbf{j}) + 2000(600 \mathbf{i} - 500 \mathbf{j}) + 1500 \mathbf{r}_A \\ 1.5 \mathbf{r}_A &= (4.5 \times 712.55 - 500 - 2 \times 600) \mathbf{i} + (-4.5 \times 260.69 + 100 + 2 \times 500) \mathbf{j} \\ \mathbf{r}_A &= (1004 \text{ m}) \mathbf{i} - (48.7 \text{ m}) \mathbf{j} \quad \blacktriangleleft \end{aligned} \quad (2)$$

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PROBLEM 14.18

In Problem 14.17, knowing that the wreckage of the small airplane was found at Point A (1200 m, 80 m) and the 1000-kg fragment of the helicopter at Point H_1 (400 m, -200 m), and assuming that all pieces hit the ground at the same time, determine the coordinates of the Point H_2 where the other fragment of the helicopter will be found.

SOLUTION

Motion of mass center G .

At collision:

$$\mathbf{v}_H = \frac{(8400 \text{ m})}{4(60 \text{ s})} \mathbf{i} = (35.00 \text{ m/s}) \mathbf{i}$$

$$\begin{aligned} \mathbf{v}_A &= \frac{(16,000 \text{ m}) \mathbf{i} - (12,000 \text{ m}) \mathbf{j}}{4(60 \text{ s})} \\ &= (66.67 \text{ m/s}) \mathbf{i} - (50 \text{ m/s}) \mathbf{j} \end{aligned}$$

Velocity of mass center:

$$\begin{aligned} (m_H + m_A) \bar{\mathbf{v}} &= m_H \mathbf{v}_H + m_A \mathbf{v}_A \\ 4500 \bar{\mathbf{v}} &= 3000(35.00 \mathbf{i}) + 1500(66.67 \mathbf{i} - 50 \mathbf{j}) \\ \bar{\mathbf{v}} &= (45.556 \text{ m/s}) \mathbf{i} - (16.667 \text{ m/s}) \mathbf{j} \end{aligned}$$

Vertical motion of G :

$$h = \frac{1}{2} g t^2 \quad t = \sqrt{\frac{2h}{g}} = \sqrt{\frac{2(1200 \text{ m})}{9.81 \text{ m/s}^2}} = 15.641 \text{ s}$$

Position of G at time of ground impact:

$$\begin{aligned} \bar{\mathbf{r}} &= \bar{\mathbf{v}} t = (45.556 \mathbf{i} - 16.667 \mathbf{j})(15.641) \\ \bar{\mathbf{r}} &= (712.55 \text{ m}) \mathbf{i} - (260.69 \text{ m}) \mathbf{j} \end{aligned} \quad (1)$$

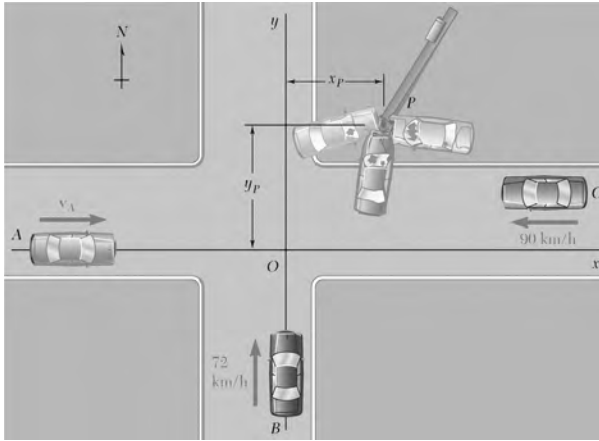
From Eq. (14.12):

$$(m_H + m_A) \bar{\mathbf{r}} = m_{H_1} \mathbf{r}_{H_1} + m_{H_2} \mathbf{r}_{H_2} + m_A \mathbf{r}_A \quad (2)$$

Substituting data:

$$\begin{aligned} 4500(712.55 \mathbf{i} - 260.69 \mathbf{j}) &= 1000(400 \mathbf{i} - 200 \mathbf{j}) + 2000 \mathbf{r}_{H_2} + 1500(1200 \mathbf{i} + 80 \mathbf{j}) \\ 2 \mathbf{r}_{H_2} &= (4.5 \times 712.55 - 400 - 1.5 \times 1200) \mathbf{i} \\ &\quad + (-4.5 \times 260.69 + 200 - 1.5 \times 80) \mathbf{j} \\ \mathbf{r}_{H_2} &= (503 \text{ m}) \mathbf{i} - (547 \text{ m}) \mathbf{j} \quad \blacktriangleleft \end{aligned}$$

PROBLEM 14.19



Car A was traveling east at high speed when it collided at Point O with car B, which was traveling north at 72 km/h. Car C, which was traveling west at 90 km/h, was 10 m east and 3 m north of Point O at the time of the collision. Because the pavement was wet, the driver of car C could not prevent his car from sliding into the other two cars, and the three cars, stuck together, kept sliding until they hit the utility pole P. Knowing that the masses of cars A, B, and C are, respectively, 1500 kg, 1300 kg, and 1200 kg, and neglecting the forces exerted on the cars by the wet pavement, solve the problems indicated.

Knowing that the coordinates of the utility pole are $x_P = 18$ m and $y_P = 13.9$ m, determine (a) the time elapsed from the first collision to the stop at P, (b) the speed of car A.

SOLUTION

Let t be the time elapsed since the first collision. No external forces in the xy plane act on the system consisting of cars A, B, and C during the impacts with one another. The mass center of the system moves at the velocity it had before the collision.

Initial mass center $\bar{\mathbf{r}}_0$:

$$(m_A + m_B + m_C)(\bar{x}_0 \mathbf{i} + \bar{y}_0 \mathbf{j}) = m_C(x_C \mathbf{i} + y_C \mathbf{j})$$

$$\bar{x}_0 = 0.3x_C = (0.3)(10) = 3 \text{ m}, \quad \bar{y}_0 = 0.3y_C = (0.3)(3) = 0.9 \text{ m}$$

Given velocities:

$$\mathbf{v}_A = v_A \mathbf{i}, \quad \mathbf{v}_B = (20 \text{ m/s}) \mathbf{j}, \quad \mathbf{v}_C = -(25 \text{ m/s}) \mathbf{i}$$

Velocity of mass center:

$$(m_A + m_B + m_C)\bar{\mathbf{v}} = m_A \mathbf{v}_A + m_B \mathbf{v}_B + m_C \mathbf{v}_C$$

$$\bar{\mathbf{v}} = 0.375 \mathbf{v}_A + 0.325 \mathbf{v}_B + 0.3 \mathbf{v}_C$$

Since the collided cars hit the pole at

$$\mathbf{r}_P = x_P \mathbf{i} + y_P \mathbf{j}$$

$$x_P \mathbf{i} + y_P \mathbf{j} = \bar{x}_0 \mathbf{i} + \bar{y}_0 \mathbf{j} + \bar{\mathbf{v}} t \quad \text{Resolve into components.}$$

$$x: x_P = \bar{x}_0 + 0.375 v_A t_P - 0.3 v_C t_P \quad (1)$$

$$y: y_P = \bar{y}_0 + 0.325 v_B t_P \quad (2)$$

PROBLEM 14.19 (Continued)

Data:

$$x_P = 18 \text{ m}, \quad y_P = 13.9 \text{ m}$$

(a) From (2),

$$13.9 = 0.9 + (0.325)(20)t_P$$

$$t_P = 2.00 \text{ s} \quad \blacktriangleleft$$

(b) From (1),

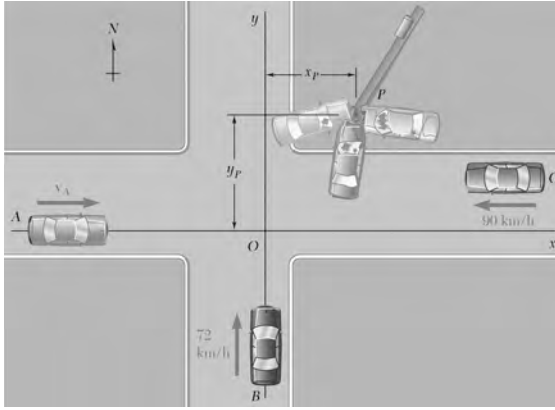
$$18 = 3 + (0.375)v_A(2.00) - (0.3)(25)(2.00)$$

$$v_A = 40 \text{ m/s}$$

$$v_A = 144.0 \text{ km/h} \quad \blacktriangleleft$$

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PROBLEM 14.20



Car A was traveling east at high speed when it collided at Point O with car B, which was traveling north at 72 km/h. Car C, which was traveling west at 90 km/h, was 10 m east and 3 m north of Point O at the time of the collision. Because the pavement was wet, the driver of car C could not prevent his car from sliding into the other two cars, and the three cars, stuck together, kept sliding until they hit the utility pole P. Knowing that the masses of cars A, B, and C are, respectively, 1500 kg, 1300 kg, and 1200 kg, and neglecting the forces exerted on the cars by the wet pavement, solve the problems indicated.

Knowing that the speed of car A was 129.6 km/h and that the time elapsed from the first collision to the stop at P was 2.4 s, determine the coordinates of the utility pole P.

SOLUTION

Let t be the time elapsed since the first collision. No external forces in the xy plane act on the system consisting of cars A, B, and C during the impacts with one another. The mass center of the system moves at the velocity it had before the collision.

Initial mass center $\bar{\mathbf{r}}_0$:

$$(m_A + m_B + m_C)(\bar{x}_0 \mathbf{i} + \bar{y}_0 \mathbf{j}) = m_C(x_C \mathbf{i} + y_C \mathbf{j})$$

$$\bar{x}_0 = 0.3x_C = (0.3)(10) = 3 \text{ m}, \quad \bar{y}_0 = 0.3y_C = (0.3)(3) = 0.9 \text{ m}$$

Given velocities:

$$\mathbf{v}_A = v_A \mathbf{i}, \quad \mathbf{v}_B = (20 \text{ m/s}) \mathbf{j}, \quad \mathbf{v}_C = -(25 \text{ m/s}) \mathbf{i}$$

Velocity of mass center:

$$(m_A + m_B + m_C) \bar{\mathbf{v}} = m_A \mathbf{v}_A + m_B \mathbf{v}_B + m_C \mathbf{v}_C$$

$$\bar{\mathbf{v}} = 0.375 \mathbf{v}_A + 0.325 \mathbf{v}_B + 0.3 \mathbf{v}_C$$

Since the collided cars hit the pole at

$$\mathbf{r}_P = x_P \mathbf{i} + y_P \mathbf{j}$$

$$x_P \mathbf{i} + y_P \mathbf{j} = \bar{x}_0 \mathbf{i} + \bar{y}_0 \mathbf{j} + \bar{\mathbf{v}} t \quad \text{Resolve into components.}$$

$$x: x_P = \bar{x}_0 + 0.375 v_A t_P - 0.3 v_C t_P \quad (1)$$

$$y: y_P = \bar{y}_0 + 0.325 v_B t_P \quad (2)$$

PROBLEM 14.20 (Continued)

Data:

$$v_A = 129.6 \text{ km/h} = 36 \text{ m/s}, \quad t_p = 2.4 \text{ s}$$

From (1),

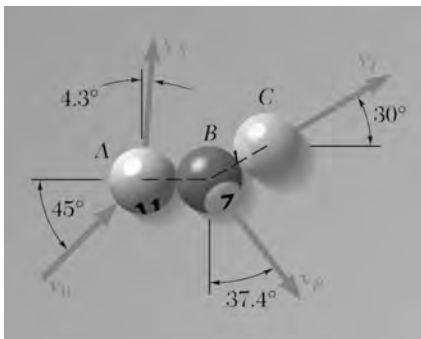
$$x_p = 3 + (0.375)(36)(2.4) - (0.3)(25)(2.4) = 17.4 \text{ m}$$

From (2),

$$y_p = 0.9 + (0.325)(20)(2.4) = 16.5 \text{ m}$$

$$x_p = 17.40 \text{ m} \quad \blacktriangleleft$$

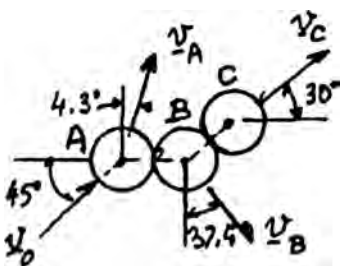
$$y_p = 16.50 \text{ m} \quad \blacktriangleleft$$



PROBLEM 14.21

In a game of pool, ball A is moving with a velocity $\mathbf{v}_0$ when it strikes balls B and C , which are at rest and aligned as shown. Knowing that after the collision the three balls move in the directions indicated, and that $v_0 = 4 \text{ m/s}$ and $v_C = 2 \text{ m/s}$, determine the magnitude of the velocity of (a) ball A , (b) ball B .

SOLUTION



Conservation of linear momentum. In x direction:

$$m(4 \text{ m/s}) \cos 45^\circ = mv_A \sin 4.3^\circ + mv_B \sin 37.4^\circ + m(2 \text{ m/s}) \cos 30^\circ$$

$$0.07498v_A + 0.60738v_B = 1.09638 \quad (1)$$

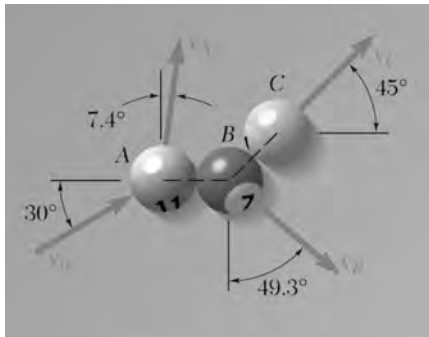
In y direction:

$$m(4 \text{ m/s}) \sin 45^\circ = mv_A \cos 4.3^\circ - mv_B \cos 37.4^\circ + m(2 \text{ m/s}) \sin 30^\circ$$

$$\text{Solving: } 0.99719v_A - 0.79441v_B = 1.82843 \quad (2)$$

$$v_A = 2.98 \text{ m/s} \quad \blacktriangleleft$$

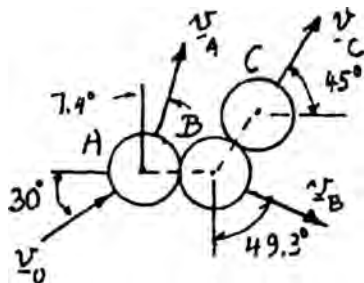
$$v_B = 1.437 \text{ m/s} \quad \blacktriangleleft$$



PROBLEM 14.22

In a game of pool, ball *A* is moving with a velocity $\mathbf{v}_0$ when it strikes balls *B* and *C*, which are at rest and aligned as shown. Knowing that after the collision the three balls move in the directions indicated and that $v_0 = 4 \text{ m/s}$ and $v_C = 2 \text{ m/s}$, determine the magnitude of the velocity of (a) ball *A*, (b) ball *B*.

SOLUTION



Conservation of linear momentum. In *x* direction:

$$\begin{aligned} m(4 \text{ m/s}) \cos 30^\circ &= mv_A \sin 7.4^\circ + mv_B \sin 49.3^\circ \\ &\quad + m(2 \text{ m/s}) \cos 45^\circ \\ 0.12880v_A + 0.75813v_B &= 2.04989 \end{aligned} \quad (1)$$

In *y* direction:

$$\begin{aligned} m(4 \text{ m/s}) \sin 30^\circ &= mv_A \cos 7.4^\circ - mv_B \cos 49.3^\circ \\ &\quad + m(2 \text{ m/s}) \sin 45^\circ \end{aligned}$$

$$\text{Solving: } 0.99167v_A - 0.65210v_B = 0.58579 \quad (2)$$

$$v_A = 2.13 \text{ m/s} \quad \blacktriangleleft$$

$$v_B = 2.34 \text{ m/s} \quad \blacktriangleleft$$

PROBLEM 14.23

An expert archer demonstrates his ability by hitting tennis balls thrown by an assistant. A 58-g tennis ball has a velocity of $(10 \text{ m/s})\mathbf{i} - (2 \text{ m/s})\mathbf{j}$ and is 10 m above the ground when it is hit by a 40-g arrow traveling with a velocity of $(50 \text{ m/s})\mathbf{j} + (70 \text{ m/s})\mathbf{k}$ where $\mathbf{j}$ is directed upwards. Determine the position P where the ball and arrow will hit the ground relative to Point O located directly under the point of impact.

SOLUTION

Assume that the ball and arrow move together after the hit.

Conservation of momentum of ball and arrow during the hit.

$$\begin{aligned}m_A \mathbf{v}_A + m_B \mathbf{v}_B &= (m_A + m_B) \bar{\mathbf{v}} \\(0.040)(50\mathbf{j} + 70\mathbf{k}) + (0.058)(10\mathbf{i} - 2\mathbf{j}) &= (0.040 + 0.058)\bar{\mathbf{v}} \\\bar{\mathbf{v}} &= (5.9184 \text{ m/s})\mathbf{i} + (19.2245 \text{ m/s})\mathbf{j} + (28.5714 \text{ m/s})\mathbf{k}\end{aligned}$$

After the hit, the ball and arrow move as a projectile.

Vertical motion.

$$\begin{aligned}y &= y_0 + (v_y)_0 t - \frac{1}{2}gt^2 \\y &= 10 + 19.2245t - \frac{1}{2}(9.8)t^2 \\y &= 0 \text{ at ground.}\end{aligned}$$

$$-4.9t^2 + 19.2245t + 10 = 0 \quad \text{Solve for } t.$$

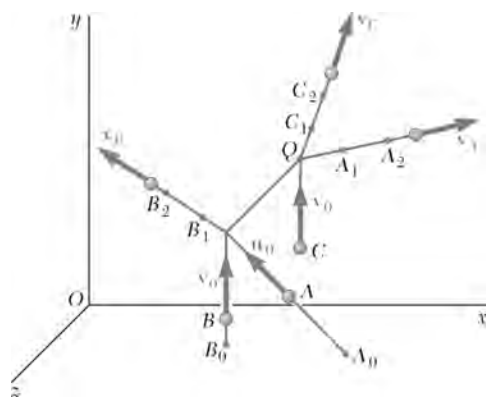
After rejecting the negative root, $t = 4.3884 \text{ s}$

Horizontal motion.

$$\begin{aligned}x &= x_0 + (v_x)_0 t \\z &= z_0 + (v_z)_0 t \\x &= 0 + (5.9184)(4.3884) \\&= 26.0 \text{ m} \\z &= 0 + (28.5714)(4.3884) \\&= 125.4 \text{ m}\end{aligned}$$

$$\mathbf{r}_P = (26.0 \text{ m})\mathbf{i} + (125.4 \text{ m})\mathbf{k} \quad \blacktriangleleft$$

PROBLEM 14.24



In a scattering experiment, an alpha particle A is projected with the velocity $\mathbf{u}_0 = -(600 \text{ m/s})\mathbf{i} + (750 \text{ m/s})\mathbf{j} - (800 \text{ m/s})\mathbf{k}$ into a stream of oxygen nuclei moving with the common velocity $\mathbf{v}_0 = (600 \text{ m/s})\mathbf{j}$. After colliding successively with nuclei B and C , particle A is observed to move along the path defined by the Points $A_1(280, 240, 120)$ and $A_2(360, 320, 160)$, while nuclei B and C are observed to move along paths defined, respectively, by $B_1(147, 220, 130)$, $B_2(114, 290, 120)$, and by $C_1(240, 232, 90)$ and $C_2(240, 280, 75)$. All paths are along straight lines and all coordinates are expressed in millimeters. Knowing that the mass of an oxygen nucleus is four times that of an alpha particle, determine the speed of each of the three particles after the collisions.

SOLUTION

Position vectors (mm):

$$\overline{A_1A_2} = 80\mathbf{i} + 80\mathbf{j} + 40\mathbf{k} \quad (A_1A_2) = 120$$

$$\overline{B_1B_2} = -33\mathbf{i} + 70\mathbf{j} - 10\mathbf{k} \quad (B_1B_2) = 78.032$$

$$\overline{C_1C_2} = 48\mathbf{j} - 15\mathbf{k} \quad (C_1C_2) = 50.289$$

Unit vectors:

Along A_1A_2 , $\lambda_A = 0.66667\mathbf{i} + 0.66667\mathbf{j} + 0.33333\mathbf{k}$

Along B_1B_2 , $\lambda_B = -0.42290\mathbf{i} + 0.89707\mathbf{j} - 0.12815\mathbf{k}$

Along C_1C_2 , $\lambda_C = 0.95448\mathbf{j} - 0.29828\mathbf{k}$

Velocity vectors after the collisions:

$$\mathbf{v}_A = v_A \lambda_A$$

$$\mathbf{v}_B = v_B \lambda_B$$

$$\mathbf{v}_C = v_C \lambda_C$$

Conservation of momentum:

$$m\mathbf{u}_0 + 4m\mathbf{v}_0 + 4m\mathbf{v}_0 = m\mathbf{v}_A + 4m\mathbf{v}_B + 4m\mathbf{v}_C$$

Divide by m and substitute data.

$$(-600\mathbf{i} + 750\mathbf{j} - 800\mathbf{k}) + 2400\mathbf{j} + 2400\mathbf{j} = v_A \lambda_A + 4v_B \lambda_B + 4v_C \lambda_C$$

Resolving into components,

$$\mathbf{i}: -600 = 0.66667v_A - 1.69160v_B$$

$$\mathbf{j}: 5550 = 0.66667v_A + 3.58828v_B + 3.81792v_C$$

$$\mathbf{k}: -800 = 0.33333v_A - 0.51260v_B - 1.19312v_C$$

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PROBLEM 14.24 (Continued)

Solving the three equations simultaneously,

$$v_A = 919.26 \text{ m/s}$$

$$v_B = 716.98 \text{ m/s}$$

$$v_C = 619.30 \text{ m/s}$$

$$v_A = 919 \text{ m/s} \quad \blacktriangleleft$$

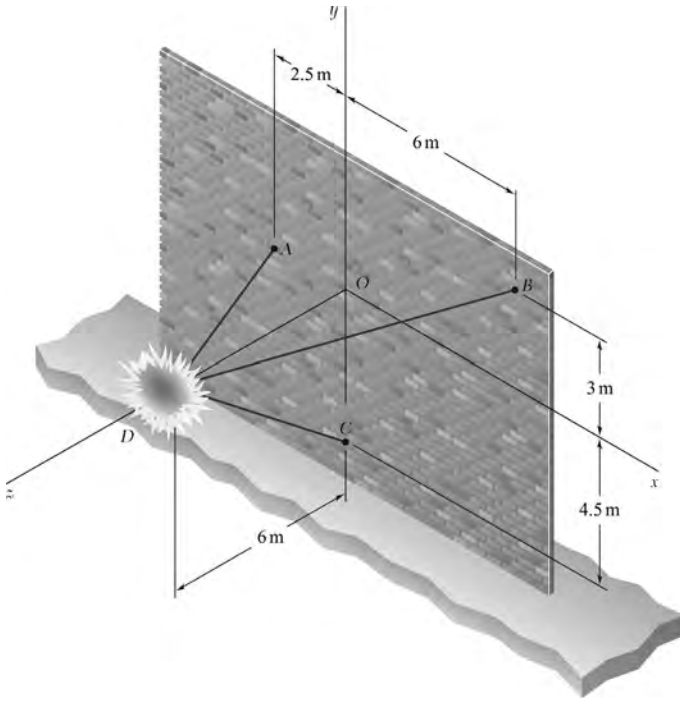
$$v_B = 717 \text{ m/s} \quad \blacktriangleleft$$

$$v_C = 619 \text{ m/s} \quad \blacktriangleleft$$

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PROBLEM 14.25

An 6-kg shell moving with a velocity $\mathbf{v}_0 = (12 \text{ m/s})\mathbf{i} - (10 \text{ m/s})\mathbf{j} - (400 \text{ m/s})\mathbf{k}$ explodes at Point D into three fragments A , B , and C weighing, respectively, 2.5-kg, 2-kg, and 1.5-kg. Knowing that the fragments hit the vertical wall at the points indicated, determine the speed of each fragment immediately after the explosion.



SOLUTION

Position vectors (m):

$$\begin{aligned} \mathbf{r}_D &= 6\mathbf{k} \\ \mathbf{r}_A &= -2.5\mathbf{i} & \mathbf{r}_{A/D} &= -2.5\mathbf{i} - 6\mathbf{k} & r_{A/D} &= 6.5 \\ \mathbf{r}_B &= 6\mathbf{i} + 3\mathbf{j} & \mathbf{r}_{B/D} &= 6\mathbf{i} + 3\mathbf{j} - 6\mathbf{k} & r_{B/D} &= 9 \\ \mathbf{r}_C &= -4.5\mathbf{j} & \mathbf{r}_{C/D} &= -4.5\mathbf{j} - 6\mathbf{k} & r_{C/D} &= 7.5 \end{aligned}$$

Unit vectors:

$$\begin{aligned} \text{Along } \mathbf{r}_{A/D}, \quad \lambda_A &= \frac{1}{6.5}(-2.5\mathbf{i} - 6\mathbf{k}) \\ \text{Along } \mathbf{r}_{B/D}, \quad \lambda_B &= \frac{1}{9}(6\mathbf{i} + 3\mathbf{j} - 6\mathbf{k}) \\ \text{Along } \mathbf{r}_{C/D}, \quad \lambda_C &= \frac{1}{7.5}(-4.5\mathbf{j} - 6\mathbf{k}) \end{aligned}$$

Assume that elevation changes due to gravity may be neglected. Then, the velocity vectors after the explosion have the directions of the unit vectors.

$$\begin{aligned} \mathbf{v}_A &= v_A \lambda_A \\ \mathbf{v}_B &= v_B \lambda_B \\ \mathbf{v}_C &= v_C \lambda_C \end{aligned}$$

PROBLEM 14.25 (Continued)

Conservation of momentum:

$$m\mathbf{v}_0 = m_A\mathbf{v}_A + m_B\mathbf{v}_B + m_C\mathbf{v}_C$$

$$\begin{aligned} 6(12\mathbf{i} - 10\mathbf{j} - 400\mathbf{k}) &= 2.5\left(\frac{v_A}{6.5}\right)(-2.5\mathbf{i} - 6\mathbf{k}) + 2\left(\frac{v_B}{9}\right)(6\mathbf{i} + 3\mathbf{j} - 6\mathbf{k}) \\ &\quad + 1.5\left(\frac{v_C}{7.5}\right)(-4.5\mathbf{j} - 6\mathbf{k}) \end{aligned}$$

Resolve into components.

$$\begin{aligned} 72 &= -\frac{25}{26}v_A + \frac{4}{3}v_B \\ -60 &= \frac{2}{3}v_B - \frac{9}{10}v_C \\ -2400 &= -\frac{30}{13}v_A - \frac{4}{3}v_B - \frac{6}{5}v_C \end{aligned}$$

Solving,

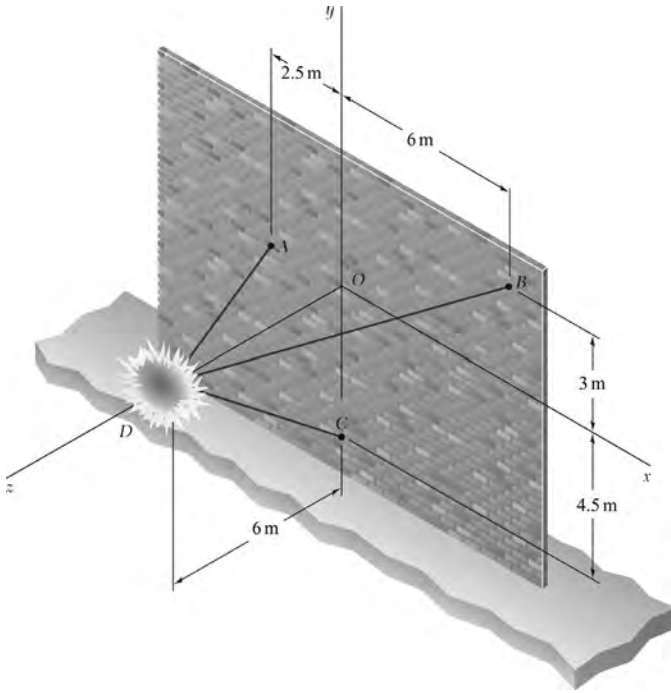
$$v_A = 563 \text{ m/s} \quad \blacktriangleleft$$

$$v_B = 460 \text{ m/s} \quad \blacktriangleleft$$

$$v_C = 407 \text{ m/s} \quad \blacktriangleleft$$

PROBLEM 14.26

An 6-kg shell moving with a velocity $\mathbf{v}_0 = (12 \text{ m/s})\mathbf{i} - (10 \text{ m/s})\mathbf{j} - (400 \text{ m/s})\mathbf{k}$ explodes at Point D into three fragments A , B , and C weighing, respectively, 2.5-kg, 2-kg, and 1.5-kg. Knowing that the fragments hit the vertical wall at the points indicated, determine the speed of each fragment immediately after the explosion.



SOLUTION

Position vectors (m):

$$\begin{aligned} \mathbf{r}_D &= 6\mathbf{k} \\ \mathbf{r}_A &= -2.5\mathbf{i} & \mathbf{r}_{A/D} &= -2.5\mathbf{i} - 6\mathbf{k} & r_{A/D} &= 6.5 \\ \mathbf{r}_B &= 6\mathbf{i} + 3\mathbf{j} & \mathbf{r}_{B/D} &= 6\mathbf{i} + 3\mathbf{j} - 6\mathbf{k} & r_{B/D} &= 9 \\ \mathbf{r}_C &= -4.5\mathbf{j} & \mathbf{r}_{C/D} &= -4.5\mathbf{j} - 6\mathbf{k} & r_{C/D} &= 7.5 \end{aligned}$$

Unit vectors:

$$\begin{aligned} \text{Along } \mathbf{r}_{A/D}, \quad \lambda_A &= \frac{1}{6.5}(-2.5\mathbf{i} - 6\mathbf{k}) \\ \text{Along } \mathbf{r}_{B/D}, \quad \lambda_B &= \frac{1}{9}(6\mathbf{i} + 3\mathbf{j} - 6\mathbf{k}) \\ \text{Along } \mathbf{r}_{C/D}, \quad \lambda_C &= \frac{1}{7.5}(-4.5\mathbf{j} - 6\mathbf{k}) \end{aligned}$$

Assume that elevation changes due to gravity may be neglected. Then, the velocity vectors after the explosion have the directions of the unit vectors.

$$\begin{aligned} \mathbf{v}_A &= v_A \lambda_A \\ \mathbf{v}_B &= v_B \lambda_B \\ \mathbf{v}_C &= v_C \lambda_C \end{aligned}$$

PROBLEM 14.26 (Continued)

Conservation of momentum:

$$m\mathbf{v}_0 = m_A\mathbf{v}_A + m_B\mathbf{v}_B + m_C\mathbf{v}_C$$

$$\begin{aligned} 6(12\mathbf{i} - 10\mathbf{j} - 400\mathbf{k}) &= 2\left(\frac{v_A}{6.5}\right)(-2.5\mathbf{i} - 6\mathbf{k}) + 1.5\left(\frac{v_B}{9}\right)(6\mathbf{i} + 3\mathbf{j} - 6\mathbf{k}) \\ &\quad + 2.5\left(\frac{v_C}{7.5}\right)(-4.5\mathbf{j} - 6\mathbf{k}) \end{aligned}$$

Resolve into components.

$$\begin{aligned} 72 &= -\frac{10}{13}v_A + v_B \\ -60 &= \frac{1}{2}v_B - \frac{3}{2}v_C \\ -2400 &= -\frac{24}{13}v_A - v_B - 2v_C \end{aligned}$$

Solving,

$$v_A = 703 \text{ m/s} \quad \blacktriangleleft$$

$$v_B = 613 \text{ m/s} \quad \blacktriangleleft$$

$$v_C = 244 \text{ m/s} \quad \blacktriangleleft$$

PROBLEM 14.27

Derive the relation

$$\mathbf{H}_O = \bar{\mathbf{r}} \times m\bar{\mathbf{v}} + \mathbf{H}_G$$

between the angular momenta $\mathbf{H}_O$ and $\mathbf{H}_G$ defined in Eqs. (14.7) and (14.24), respectively. The vectors $\bar{\mathbf{r}}$ and $\bar{\mathbf{v}}$ define, respectively, the position and velocity of the mass center G of the system of particles relative to the newtonian frame of reference $Oxyz$, and m represents the total mass of the system.

SOLUTION

From Eq. (14.7),

$$\begin{aligned}\mathbf{H}_O &= \sum_{i=1}^n (\mathbf{r}_i \times m_i \mathbf{v}_i) \\ &= \sum_{i=1}^n [(\bar{\mathbf{r}} + \mathbf{r}'_i) \times m_i \mathbf{v}_i] \\ &= \bar{\mathbf{r}} \times \sum_{i=1}^n (m_i \mathbf{v}_i) + \sum_{i=1}^n (\mathbf{r}'_i \times m_i \mathbf{v}_i) \\ &= \bar{\mathbf{r}} \times m\bar{\mathbf{v}} + \mathbf{H}_G\end{aligned}$$

PROBLEM 14.28

Show that Eq. (14.23) may be derived directly from Eq. (14.11) by substituting for $\mathbf{H}_O$ the expression given in Problem 14.27.

SOLUTION

From Eq. (14.7),

$$\begin{aligned}\mathbf{H}_O &= \sum_{i=1}^n (\mathbf{r}_i \times m_i \mathbf{v}_i) \\ &= \sum_{i=1}^n [(\bar{\mathbf{r}} + \mathbf{r}'_i) \times m_i \mathbf{v}_i] \\ &= \bar{\mathbf{r}} \times \sum_{i=1}^n (m_i \mathbf{v}_i) + \sum_{i=1}^n (\mathbf{r}'_i \times m_i \mathbf{v}_i) \\ &= \bar{\mathbf{r}} \times m \bar{\mathbf{v}} + \mathbf{H}_G\end{aligned}$$

Differentiating,

$$\dot{\mathbf{H}}_O = \dot{\bar{\mathbf{r}}} \times m \bar{\mathbf{v}} + \bar{\mathbf{r}} \times m \dot{\bar{\mathbf{v}}} + \dot{\mathbf{H}}_G$$

Using Eq. (14.11),

$$\begin{aligned}\Sigma \mathbf{M}_O &= \dot{\bar{\mathbf{r}}} \times m \bar{\mathbf{v}} + \bar{\mathbf{r}} \times m \dot{\bar{\mathbf{v}}} + \dot{\mathbf{H}}_G \\ &= \bar{\mathbf{v}} \times m \bar{\mathbf{v}} + \bar{\mathbf{r}} \times m \bar{\mathbf{a}} + \dot{\mathbf{H}}_G \\ &= 0 + \bar{\mathbf{r}} \times \left(\sum_{i=1}^n \mathbf{F}_i \right) + \dot{\mathbf{H}}_G\end{aligned}\tag{1}$$

But

$$\sum_{i=1}^n \mathbf{M}_O = \sum_{i=1}^n \mathbf{M}_G + \sum_{i=1}^n \bar{\mathbf{r}} \times \left(\sum_{i=1}^n \mathbf{F}_i \right)$$

Subtracting $\bar{\mathbf{r}} \times \left(\sum_{i=1}^n \mathbf{F}_i \right)$ from each side of Eq. (1) gives

$$\Sigma \mathbf{M}_G = \dot{\mathbf{H}}_G$$

PROBLEM 14.29

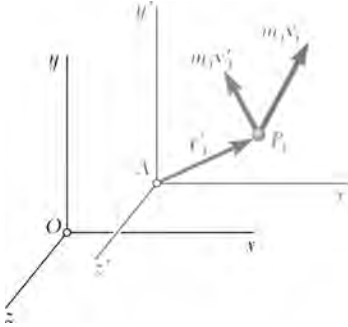
Consider the frame of reference $Ax'y'z'$ in translation with respect to the newtonian frame of reference $Oxyz$. We define the angular momentum $\mathbf{H}_A$ of a system of n particles about A as the sum

$$\mathbf{H}_A = \sum_{i=1}^n \mathbf{r}'_i \times m_i \mathbf{v}'_i \quad (1)$$

of the moments about A of the momenta $m_i \mathbf{v}'_i$ of the particles in their motion relative to the frame $Ax'y'z'$. Denoting by $\mathbf{H}_A$ the sum

$$\mathbf{H}_A = \sum_{i=1}^n \mathbf{r}'_i \times m_i \mathbf{v}_i \quad (2)$$

of the moments about A of the momenta $m_i \mathbf{v}_i$ of the particles in their motion relative to the newtonian frame $Oxyz$, show that $\mathbf{H}_A = \mathbf{H}'_A$ at a given instant if, and only if, one of the following conditions is satisfied at that instant: (a) A has zero velocity with respect to the frame $Oxyz$, (b) A coincides with the mass center G of the system, (c) the velocity $\mathbf{v}_A$ relative to $Oxyz$ is directed along the line AG .

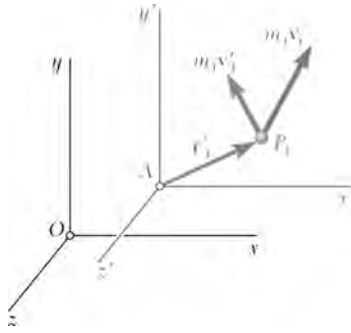


SOLUTION

$$\begin{aligned} \mathbf{v}_i &= \mathbf{v}_A + \mathbf{v}'_i \\ \mathbf{H}_A &= \sum_{i=1}^n \mathbf{r}'_i \times m_i \mathbf{v}_i \\ &= \sum_{i=1}^n \mathbf{r}'_i \times m_i (\mathbf{v}_A + \mathbf{v}'_i) \\ &= \sum_{i=1}^n (\mathbf{r}'_i \times m_i \mathbf{v}_A) + \sum_{i=1}^n \mathbf{r}'_i \times m_i \mathbf{v}'_i \\ &= \sum_{i=1}^n (m_i \mathbf{r}'_i) \times \mathbf{v}_A + \mathbf{H}'_A \\ &= \sum_{i=1}^n m_i (\mathbf{r}_i - \mathbf{r}_A) \times \mathbf{v}_A + \mathbf{H}'_A \\ &= m(\bar{\mathbf{r}} - \mathbf{r}_A) \times \mathbf{v}_A + \mathbf{H}'_A \\ \mathbf{H}_A &= \mathbf{H}'_A \quad \text{if, and only if,} \quad m(\bar{\mathbf{r}} - \mathbf{r}_A) \times \mathbf{v}_A = 0 \end{aligned}$$

This condition is satisfied if

- | | | | |
|-----|--------------------|---|---|
| (a) | $\mathbf{v}_A = 0$ | Point A has zero velocity. | |
| or | (b) | $\bar{\mathbf{r}} = \mathbf{r}_A$ | Point A coincides with the mass center. |
| or | (c) | $\mathbf{v}_A$ is parallel to $\bar{\mathbf{r}} - \mathbf{r}_A$. | Velocity $\mathbf{v}_A$ is directed along line AG . |



PROBLEM 14.30

Show that the relation $\Sigma \mathbf{M}_A = \dot{\mathbf{H}}'_A$, where $\mathbf{H}'_A$ is defined by Eq. (1) of Problem 14.29 and where $\Sigma \mathbf{M}_A$ represents the sum of the moments about A of the external forces acting on the system of particles, is valid if, and only if, one of the following conditions is satisfied: (a) the frame $Ax'y'z'$ is itself a newtonian frame of reference, (b) A coincides with the mass center G, (c) the acceleration $\mathbf{a}_A$ of A relative to $Oxyz$ is directed along the line AG.

SOLUTION

From equation (1),

$$\mathbf{H}'_A = \sum_{i=1}^n (\mathbf{r}'_i \times m_i \mathbf{v}'_i)$$

$$\mathbf{H}'_A = \sum_{i=1}^n [(\mathbf{r}_i - \mathbf{r}_A) \times m_i (\mathbf{v}_i - \mathbf{v}_A)]$$

Differentiate with respect to time.

$$\dot{\mathbf{H}}'_A = \sum_{i=1}^n [(\dot{\mathbf{r}}_i - \dot{\mathbf{r}}_A) \times m_i (\mathbf{v}_i - \mathbf{v}_A)] + \sum_{i=1}^n [(\mathbf{r}_i - \mathbf{r}_A) \times m_i (\dot{\mathbf{v}}_i - \dot{\mathbf{v}}_A)]$$

But

$$\dot{\mathbf{r}}_i = \mathbf{v}_i$$

$$\dot{\mathbf{v}}_i = \mathbf{a}_i$$

$$\dot{\mathbf{r}}_A = \mathbf{v}_A$$

and

$$\dot{\mathbf{v}}_A = \mathbf{a}_A$$

Hence,

$$\begin{aligned} \dot{\mathbf{H}}'_A &= 0 + \sum_{i=1}^n [(\mathbf{r}_i - \mathbf{r}_A) \times m_i (\mathbf{a}_i - \mathbf{a}_A)] \\ &= \sum_{i=1}^n [(\mathbf{r}_i - \mathbf{r}_A) \times (\mathbf{F}_i - m_i \mathbf{a}_A)] \\ &= \sum_{i=1}^n [(\mathbf{r}_i - \mathbf{r}_A) \times \mathbf{F}_i] - \sum_{i=1}^n [m_i (\mathbf{r}_i - \mathbf{r}_A)] \times \mathbf{a}_A \\ &= \mathbf{M}_A - m(\bar{\mathbf{r}} - \mathbf{r}_A) \times \mathbf{a}_A \end{aligned}$$

$$\dot{\mathbf{H}}'_A = \mathbf{M}_A \quad \text{if, and only if,} \quad m(\bar{\mathbf{r}} - \mathbf{r}_A) \times \mathbf{a}_A = 0$$

This condition is satisfied if

(a) $\mathbf{a}_A = 0$

The frame is newtonian.

or (b) $\bar{\mathbf{r}} = \mathbf{r}_A$

Point A coincides with the mass center.

or (c) $\mathbf{a}_A$ is parallel to $\bar{\mathbf{r}} - \mathbf{r}_A$.

Acceleration $\mathbf{a}_A$ is directed along line AG.

PROBLEM 14.31



Assuming that the airline employee of Problem 14.1 first tosses the 15-kg suitcase on the baggage carrier, determine the energy lost (a) as the first suitcase hits the carrier, (b) as the second suitcase hits the carrier.

PROBLEM 14.1 An airline employee tosses two suitcases, of mass 15 kg and 20 kg, respectively, onto a 25-kg baggage carrier in rapid succession. Knowing that the carrier is initially at rest and that the employee imparts a 3-m/s horizontal velocity to the 15-kg suitcase and a 2-m/s horizontal velocity to the 20-kg suitcase, determine the final velocity of the baggage carrier if the first suitcase tossed onto the carrier is (a) the 15-kg suitcase, (b) the 20-kg suitcase.

SOLUTION

There are no horizontal external forces acting during the impacts. The baggage carrier is to coast between impacts.

Let v_1 be the common velocity of suitcase A and the carrier after the first impact and v_2 be the common velocity of the two suitcases and the carrier of the second impact.

(a) 15-kg suitcase tossed on carrier first:

Conservation of momentum:

$$(15)(3) = (40)v_1 \\ v_1 = 1.125 \text{ m/s} \rightarrow$$

$$\frac{(15 \text{ kg})(3 \text{ m/s})}{\boxed{A}} = \frac{(15 \text{ kg} + 25 \text{ kg}) v_1}{\boxed{A} \text{ on cart}}$$

Initial kinetic energy:

$$T_0 = \frac{1}{2} m_A v_0^2 \\ = \frac{1}{2} (15 \text{ kg})(3 \text{ m/s})^2 \\ = 67.5$$

Final kinetic energy:

$$T_1 = \frac{1}{2} (m_A + m_C) v_1^2 \\ = \frac{1}{2} (40 \text{ kg})(1.125 \text{ m/s}^2) \\ = 25.3125 \text{ J}$$

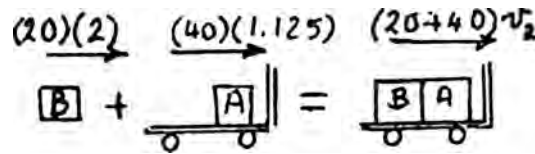
$$T_0 - T_1 = 42.1875 \text{ J}$$

$$\text{energy lost} = 42.2 \text{ J} \blacktriangleleft$$



PROBLEM 14.31 (Continued)

(b) 20-kg suitcase tossed next:



$$(20)(2) + (40)(1.125) = 60v_2$$

$$v_2 = 1.417 \text{ m/s} \rightarrow$$

Initial kinetic energy:

$$T_0' = T_1 + \frac{1}{2}m_B v_0^2$$

$$= 25.3125 \text{ J} + \frac{1}{2}(20 \text{ kg})(2 \text{ m/s})^2$$

$$= 65.3125 \text{ J}$$

Final kinetic energy:

$$T_2 = \frac{1}{2}(m_A + m_B + m_C)v_2^2$$

$$= \frac{1}{2}(60 \text{ kg})(1.417 \text{ m/s})^2$$

$$= 60.2083 \text{ J}$$

$$T_0' - T_2 = 5.10 \text{ J}$$

$$\text{energy lost} = 5.10 \text{ J} \blacktriangleleft$$

PROBLEM 14.32

Determine the energy loss as a result of the series of collisions described in Problem 14.7.

SOLUTION

Summary of results from Problem 14.7:

$$m_A = 240 \text{ kg}, \quad m_B = 260 \text{ kg}, \quad m_C = 235 \text{ kg}$$
$$\mathbf{v}_A = 2 \text{ m/s} \rightarrow, \quad \mathbf{v}_B = 0, \quad \mathbf{v}_C = 1.5 \text{ m/s} \leftarrow.$$

(a) Cars A and C hit B at the same time.

$$\mathbf{v}'_A = 1.288 \text{ m/s} \leftarrow, \quad \mathbf{v}'_B = 0.312 \text{ m/s} \rightarrow, \quad \mathbf{v}'_C = 1.512 \text{ m/s} \rightarrow$$

Initial kinetic energy:

$$T_1 = \frac{1}{2} m_A v_A^2 + \frac{1}{2} m_B v_B^2 + \frac{1}{2} m_C v_C^2$$
$$T_1 = \frac{1}{2} (240)(2)^2 + 0 + \frac{1}{2} (235)(1.5)^2$$
$$= 744.375 \text{ J}$$

Final kinetic energy:

$$T_2 = \frac{1}{2} m_A (v'_A)^2 + \frac{1}{2} m_B (v'_B)^2 + \frac{1}{2} m_C (v'_C)^2$$
$$= \frac{1}{2} (240)(1.288)^2 + \frac{1}{2} (260)(0.312)^2 + \frac{1}{2} (235)(1.512)^2$$
$$= 480.35 \text{ J}$$

$$T_1 - T_2 = 264.03 \text{ J} \qquad \text{energy loss} = 264 \text{ J} \blacktriangleleft$$

(b) Car A hits car B. Then B and C hit. Then A and B hit. The final velocities (from Problem 14.7) are:

$$\mathbf{v}'''_A = 0.956 \text{ m/s} \leftarrow, \quad \mathbf{v}'''_B = 0.0296 \text{ m/s} \leftarrow, \quad \mathbf{v}''_C = 1.552 \text{ m/s} \rightarrow$$

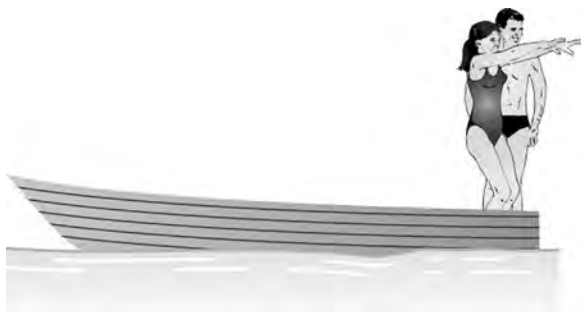
Initial kinetic energy:

$$T_1 = 744.375 \text{ J}$$

Final kinetic energy:

$$T'_2 = \frac{1}{2} m_A (v'''_A)^2 + \frac{1}{2} m_B (v'''_B)^2 + \frac{1}{2} m_C (v''_C)^2$$
$$T'_2 = \frac{1}{2} (240)(0.956)^2 + \frac{1}{2} (260)(0.0296)^2 + \frac{1}{2} (235)(1.552)^2$$
$$= 392.8 \text{ J}$$

$$T_1 - T'_2 = 351.57 \text{ J} \qquad \text{energy loss} = 352 \text{ J} \blacktriangleleft$$



PROBLEM 14.33

In Problem 14.3, determine the work done by the woman and by the man as each dives from the boat, assuming that the woman dives first.

SOLUTION

Woman dives first.

Conservation of momentum:

$$60(5 - v_1) - (150 + 90)v_1 = 0$$

$$v_1 = 1 \text{ m/s} \leftarrow$$

$$5 - v_1 = 4 \text{ m/s} \rightarrow$$

Kinetic energy before dive:

$$T_0 = 0$$

Kinetic energy after dive:

$$T_1 = \frac{1}{2}(150 + 90)(1)^2 + \frac{1}{2}(60)(4)^2$$

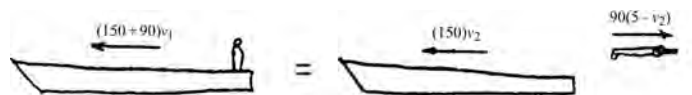
$$= 600 \text{ J}$$

Work of woman:

$$T_1 - T_0 = 600 \text{ J}$$

$$T_1 - T_0 = 600 \text{ J} \blacktriangleleft$$

Man dives next. Conservation of momentum:



$$-240v_1 = -150v_2 + 90(5 - v_2)$$

$$v_2 = \frac{240v_1 + 450}{240} = 2.875 \text{ m/s} \leftarrow$$

$$5 - 2.875 = 2.125 \text{ m/s} \rightarrow$$

Kinetic energy before dive:

$$T'_1 = \frac{1}{2}(150 + 90)(1)^2$$

$$= 120 \text{ J}$$

Kinetic energy after dive:

$$T'_2 = \frac{1}{2}(150)(2.875)^2 + \frac{1}{2}(90)(2.125)^2$$

$$= 823.125 \text{ J}$$

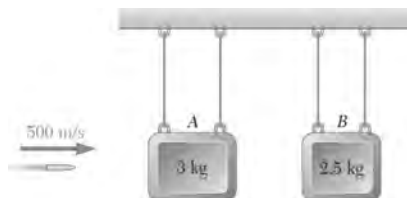
Work of man:

$$T'_2 - T'_1 = 703.125 \text{ J}$$

$$T'_2 - T'_1 = 703 \text{ J} \blacktriangleleft$$

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PROBLEM 14.34

In Problem 14.5, determine the energy lost as the bullet (a) passes through block A, (b) becomes embedded in block B.

SOLUTION

The masses are m for the bullet and m_A and m_B for the blocks.

- (a) The bullet passes through block A and embeds in block B. Momentum is conserved.

Initial momentum: $mv_0 + m_A(0) + m_B(0) = mv_0$

Final momentum: $mv_B + m_A v_A + m_B v_B$

Equating, $mv_0 = mv_B + m_A v_A + m_B v_B$

$$m = \frac{m_A v_A + m_B v_B}{v_0 - v_B} = \frac{(3)(1.5) + (2.5)(2.7)}{(450 - 2.7)} = 0.0252 \text{ kg}$$

$$m = 25.2 \text{ g} \quad \blacktriangleleft$$

- (b) The bullet passes through block A. Momentum is conserved.

Initial momentum: $mv_0 + m_A(0) = mv_0$

Final momentum: $mv_1 + m_A v_A$

Equating, $mv_0 = mv_1 + m_A v_A$

$$v_1 = \frac{mv_0 - m_A v_A}{m} = v_0 - \frac{m_A v_A}{m} = 450 - \frac{3(1.5)}{0.02515} = 271.07 \text{ m/s}$$

$$v_1 = 271 \text{ m/s} \quad \rightarrow \blacktriangleleft$$

The masses are:

$$m = 0.0252 \text{ kg}$$

$$m_A = 3 \text{ kg}$$

$$m_B = 2.5 \text{ kg}$$

PROBLEM 14.34 (Continued)

(a) *Bullet passes through block A. Kinetic energies.*

$$\begin{aligned}\text{Before:} \quad T_0 &= \frac{1}{2}mv_0^2 \\ &= \frac{1}{2}(0.0252)(450)^2 \\ &= 2551.5 \text{ J}\end{aligned}$$

$$\begin{aligned}\text{After:} \quad T_1 &= \frac{1}{2}mv_1^2 + \frac{1}{2}m_A v_A^2 \\ &= \frac{1}{2}(0.0252)(271.07)^2 + \frac{1}{2}(3)(1.5)^2 \\ &= 929.21 \text{ J}\end{aligned}$$

$$\begin{aligned}\text{Lost:} \quad T_0 - T_1 &= (2551.5 - 929.21) \\ &= 1622.29 \text{ J} \quad \text{energy lost} = 1622 \text{ J} \blacktriangleleft\end{aligned}$$

(b) *Bullet becomes embedded in block B. Kinetic energies.*

$$\begin{aligned}\text{Before:} \quad T_2 &= \frac{1}{2}mv_1^2 \\ &= \frac{1}{2}(0.0252)(271.07)^2 \\ &= 925.835 \text{ J}\end{aligned}$$

$$\begin{aligned}\text{After:} \quad T_3 &= \frac{1}{2}(m + m_B)v_B^2 \\ &= \frac{1}{2}(0.0252 + 2.5)(2.7)^2 \\ &= 9.2043 \text{ J}\end{aligned}$$

$$\begin{aligned}\text{Lost:} \quad T_2 - T_3 &= 925.835 - 9.2043 \text{ J} \\ &= 916.631 \text{ J} \quad \text{energy lost} = 917 \text{ J} \blacktriangleleft\end{aligned}$$

PROBLEM 14.35

Two automobiles A and B, of mass m_A and m_B , respectively, are traveling in opposite directions when they collide head on. The impact is assumed perfectly plastic, and it is further assumed that the energy absorbed by each automobile is equal to its loss of kinetic energy with respect to a moving frame of reference attached to the mass center of the two-vehicle system. Denoting by E_A and E_B , respectively, the energy absorbed by automobile A and by automobile B, (a) show that $E_A/E_B = m_B/m_A$, that is, the amount of energy absorbed by each vehicle is inversely proportional to its mass, (b) compute E_A and E_B , knowing that $m_A = 1600$ kg and $m_B = 900$ kg and that the speeds of A and B are, respectively, 90 km/h and 60 km/h.



SOLUTION

Velocity of mass center:

$$(m_A + m_B)\bar{\mathbf{v}} = m_A \mathbf{v}_A + m_B \mathbf{v}_B$$

$$\bar{\mathbf{v}} = \frac{m_A \mathbf{v}_A + m_B \mathbf{v}_B}{m_A + m_B}$$

Velocities relative to the mass center:

$$\mathbf{v}'_A = \mathbf{v}_A - \bar{\mathbf{v}} = \mathbf{v}_A - \frac{m_A \mathbf{v}_A + m_B \mathbf{v}_B}{m_A + m_B} = \frac{m_B(\mathbf{v}_A + \mathbf{v}_B)}{m_A + m_B}$$

$$\mathbf{v}'_B = \mathbf{v}_B - \bar{\mathbf{v}} = \mathbf{v}_B - \frac{m_A \mathbf{v}_A + m_B \mathbf{v}_B}{m_A + m_B} = \frac{m_A(\mathbf{v}_A + \mathbf{v}_B)}{m_A + m_B}$$

Energies:

$$E_A = \frac{1}{2} m_A \mathbf{v}'_A \cdot \mathbf{v}'_A = \frac{m_A m_B^2 (\mathbf{v}_A + \mathbf{v}_B) \cdot (\mathbf{v}_A + \mathbf{v}_B)}{2(m_A + m_B)^2}$$

$$E_B = \frac{1}{2} m_B \mathbf{v}'_B \cdot \mathbf{v}'_B = \frac{m_A^2 m_B (\mathbf{v}_A + \mathbf{v}_B) \cdot (\mathbf{v}_A + \mathbf{v}_B)}{2(m_A + m_B)^2}$$

(a) Ratio:

$$\frac{E_A}{E_B} = \frac{m_B}{m_A} \quad \blacktriangleleft$$

(b)

$$\mathbf{v}_A = 90 \text{ km/h} = 25 \text{ m/s} \rightarrow$$

$$\mathbf{v}_B = 60 \text{ km/h} = 16.667 \text{ m/s} \leftarrow$$

$$\mathbf{v}_A + \mathbf{v}_B = 41.667 \text{ m/s} \rightarrow$$

$$E_A = \frac{(1600)(900)^2(41.667)^2}{(2)(2500)^2} = 180.0 \times 10^3 \text{ J} \quad E_A = 180.0 \text{ kJ} \quad \blacktriangleleft$$

$$E_B = \frac{(1600)^2(900)(41.667)^2}{(2)(2500)^2} = 320 \times 10^3 \text{ J} \quad E_B = 320 \text{ kJ} \quad \blacktriangleleft$$

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PROBLEM 14.36

It is assumed that each of the two automobiles involved in the collision described in Problem 14.35 had been designed to safely withstand a test in which it crashed into a solid, immovable wall at the speed v_0 . The severity of the collision of Problem 14.35 may then be measured for each vehicle by the ratio of the energy absorbed in the collision to the energy it absorbed in the test. On that basis, show that the collision described in Problem 14.35 is $(m_A/m_B)^2$ times more severe for automobile B than for automobile A.



SOLUTION

Velocity of mass center:

$$(m_A + m_B)\bar{\mathbf{v}} = m_A \mathbf{v}_A + m_B \mathbf{v}_B$$

$$\bar{\mathbf{v}} = \frac{m_A \mathbf{v}_A + m_B \mathbf{v}_B}{m_A + m_B}$$

Velocities relative to the mass center:

$$\mathbf{v}'_A = \mathbf{v}_A - \bar{\mathbf{v}} = \mathbf{v}_A - \frac{m_A \mathbf{v}_A + m_B \mathbf{v}_B}{m_A + m_B} = \frac{m_B(\mathbf{v}_A + \mathbf{v}_B)}{m_A + m_B}$$

$$\mathbf{v}'_B = \mathbf{v}_B - \bar{\mathbf{v}} = \mathbf{v}_B - \frac{m_A \mathbf{v}_A + m_B \mathbf{v}_B}{m_A + m_B} = \frac{m_A(\mathbf{v}_A + \mathbf{v}_B)}{m_A + m_B}$$

Energies:

$$E_A = \frac{1}{2} m_A \mathbf{v}'_A \cdot \mathbf{v}'_A = \frac{m_A m_B^2 (\mathbf{v}_A + \mathbf{v}_B) \cdot (\mathbf{v}_A + \mathbf{v}_B)}{2(m_A + m_B)^2}$$

$$E_B = \frac{1}{2} m_B \mathbf{v}'_B \cdot \mathbf{v}'_B = \frac{m_A^2 m_B (\mathbf{v}_A + \mathbf{v}_B) \cdot (\mathbf{v}_A + \mathbf{v}_B)}{2(m_A + m_B)^2}$$

Energies from tests:

$$(E_A)_0 = \frac{1}{2} m_A v_0^2, \quad (E_B)_0 = \frac{1}{2} m_B v_0^2$$

Severities:

$$S_A = \frac{E_A}{(E_A)_0} = \frac{m_B^2 (\mathbf{v}_A + \mathbf{v}_B) \cdot (\mathbf{v}_A + \mathbf{v}_B)}{(m_A + m_B)^2 v_0^2}$$

$$S_B = \frac{E_B}{(E_B)_0} = \frac{m_A^2 (\mathbf{v}_A + \mathbf{v}_B) \cdot (\mathbf{v}_A + \mathbf{v}_B)}{(m_A + m_B)^2 v_0^2}$$

Ratio:

$$\frac{S_A}{S_B} = \frac{m_B^2}{m_A^2} \quad \blacktriangleleft$$

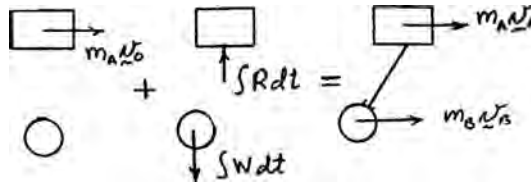
PROBLEM 14.37

Solve Sample Problem 14.4, assuming that cart A is given an initial horizontal velocity v_0 while ball B is at rest.

SOLUTION

- (a) Velocity of B at maximum elevation. At maximum elevation, ball B is at rest relative to cart A. $\mathbf{v}_B = \mathbf{v}_A$

Use impulse-momentum principle.



x components:

$$\begin{aligned} m_A v_0 + 0 &= m_A v_A + m_B v_B \\ &= (m_A + m_B) v_B \end{aligned}$$

$$v_B = \frac{m_A v_0}{m_A + m_B} \rightarrow \blacktriangleleft$$

- (b) Conservation of energy:

$$T_1 = \frac{1}{2} m_A v_0^2, \quad V_1 = 0$$

$$T_2 = \frac{1}{2} m_A v_A^2 + \frac{1}{2} m_B v_B^2$$

$$= \frac{1}{2} (m_A + m_B) v_B^2$$

$$= \frac{m_A^2 v_0^2}{2(m_A + m_B)}$$

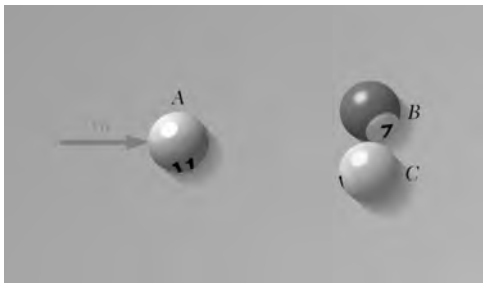
$$V_2 = m_B g h$$

$$T_2 + V_2 = T_1 + V_1$$

$$\frac{m_A^2 v_0^2}{2(m_A + m_B)} + m_B g h = \frac{1}{2} m_A v_0^2$$

$$h = \frac{1}{2m_B g} \left[m_A v_0^2 - \frac{m_A^2 v_0^2}{m_A + m_B} \right]$$

$$h = \frac{m_A}{m_A + m_B} \frac{v_0^2}{2g} \blacktriangleleft$$



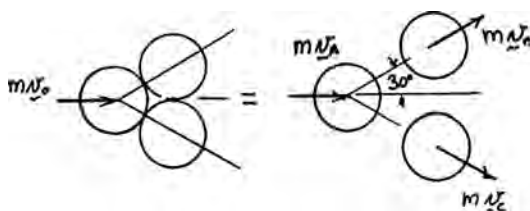
PROBLEM 14.38

In a game of pool, ball A is moving with the velocity $\mathbf{v}_0 = v_0 \mathbf{i}$ when it strikes balls B and C, which are at rest side by side. Assuming frictionless surfaces and perfectly elastic impact (i.e., conservation of energy), determine the final velocity of each ball, assuming that the path of A is (a) perfectly centered and that A strikes B and C simultaneously, (b) not perfectly centered and that A strikes B slightly before it strikes C.

SOLUTION

(a) A strikes B and C simultaneously.

During the impact, the contact impulses make 30° angles with the velocity $\mathbf{v}_0$.



Thus,

$$\mathbf{v}_B = v_B (\cos 30^\circ \mathbf{i} + \sin 30^\circ \mathbf{j})$$

$$\mathbf{v}_C = v_C (\cos 30^\circ \mathbf{i} - \sin 30^\circ \mathbf{j})$$

By symmetry,

$$\mathbf{v}_A = v_A \mathbf{i}$$

Conservation of momentum: $m\mathbf{v}_0 = m\mathbf{v}_A + m\mathbf{v}_B + m\mathbf{v}_C$

$$y \text{ component: } 0 = 0 + mv_B \sin 30^\circ - mv_C \sin 30^\circ \quad v_C = v_B$$

$$x \text{ component: } mv_0 = mv_A + mv_B \cos 30^\circ + mv_C \cos 30^\circ$$

$$v_B + v_C = \frac{v_0 - v_A}{\cos 30^\circ} = \frac{2}{\sqrt{3}}(v_0 - v_A)$$

$$v_B = v_C = \frac{v_0 - v_A}{\sqrt{3}}$$

$$\text{Conservation of energy: } \frac{1}{2}mv_0^2 = \frac{1}{2}mv_A^2 + \frac{1}{2}mv_B^2 + \frac{1}{2}mv_C^2$$

$$v_0^2 = v_A^2 + \frac{2}{3}(v_0 - v_A)^2$$

$$v_0^2 - v_A^2 = (v_0 - v_A)(v_0 + v_A) = \frac{2}{3}(v_0 - v_A)^2$$

PROBLEM 14.38 (Continued)

$$v_0 + v_A = \frac{2}{3}(v_0 - v_A) \quad \frac{1}{3}v_0 = -\frac{5}{3}v_A \quad v_A = -\frac{1}{5}v_0$$

$$v_B = v_C = \frac{6}{5\sqrt{3}}v_0 = \frac{2\sqrt{3}}{5}v_0$$

$$\mathbf{v}_A = 0.200v_0 \leftarrow \blacktriangleleft$$

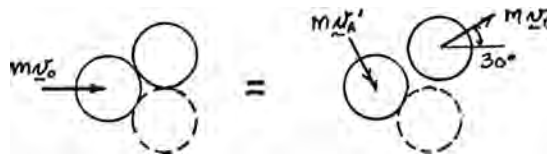
$$\mathbf{v}_B = 0.693v_0 \nearrow 30^\circ \blacktriangleleft$$

$$\mathbf{v}_C = 0.693v_0 \searrow 30^\circ \blacktriangleleft$$

(b) A strikes B before it strikes C.

First impact: A strikes B.

During the impact, the contact impulse makes a 30° angle with the velocity $\mathbf{v}_0$.



Thus, $\mathbf{v}_B = v_B(\cos 30^\circ \mathbf{i} + \sin 30^\circ \mathbf{j})$

Conservation of momentum: $m\mathbf{v}_0 = m\mathbf{v}_A + m\mathbf{v}_B$

y component: $0 = m(v'_A)_y + mv_B \sin 30^\circ \quad (v'_A)_y = -v_B \sin 30^\circ$

x component: $v_0 = m(v'_A)_x + mv_B \cos 30^\circ \quad (v'_A)_x = v_0 - v_B \cos 30^\circ$

Conservation of energy:

$$\begin{aligned} \frac{1}{2}mv_0^2 &= \frac{1}{2}m(v'_A)_x^2 + \frac{1}{2}m(v'_A)_y^2 + \frac{1}{2}mv_B^2 \\ &= \frac{1}{2}m(v_0 - v_B \cos 30^\circ)^2 + \frac{1}{2}(v_B \sin 30^\circ)^2 + \frac{1}{2}v_B^2 \\ &= \frac{1}{2}m(v_0^2 - 2v_0v_B \cos 30^\circ + v_B^2 \cos^2 30^\circ + v_B^2 \sin^2 30^\circ + v_B^2) \end{aligned}$$

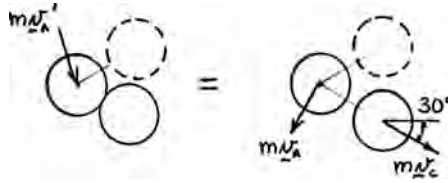
$$v_B = v_0 \cos 30^\circ = \frac{\sqrt{3}}{2}v_0, \quad (v'_A)_x = v_0 \sin^2 30^\circ = \frac{1}{4}v_0,$$

$$(v'_A)_y = -v_0 \cos 30^\circ \sin 30^\circ = -\frac{\sqrt{3}}{4}v_0$$

PROBLEM 14.38 (Continued)

Second impact: A strikes C.

During the impact, the contact impulse makes a 30° angle with the velocity $\mathbf{v}_0$.



Thus,

$$\mathbf{v}_C = v_C (\cos 30^\circ \mathbf{i} - \sin 30^\circ \mathbf{j})$$

Conservation of momentum: $m\mathbf{v}'_A = m\mathbf{v}_A + m\mathbf{v}_C$

x component: $m(v'_A)_x = m(v_A)_x + mv_C \cos 30^\circ$,

$$(v_A)_x = (v'_A)_x - v_C \cos 30^\circ = \frac{1}{4}v_0 - v_C \cos 30^\circ$$

y component: $m(v'_A)_y = m(v_A)_y - mv_C \sin 30^\circ$

$$(v_A)_y = (v'_A)_y + v_C \sin 30^\circ = -\frac{\sqrt{3}}{4}v_0 + v_C \sin 30^\circ$$

Conservation of energy:

$$\begin{aligned} \frac{1}{2}m(v'_A)_x^2 + \frac{1}{2}m(v'_A)_y^2 &= \frac{1}{2}m(v_A)_x^2 + \frac{1}{2}m(v_A)_y^2 + \frac{1}{2}mv_C^2 \\ \frac{1}{2}m\left[\frac{1}{16}v_0^2 + \frac{3}{16}v_0^2\right] &= \frac{1}{2}m\left[\left(\frac{1}{4}v_0 - v_C \cos 30^\circ\right)^2 + \left(-\frac{\sqrt{3}}{4}v_0 + v_C \sin 30^\circ\right)^2 + v_C^2\right] \end{aligned}$$

$$\begin{aligned} &= \frac{1}{2}m\left[\frac{1}{16}v_0^2 - \frac{1}{2}v_0v_C \cos 30^\circ + v_C^2 \cos^2 30^\circ\right. \\ &\quad \left.+ \frac{3}{16}v_0^2 - \frac{\sqrt{3}}{2}v_0v_C \sin 30^\circ + v_C^2 \sin^2 30^\circ + v_C^2\right] \end{aligned}$$

$$0 = -v_0v_C\left(\frac{1}{2}\cos 30^\circ + \frac{\sqrt{3}}{2}\sin 30^\circ\right) + 2v_C^2$$

$$v_C = v_0\left(\frac{1}{4}\cos 30^\circ + \frac{\sqrt{3}}{4}\sin 30^\circ\right) = \frac{\sqrt{3}}{4}v_0$$

$$(v_A)_x = \frac{1}{4}v_0 - \frac{\sqrt{3}}{4}v_0 \cos 30^\circ = -\frac{1}{8}v_0$$

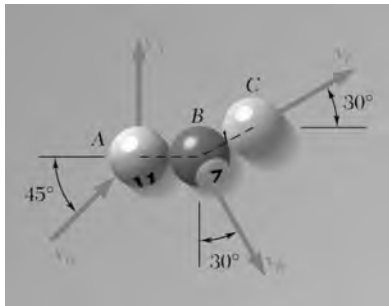
$$(v_A)_y = -\frac{\sqrt{3}}{4}v_0 + \frac{\sqrt{3}}{4}v_0 \sin 30^\circ = -\frac{\sqrt{3}}{8}v_0$$

$$\mathbf{v}_A = 0.250v_0 \nearrow 60^\circ \blacktriangleleft$$

$$\mathbf{v}_B = 0.866v_0 \nearrow 30^\circ \blacktriangleleft$$

$$\mathbf{v}_C = 0.433v_0 \nwarrow 30^\circ \blacktriangleleft$$

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PROBLEM 14.39

In a game of pool, ball A is moving with a velocity $\mathbf{v}_0$ of magnitude $v_0 = 5$ m/s when it strikes balls B and C , which are at rest and aligned as shown. Knowing that after the collision the three balls move in the directions indicated and assuming frictionless surfaces and perfectly elastic impact (i.e., conservation of energy), determine the magnitudes of the velocities $\mathbf{v}_A$, $\mathbf{v}_B$, and $\mathbf{v}_C$.

SOLUTION

Velocity vectors: $\mathbf{v}_0 = v_0(\cos 45^\circ \mathbf{i} + \sin 45^\circ \mathbf{j})$ $v_0 = 5$ m/s

$\mathbf{v}_A = v_A \mathbf{j}$

$\mathbf{v}_B = v_B(\sin 30^\circ \mathbf{i} - \cos 30^\circ \mathbf{j})$

$\mathbf{v}_C = v_C(\cos 30^\circ \mathbf{i} + \sin 30^\circ \mathbf{j})$

Conservation of momentum:

$$m\mathbf{v}_0 = m\mathbf{v}_A + m\mathbf{v}_B + m\mathbf{v}_C$$

Divide by m and resolve into components.

$$\mathbf{i}: v_0 \cos 45^\circ = v_B \sin 30^\circ + v_C \cos 30^\circ$$

$$\mathbf{j}: v_0 \sin 45^\circ = v_A - v_B \cos 30^\circ + v_C \sin 30^\circ$$

Solving for v_B and v_C ,

$$v_B = -0.25882v_0 + 0.86603v_A$$

$$v_C = 0.96593v_0 - 0.5v_A$$

Conservation of energy: $\frac{1}{2}mv_0^2 = \frac{1}{2}mv_A^2 + \frac{1}{2}mv_B^2 + \frac{1}{2}mv_C^2$

Divide by $\frac{1}{2}m$ and substitute for v_B and v_C .

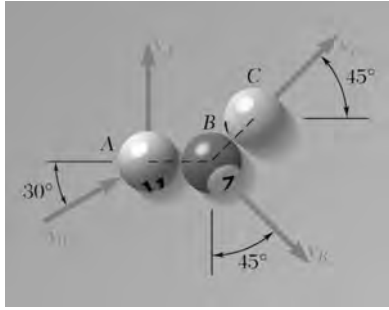
$$v_0^2 = v_A^2 + (-0.25882v_0 + 0.86603v_A)^2 + (0.96593v_0 - 0.5v_A)^2$$

$$= 2v_A^2 + v_0^2 - 1.41422v_0v_A$$

$$v_A = 0.70711v_0 = 3.5355 \text{ m/s} \quad v_A = 3.54 \text{ m/s} \quad \blacktriangleleft$$

$$v_B = 0.35355v_0 = 1.7677 \text{ m/s} \quad v_B = 1.768 \text{ m/s} \quad \blacktriangleleft$$

$$v_C = 0.61237v_0 = 3.06185 \text{ m/s} \quad v_C = 3.06 \text{ m/s} \quad \blacktriangleleft$$



PROBLEM 14.40

In a game of pool, ball A is moving with a velocity $\mathbf{v}_0$ of magnitude $v_0 = 5$ m/s when it strikes balls B and C , which are at rest and aligned as shown. Knowing that after the collision the three balls move in the directions indicated and assuming frictionless surfaces and perfectly elastic impact (that is, conservation of energy), determine the magnitudes of the velocities $\mathbf{v}_A$, $\mathbf{v}_B$, and $\mathbf{v}_C$.

SOLUTION

Velocity vectors:

$$\mathbf{v}_0 = v_0(\cos 30^\circ \mathbf{i} + \sin 30^\circ \mathbf{j}) \quad v_0 = 5 \text{ m/s}$$

$$\mathbf{v}_A = v_A \mathbf{j}$$

$$\mathbf{v}_B = v_B(\sin 45^\circ \mathbf{i} - \cos 45^\circ \mathbf{j})$$

$$\mathbf{v}_C = v_C(\cos 45^\circ \mathbf{i} + \sin 45^\circ \mathbf{j})$$

Conservation of momentum:

$$m\mathbf{v}_0 = m\mathbf{v}_A + m\mathbf{v}_B + m\mathbf{v}_C$$

Divide by m and resolve into components.

$$\mathbf{i}: v_0 \cos 30^\circ = v_B \sin 45^\circ + v_C \cos 45^\circ$$

$$\mathbf{j}: v_0 \sin 30^\circ = v_A - v_B \cos 45^\circ + v_C \sin 45^\circ$$

Solving for v_B and v_C ,

$$v_B = 0.25882v_0 + 0.70711v_A$$

$$v_C = 0.96593v_0 - 0.70711v_A$$

Conservation of energy:

$$\frac{1}{2}mv_0^2 = \frac{1}{2}mv_A^2 + \frac{1}{2}mv_B^2 + \frac{1}{2}mv_C^2$$

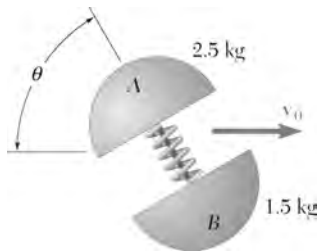
Divide by m and substitute for v_B and v_C .

$$\begin{aligned} v_0^2 &= v_A^2 + (0.25882v_0 + 0.70711v_A)^2 \\ &\quad + (0.96593v_0 - 0.70711v_A)^2 \\ &= v_0^2 - v_0v_A + 2v_A^2 \end{aligned}$$

$$v_A = 0.5v_0 = 2.500 \text{ m/s} \quad v_A = 2.50 \text{ m/s} \quad \blacktriangleleft$$

$$v_B = 0.61237v_0 = 3.06185 \text{ m/s} \quad v_B = 3.06 \text{ m/s} \quad \blacktriangleleft$$

$$v_C = 0.61237v_0 = 3.06185 \text{ m/s} \quad v_C = 3.06 \text{ m/s} \quad \blacktriangleleft$$



PROBLEM 14.41

Two hemispheres are held together by a cord which maintains a spring under compression (the spring is not attached to the hemispheres). The potential energy of the compressed spring is 120 J and the assembly has an initial velocity $\mathbf{v}_0$ of magnitude $v_0 = 8$ m/s. Knowing that the cord is severed when $\theta = 30^\circ$, causing the hemispheres to fly apart, determine the resulting velocity of each hemisphere.

SOLUTION

Use a frame of reference moving with the mass center.

Conservation of momentum:

$$0 = -m_A v'_A + m_B v'_B$$

$$v'_A = \frac{m_B}{m_A} v'_B$$

Conservation of energy:

$$\begin{aligned} V &= \frac{1}{2} m_A (v'_A)^2 + \frac{1}{2} m_B (v'_B)^2 \\ &= \frac{1}{2} m_A \left(\frac{m_B}{m_A} v'_B \right)^2 + \frac{1}{2} m_B (v'_B)^2 \\ &= \frac{m_B (m_A + m_B)}{2m_A} (v'_B)^2 \end{aligned}$$

$$v'_B = \sqrt{\frac{2m_A V}{m_B (m_A + m_B)}}$$

Data:

$$m_A = 2.5 \text{ kg} \quad m_B = 1.5 \text{ kg}$$

$$V = 120 \text{ J}$$

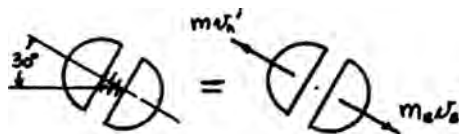
$$v'_B = \sqrt{\frac{(2)(2.5)(120)}{(1.5)(40)}} = 10 \quad v'_B = 10 \text{ m/s} \searrow 30^\circ$$

$$v'_A = \frac{1.5}{2.5} (10) = 6 \quad v'_A = 6 \text{ m/s} \nearrow 30^\circ$$

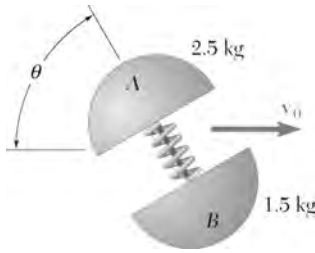
Velocities of A and B.

$$\mathbf{v}_A = [8 \text{ m/s} \rightarrow] + [6 \text{ m/s} \nearrow 30^\circ] \quad \mathbf{v}_A = 4.11 \text{ m/s} \nearrow 46.9^\circ \swarrow$$

$$\mathbf{v}_B = [8 \text{ m/s} \rightarrow] + [10 \text{ m/s} \searrow 30^\circ] \quad \mathbf{v}_B = 17.39 \text{ m/s} \searrow 16.7^\circ \swarrow$$



PROBLEM 14.42



Solve Problem 14.41, knowing that the cord is severed when $\theta = 120^\circ$.

PROBLEM 14.41 Two hemispheres are held together by a cord which maintains a spring under compression (the spring is not attached to the hemispheres). The potential energy of the compressed spring is 120 J and the assembly has an initial velocity $\mathbf{v}_0$ of magnitude $v_0 = 8$ m/s. Knowing that the cord is severed when $\theta = 30^\circ$, causing the hemispheres to fly apart, determine the resulting velocity of each hemisphere.

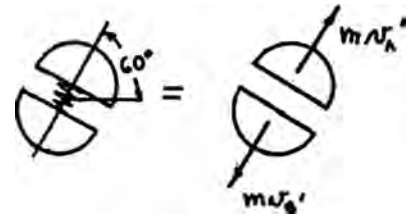
SOLUTION

Use a frame of reference moving with the mass center.

Conservation of momentum:

$$0 = -m_A v'_A + m_B v'_B$$

$$v'_A = \frac{m_B}{m_A} v'_B$$



Conservation of energy:

$$V = \frac{1}{2} m_A (v'_A)^2 + \frac{1}{2} m_B (v'_B)^2$$

$$= \frac{1}{2} m_A \left(\frac{m_B}{m_A} v'_B \right)^2 + \frac{1}{2} m_B (v'_B)^2$$

$$= \frac{m_B (m_A + m_B)}{2 m_A} (v'_B)^2$$

$$v'_B = \sqrt{\frac{2 m_A V}{m_B (m_A + m_B)}}$$

Data:

$$m_A = 2.5 \text{ kg} \quad m_B = 1.5 \text{ kg}$$

$$V = 120 \text{ J}$$

$$v'_B = \sqrt{\frac{(2)(2.5)(120)}{(1.5)(40)}} = 10 \quad v'_B = 10 \text{ m/s} \nearrow 60^\circ$$

$$v'_A = \frac{1.5}{2.5} (10) = 6 \quad v'_A = 6 \text{ m/s} \searrow 60^\circ$$

Velocities of A and B.

$$\mathbf{v}_A = [8 \text{ m/s} \rightarrow] + [6 \text{ ft/s} \nearrow 60^\circ]$$

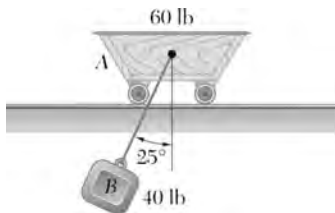
$$\mathbf{v}_A = 12.17 \text{ m/s} \nearrow 25.3^\circ \blacktriangleleft$$

$$\mathbf{v}_B = [8 \text{ m/s} \rightarrow] + [10 \text{ ft/s} \searrow 60^\circ]$$

$$\mathbf{v}_B = 9.17 \text{ m/s} \searrow 70.9^\circ \blacktriangleleft$$

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PROBLEM 14.43

A 20-kg block B is suspended from a 2-m cord attached to a 30-kg cart A , which may roll freely on a frictionless, horizontal track. If the system is released from rest in the position shown, determine the velocities of A and B as B passes directly under A .

SOLUTION

Conservation of linear momentum.

Since block and cart are initially at rest,

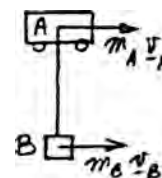
$$L_0 = 0$$

Thus, as B passes under A ,

$$L = m_A v_A + m_B v_B = 0$$

$$\rightarrow m_A v_A + m_B v_B = 0$$

$$v_A = -\frac{m_B}{m_A} v_B \quad (1)$$



Conservation of energy.

Initially,

$$T_0 = 0$$

$$V_0 = m_B g l (1 - \cos \theta)$$

As B passes under A ,

$$T = \frac{1}{2} m_A v_A^2 + \frac{1}{2} m_B v_B^2$$

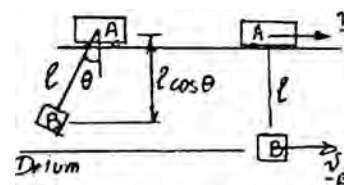
$$V = 0$$

Thus,

$$T_0 + V_0 = T + V: m_B g l (1 - \cos \theta) = \frac{1}{2} m_A v_A^2 + \frac{1}{2} m_B v_B^2$$

Substituting for v_A from (1) and multiplying by 2:

$$\begin{aligned} 2m_B g l (1 - \cos \theta) &= m_A \left(\frac{m_B^2}{m_A^2} v_B^2 \right) + m_B v_B^2 \\ &= \left(\frac{m_B^2}{m_A} + m_B \right) v_B^2 = m_B \frac{m_B + m_A}{m_A} v_B^2 \\ v_B &= \sqrt{\frac{2m_A}{m_A + m_B} g l (1 - \cos \theta)} \quad (2) \end{aligned}$$



PROBLEM 14.43 (Continued)

Given data:

$$m_A = 30 \text{ kg} \quad m_B = 20 \text{ kg}, \quad l = 2 \text{ m}$$

$$g = 9.81 \text{ m/s}^2 \quad \theta = 25^\circ$$

$$\frac{2m_A}{m_A + m_B} = \frac{2 \times 30}{30 + 20} = 1.2$$

From Eq. (2),

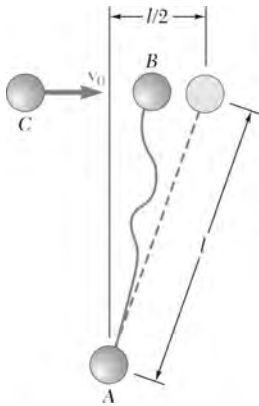
$$v_B = \sqrt{(1.2)(9.81)(2)(1 - \cos 25^\circ)} \\ = 1.48522 \text{ m/s}$$

$$\mathbf{v}_B = 1.485 \text{ m/s} \rightarrow \blacktriangleleft$$

From Eq. (1),

$$v_A = -\frac{m_B}{m_A} v_B = -\frac{20}{30}(1.48522)$$

$$\mathbf{v}_A = 0.990 \text{ m/s} \leftarrow \blacktriangleleft$$



PROBLEM 14.44

Three spheres, each of mass m , can slide freely on a frictionless, horizontal surface. Spheres A and B are attached to an inextensible, inelastic cord of length l and are at rest in the position shown when sphere B is struck squarely by sphere C, which is moving to the right with a velocity $\mathbf{v}_0$. Knowing that the cord is slack when sphere B is struck by sphere C, and assuming perfectly elastic impact between B and C, determine (a) the velocity of each sphere immediately after the cord becomes taut, (b) the fraction of the initial kinetic energy of the system which is dissipated when the cord becomes taut.

SOLUTION

(a) Determination of velocities.

Impact of C and B.

Conservation of momentum:

$$m\mathbf{v}_0 = m\mathbf{v}_C + m\mathbf{v}_1$$

$$v_C + v_1 = v_0 \quad (1)$$

Conservation of energy (perfectly elastic impact):

$$\frac{1}{2}mv_0^2 = \frac{1}{2}mv_C^2 + \frac{1}{2}mv_1^2 \quad v_C^2 + v_1^2 = v_0^2 \quad (2)$$

Square Eq. (1): $v_C^2 + 2v_Cv_1 + v_1^2 = v_0^2$

Subtract Eq. (2): $2v_Cv_1 = 0$

$v_1 = 0$ corresponds to initial conditions and should be eliminated. Therefore,

From Eq. (1): $v_1 = v_0$

Cord AB becomes taut.

Because cord is inextensible, component of $\mathbf{v}_B$ along AB must be equal to $\mathbf{v}_A$.

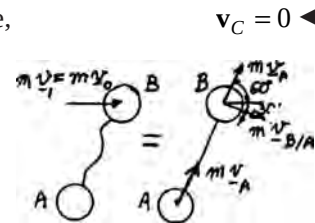
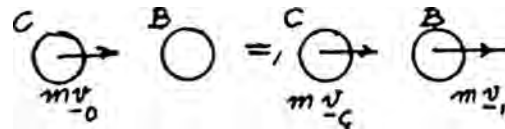
Conservation of momentum:

$$m\mathbf{v}_0 = 2m\mathbf{v}_A + m\mathbf{v}_{B/A}$$

+↑ y comp: $0 = 2mv_A \sin 60^\circ - mv_{B/A} \sin 30^\circ$

$$v_{B/A} = 2\sqrt{3}v_A \quad (3)$$

+→ x comp: $m\mathbf{v}_0 = 2mv_A \cos 60^\circ + mv_{B/A} \cos 30^\circ$



PROBLEM 14.44 (Continued)

Dividing by m and substituting for $v_{B/A}$ from Eq. (3):

$$v_0 = 2v_A(0.5) + (2\sqrt{3}v_A)\left(\frac{\sqrt{3}}{2}\right)$$

$$v_0 = 4v_A \quad v_A = 0.250v_0$$

$$v_A = 0.250v_0 \nearrow 60^\circ \blacktriangleleft$$

Carrying into Eq. (3):

$$v_{B/A} = 2\sqrt{3}(0.250v_0) = 0.866v_0$$

Thus,

$$\mathbf{v}_B = \mathbf{v}_A + \mathbf{v}_{B/A}$$

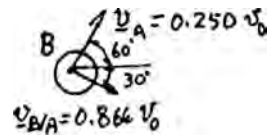
$$= 0.250v_0 \nearrow 60^\circ + 0.866v_0 \searrow 30^\circ$$

$$\mathbf{v}_B = (0.250v_0 \cos 60^\circ + 0.866v_0 \cos 30^\circ)\mathbf{i} \\ + (0.250v_0 \sin 60^\circ - 0.866v_0 \sin 30^\circ)\mathbf{j}$$

$$\mathbf{v}_B = 0.875v_0\mathbf{i} - 0.2165\mathbf{j}$$

$$v_B = 0.90139v_0 \searrow 13.90^\circ$$

$$v_B = 0.901v_0 \searrow 13.9^\circ \blacktriangleleft$$



(b) Fraction of kinetic energy lost.

$$T_0 = \frac{1}{2}mv_0^2$$

$$T_{\text{final}} = \frac{1}{2}mv_A^2 + \frac{1}{2}mv_B^2 + \frac{1}{2}mv_C^2 \\ = \frac{1}{2}m(0.250v_0)^2 + \frac{1}{2}m(0.90139v_0)^2 + \frac{1}{2}m(0) \\ = \frac{1}{2}m(0.875)v_0^2$$

$$\text{Kinetic energy lost} = T_0 - T_{\text{final}} = \frac{1}{2}m(1 - 0.875)v_0^2 = \frac{1}{2} - \frac{1}{8}mv_0^2$$

$$\text{Fraction of kinetic energy lost} = \frac{1}{8} \blacktriangleleft$$



PROBLEM 14.45

A 360-kg space vehicle traveling with a velocity $\mathbf{v}_0 = (450 \text{ m/s})\mathbf{k}$ passes through the origin O . Explosive charges then separate the vehicle into three parts A , B , and C , with masses of 60 kg, 120 kg, and 180 kg, respectively. Knowing that shortly thereafter the positions of the three parts are, respectively, $A(72, 72, 648)$, $B(180, 396, 972)$, and $C(-144, -288, 576)$, where the coordinates are expressed in meters, that the velocity of B is $\mathbf{v}_B = (150 \text{ m/s})\mathbf{i} + (330 \text{ m/s})\mathbf{j} + (660 \text{ m/s})\mathbf{k}$, and that the x component of the velocity of C is -120 m/s , determine the velocity of Part A .

SOLUTION

Position vectors (meters):

$$\begin{aligned}\mathbf{r}_A &= 72\mathbf{i} + 72\mathbf{j} + 648\mathbf{k} \\ \mathbf{r}_B &= 180\mathbf{i} + 396\mathbf{j} + 972\mathbf{k} \\ \mathbf{r}_C &= -144\mathbf{i} - 288\mathbf{j} + 576\mathbf{k}\end{aligned}$$

Since there are no external forces, linear momentum is conserved.

$$\begin{aligned}(m_A + m_B + m_C)\mathbf{v}_0 &= m_A\mathbf{v}_A + m_B\mathbf{v}_B + m_C\mathbf{v}_C \\ \mathbf{v}_A &= \frac{m_A + m_B + m_C}{m_A}\mathbf{v}_0 - \frac{m_B}{m_A}\mathbf{v}_B - \frac{m_C}{m_A}\mathbf{v}_C = 6\mathbf{v}_0 - 2\mathbf{v}_B - 3\mathbf{v}_C \quad (1) \\ &= (6)(450\mathbf{k}) - (2)(150\mathbf{i} + 330\mathbf{j} + 660\mathbf{k}) - (3)[-120\mathbf{i} + (v_C)_y\mathbf{j} + (v_C)_z\mathbf{k}] \\ &= -3(v_C)_y\mathbf{j} - 3(v_C)_z\mathbf{k} + 60\mathbf{i} - 660\mathbf{j} + 1380\mathbf{k} \\ (v_A)_x &= 60, \quad (v_C)_y = -3(v_C)_y - 660, \quad (v_A)_z = -3(v_C)_z + 1380\end{aligned}$$

Conservation of angular momentum about O :

$$(\mathbf{H}_O)_2 = (\mathbf{H}_O)_1$$

Since the vehicle passes through the origin, $(\mathbf{H}_O)_1 = 0$.

$$(\mathbf{H}_O)_2 = \mathbf{r}_A \times (m_A\mathbf{v}_A) + \mathbf{r}_B \times (m_B\mathbf{v}_B) + \mathbf{r}_C \times (m_C\mathbf{v}_C) = 0$$

Divide by m_A .

$$\begin{aligned}\mathbf{r}_A \times \mathbf{v}_A + \frac{m_B}{m_A}\mathbf{r}_B \times \mathbf{v}_B + \frac{m_C}{m_A}\mathbf{r}_C \times \mathbf{v}_C &= \mathbf{r}_A \times \mathbf{v}_A + 2\mathbf{r}_B \times \mathbf{v}_B + 3\mathbf{r}_C \times \mathbf{v}_C \\ &= \mathbf{r}_A \times (6\mathbf{v}_0 - 2\mathbf{v}_B - 3\mathbf{v}_C) + 2\mathbf{r}_B \times \mathbf{v}_B + 3\mathbf{r}_C \times \mathbf{v}_C \\ &= 3(\mathbf{r}_C - \mathbf{r}_A) \times \mathbf{v}_C + 6\mathbf{r}_A \times \mathbf{v}_0 + 2(\mathbf{r}_B - \mathbf{r}_A) \times \mathbf{v}_B \\ &= (-648\mathbf{i} - 1080\mathbf{j} - 216\mathbf{k}) \times \mathbf{v}_C + (432\mathbf{i} + 432\mathbf{j} + 3894\mathbf{k}) \times \mathbf{v}_0 \\ &\quad + (216\mathbf{i} + 648\mathbf{j} + 648\mathbf{k}) \times \mathbf{v}_B = 0\end{aligned}$$

PROBLEM 14.45 (Continued)

$$\begin{vmatrix} \mathbf{i} & \mathbf{j} & \mathbf{k} \\ -648 & -1080 & -216 \\ (v_C)_x & (v_C)_y & (v_C)_z \end{vmatrix} + \begin{vmatrix} \mathbf{i} & \mathbf{j} & \mathbf{k} \\ 432 & 432 & 3894 \\ 0 & 0 & 450 \end{vmatrix} + \begin{vmatrix} \mathbf{i} & \mathbf{j} & \mathbf{k} \\ 216 & 648 & 648 \\ 150 & 330 & 660 \end{vmatrix} = 0$$

Resolve into components.

$$\mathbf{i}: \quad 216(v_C)_y - 1080(v_C)_z + 194,400 + 213,840 = 0 \quad (2)$$

$$\mathbf{j}: \quad 648(v_C)_z - 216(v_C)_x - 194,400 - 45,360 = 0 \quad (3)$$

$$\mathbf{k}: \quad 1080(v_C)_x - 648(v_C)_y + 0 - 25,920 = 0 \quad (4)$$

$$\text{Set} \quad (v_C)_x = -120 \text{ m/s}$$

$$\text{From Eq. (4),} \quad (v_C)_y = -240 \text{ m/s}$$

$$\text{From Eq. (3),} \quad (v_C)_z = 330 \text{ m/s}$$

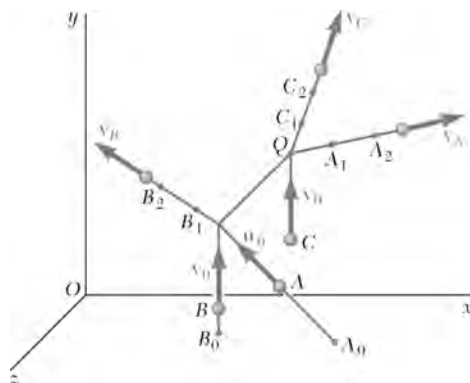
$$\begin{aligned} \text{From Eq. (1),} \quad \mathbf{v}_A &= (6)(450\mathbf{k}) - (2)(150\mathbf{i} + 330\mathbf{j} + 660\mathbf{k}) - 3(-120\mathbf{i} - 240\mathbf{j} + 330\mathbf{k}) \\ &= 60\mathbf{i} + 60\mathbf{j} + 390\mathbf{k} \end{aligned}$$

$$\mathbf{v}_A = (60.0 \text{ m/s})\mathbf{i} + (60.0 \text{ m/s})\mathbf{j} + (390 \text{ m/s})\mathbf{k} \quad \blacktriangleleft$$

PROBLEM 14.46

In the scattering experiment of Problem 14.24, it is known that the alpha particle is projected from $A_0(300, 0, 300)$ and that it collides with the oxygen nucleus C at $Q(240, 200, 100)$, where all coordinates are expressed in millimeters. Determine the coordinates of Point B_0 where the original path of nucleus B intersects the xz plane. (Hint: Express that the angular momentum of the three particles about Q is conserved.)

PROBLEM 14.24 In this scattering experiment, an alpha particle A is projected with the velocity $\mathbf{u}_0 = -(600 \text{ m/s})\mathbf{i} + (750 \text{ m/s})\mathbf{j} - (800 \text{ m/s})\mathbf{k}$ into a stream of oxygen nuclei moving with the common velocity $\mathbf{v}_0 = (600 \text{ m/s})\mathbf{j}$. After colliding successively with nuclei B and C , particle A is observed to move along the path defined by the Points $A_1(280, 240, 120)$ and $A_2(360, 320, 160)$, while nuclei B and C are observed to move along paths defined, respectively, by $B_1(147, 220, 130)$, $B_2(114, 290, 120)$, and by $C_1(240, 232, 90)$ and $C_2(240, 280, 75)$. All paths are along straight lines and all coordinates are expressed in millimeters. Knowing that the mass of an oxygen nucleus is four times that of an alpha particle, determine the speed of each of the three particles after the collisions.



SOLUTION

Conservation of angular momentum about Q :

$$\begin{aligned} \overline{QA}_0 \times (m\mathbf{u}_0) + \overline{QB}_0 \times (4m\mathbf{v}_0) + \overline{QC}_0 \times (4m\mathbf{v}_0) &= \overline{QA}_1 \times (m\mathbf{v}_A) + \overline{QB}_1 \times (4m\mathbf{v}_B) + \overline{QC}_1 \times (4m\mathbf{v}_C) \\ \overline{QA}_0 \times (m\mathbf{u}_0) + \overline{QB}_0 \times (4m\mathbf{v}_0) + 0 &= 0 + \overline{QB}_1 \times (4m\mathbf{v}_B) + 0 \end{aligned} \quad (1)$$

where

$$\begin{aligned} \overline{QA}_0 &= \mathbf{r}_{A_0} - \mathbf{r}_Q = (300\mathbf{i} + 300\mathbf{k}) - (240\mathbf{i} + 200\mathbf{j} + 100\mathbf{k}) \\ &= (60 \text{ mm})\mathbf{i} - (200 \text{ mm})\mathbf{j} + (200 \text{ mm})\mathbf{k} \end{aligned}$$

$$\overline{QB}_0 = (\Delta x)\mathbf{i} + (\Delta y)\mathbf{j} + (\Delta z)\mathbf{k}$$

$$\begin{aligned} \overline{QB}_1 &= \mathbf{r}_{B_1} - \mathbf{r}_Q = (147\mathbf{i} + 220\mathbf{j} + 130\mathbf{k}) - (240\mathbf{i} + 200\mathbf{j} + 100\mathbf{k}) \\ &= (-93 \text{ mm})\mathbf{i} + (20 \text{ mm})\mathbf{j} + (30 \text{ mm})\mathbf{k} \end{aligned}$$

$$\mathbf{u}_0 = -(600 \text{ m/s})\mathbf{i} + (750 \text{ m/s})\mathbf{j} - (800 \text{ m/s})\mathbf{k} \quad \mathbf{v}_0 = (600 \text{ m/s})\mathbf{j}$$

and from the solution to Problem 14.27,

$$\begin{aligned} \mathbf{v}_B &= v_B \lambda_B = (716.98)(-0.42290\mathbf{i} + 0.89707\mathbf{j} - 0.12815\mathbf{k}) \\ &= -(303.21 \text{ m/s})\mathbf{i} + (643.18 \text{ m/s})\mathbf{j} - (91.88 \text{ m/s})\mathbf{k} \end{aligned}$$

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PROBLEM 14.46 (Continued)

Calculating each term and dividing by m ,

$$\overline{QA}_0 \times \mathbf{u}_0 = \begin{vmatrix} \mathbf{i} & \mathbf{j} & \mathbf{k} \\ 60 & -200 & 200 \\ -600 & 750 & -800 \end{vmatrix} = 10,000\mathbf{i} - 72,000\mathbf{j} - 75,000\mathbf{k}$$

$$\begin{aligned} \overline{QB}_0 \times (4\mathbf{v}_0) &= [(\Delta x)\mathbf{i} + (\Delta y)\mathbf{j} + (\Delta z)\mathbf{k}] \times (2400\mathbf{j}) \\ &= -2400(\Delta z)\mathbf{i} + 2400(\Delta x)\mathbf{k} \end{aligned}$$

$$\overline{QB}_1 \times (4\mathbf{v}_B) = \begin{vmatrix} \mathbf{i} & \mathbf{j} & \mathbf{k} \\ -93 & 20 & 30 \\ -1212.84 & 2572.72 & -367.52 \end{vmatrix} = -84,532\mathbf{i} - 70,565\mathbf{j} - 215,006\mathbf{k}$$

Collect terms and resolve into components.

$$\mathbf{i}: \quad 10,000 - 2400(\Delta z) = -84,532 \quad \Delta z = 39.388 \text{ mm}$$

$$\mathbf{k}: \quad -75,000 + 2400(\Delta x) = -215,006 \quad \Delta x = -58.336 \text{ mm}$$

Coordinates:

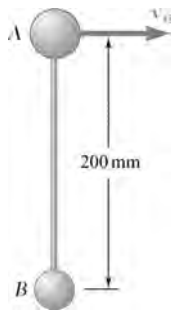
$$x_{B_0} = x_Q + \Delta x = 240 - 58.336$$

$$x_{B_0} = 181.7 \text{ mm} \quad \blacktriangleleft$$

$$y_{B_0} = 0 \quad \blacktriangleleft$$

$$z_{B_0} = z_Q + \Delta z = 100 + 39.388$$

$$z_{B_0} = 139.4 \text{ mm} \quad \blacktriangleleft$$



PROBLEM 14.47

Two small spheres A and B, weighing 2.5 kg and 1 kg, respectively, are connected by a rigid rod of negligible weight. The two spheres are resting on a horizontal, frictionless surface when A is suddenly given the velocity $\mathbf{v}_0 = (3 \text{ m/s})\mathbf{i}$. Determine (a) the linear momentum of the system and its angular momentum about its mass center G, (b) the velocities of A and B after the rod AB has rotated through 180° .

SOLUTION

Position of mass center.

$$AG + BG = 200 \text{ mm} = 0.2 \text{ m}$$

$$AG(2.5 \text{ kg}) = BG(1 \text{ kg}) \quad BG = 2.5AG \quad 3.5AG = 0.2 \quad AG = \frac{2}{35} \text{ m}$$

$$BG = \frac{1}{7} \text{ m}$$

(a) *Linear and angular momentum*

$$L = m_A \mathbf{v}_0 = (2.5 \text{ kg})(3 \text{ m/s})\mathbf{i} = 7.5 \text{ kg} \cdot \text{m/s}$$

$$\mathbf{L} = (7.50 \text{ kg} \cdot \text{m/s})\mathbf{i} \quad \blacktriangleleft$$

$$\mathbf{H}_G = \overline{GA} \times m_A \mathbf{v}_0 = \left(\frac{2}{35} \text{ m} \right) \mathbf{j} \times (7.50 \text{ kg} \cdot \text{m/s})\mathbf{i}$$

$$= -\left(\frac{3}{7} \text{ kg} \cdot \text{m}^2/\text{s} \right) \mathbf{k}$$

$$= -(0.42857 \text{ kg} \cdot \text{m}^2/\text{s})\mathbf{k}$$

$$\mathbf{H}_G = -(0.429 \text{ kg} \cdot \text{m}^2/\text{s})\mathbf{k} \quad \blacktriangleleft$$

(b) *Velocities of A and B after 180° rotation*

Conservation of linear momentum:

$$m_A \mathbf{v}_0 = m_A \mathbf{v}'_A + m_B \mathbf{v}'_B$$

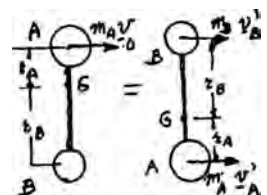
$$2.5 \mathbf{v}_0 = 2.5 \mathbf{v}'_A + \mathbf{v}'_B$$

$$2.5 \mathbf{v}'_A + \mathbf{v}'_B = 7.5$$

Conservation of angular momentum about G' :

$$\curvearrowright r_A m_A \mathbf{v}_0 = -r_A m_A \mathbf{v}'_A + r_B m_B \mathbf{v}'_B$$

$$\left(\frac{2}{35} \text{ m} \right) (2.5)(3) = -\left(\frac{2}{35} \text{ m} \right) (2.5) \mathbf{v}'_A + \left(\frac{1}{7} \right) (1)(\mathbf{v}'_B)$$



(1)

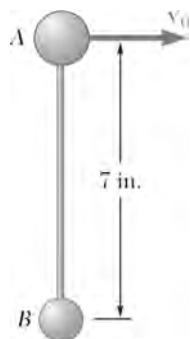
PROBLEM 14.47 (Continued)

$$\begin{aligned} -\frac{v'_A}{7} + \frac{v'_B}{7} &= \frac{3}{7} \\ -v'_A + v'_B &= 3 \end{aligned} \quad (2)$$

From Eq. (1):

Solving: $v'_A = \frac{9}{7} \text{ m/s}, \quad v'_B = \frac{30}{7} \text{ m/s}$

$$\mathbf{v}'_A = (1.286 \text{ m/s})\mathbf{i}; \quad \mathbf{v}'_B = (4.29 \text{ m/s})\mathbf{i} \quad \blacktriangleleft$$



PROBLEM 14.48

Solve Problem 14.47, assuming that it is B which is suddenly given the velocity $\mathbf{v}_0 = (3 \text{ m/s})\mathbf{i}$.

PROBLEM 14.47 Two small spheres A and B, weighing 2.5 kg and 1 kg, respectively, are connected by a rigid rod of negligible weight. The two spheres are resting on a horizontal, frictionless surface when A is suddenly given the velocity $\mathbf{v}_0 = (3 \text{ m/s})\mathbf{i}$. Determine (a) the linear momentum of the system and its angular momentum about its mass center G, (b) the velocities of A and B after the rod AB has rotated through 180° .

SOLUTION

Position of mass center.

$$AG + BG = 200 \text{ mm} = 0.2 \text{ m}$$

$$AG (2.5 \text{ kg}) = BG (1 \text{ kg}) \quad BG = 2.5AG \quad 3.5AG = 0.2 \quad AG = \frac{2}{35} \text{ m}$$

$$BG = \frac{1}{7} \text{ m}$$

(a) *Linear and angular momentum*

$$L = m_B \mathbf{v}_0 = (1 \text{ kg})(3 \text{ m/s})\mathbf{i} = (3 \text{ kg} \cdot \text{m/s})\mathbf{i}$$

$$\mathbf{L} = (3 \text{ kg} \cdot \text{m/s})\mathbf{i} \quad \blacktriangleleft$$

$$\begin{aligned} \mathbf{H}_G &= \overline{GB} \times m_B \mathbf{v}_0 = -\left(\frac{1}{7} \text{ m}\right) \mathbf{j} \times (3 \text{ kg} \cdot \text{m/s})\mathbf{i} \\ &= +\left(\frac{3}{7} \text{ kg} \cdot \text{m}^2/\text{s}\right) \mathbf{k} \end{aligned}$$

$$\mathbf{H}_G = +(0.429 \text{ kg} \cdot \text{m}^2/\text{s})\mathbf{k} \quad \blacktriangleleft$$

(b) *Velocities of A and B after 180° rotation*

Conservation of linear momentum:

$$m_B v_0 = m_A v'_A + m_B v'_B$$

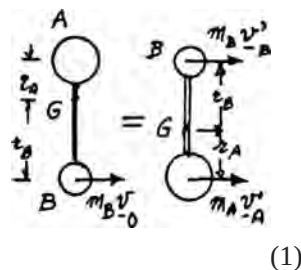
$$1(3) = (2.5)v'_A + v'_B$$

$$2.5v'_A + v'_B = 3$$

Conservation of angular momentum about G:

$$\curvearrowright r_B m_B v_0 = r_A m_A v'_A - r_B m_B v'_B$$

$$\left(\frac{1}{7}\right)(1)(3) = \left(\frac{2}{35}\right)(2.5)v'_A - \left(\frac{1}{7}\right)(1)v'_B$$



(1)

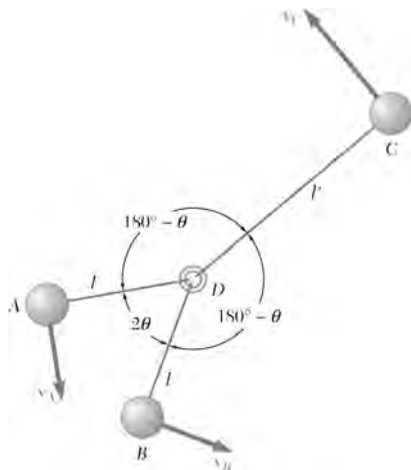
PROBLEM 14.48 (Continued)

$$\begin{aligned}\frac{v'_A}{7} - \frac{v'_B}{7} &= \frac{3}{7} \\ v'_A - v'_B &= 3\end{aligned}\tag{2}$$

Solving:

$$v'_A = \frac{12}{7} \text{ m/s}, \quad v'_B = -\frac{9}{7} \text{ m/s}$$

$$\mathbf{v}'_A = (1.714 \text{ m/s})\mathbf{i}; \quad \mathbf{v}'_B = -(1.259 \text{ m/s})\mathbf{i} \quad \blacktriangleleft$$



PROBLEM 14.49

Three identical spheres A, B, and C, which can slide freely on a frictionless horizontal surface, are connected by means of inextensible, inelastic cords to a small ring D located at the mass center of the three spheres ($l' = 2l \cos \theta$). The spheres are rotating initially about ring D, which is at rest, at speeds proportional to their distances from D. We denote by v_0 the original speed of A and B and assume that $\theta = 30^\circ$. Suddenly, cord CD breaks, causing sphere C to slide away. Considering the motion of spheres A and B and of ring D after the other two cords have again become taut, determine (a) the speed of ring D, (b) the relative speed at which spheres A and B rotate about D, (c) the percent of energy of the original system which is dissipated when cords AD and BD again become taut.

SOLUTION

We consider the following two positions of the spheres A and B and the ring D.

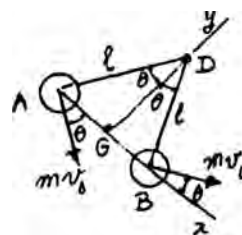
Position 1: Immediately after cord CD breaks linear momentum:

$$\mathbf{L} = (2mv_0 \cos \theta) \mathbf{i} \quad (1)$$

Angular momentum about G:

$$H_G = 2(l \sin \theta)(mv_0 \sin \theta) \mathbf{k}$$

$$H_G = (2lmv_0 \sin^2 \theta) \mathbf{k} \quad (2)$$



Position 2: After cords AD and BD become taut:

(a) Speed of mass center (now located at D).

Recalling Eq. (1):

$$\begin{aligned} L &= (2m)\bar{v} = (2mv_0 \cos \theta) \mathbf{i} & \bar{v} &= (v_0 \cos \theta) \mathbf{i} \\ v_D &= \bar{v} = v_0 \cos \theta \end{aligned} \quad (3)$$

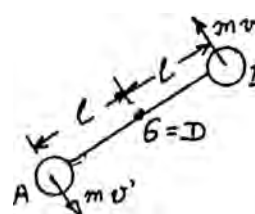
(b) Relative speed v' at which A and B rotate about D.

Angular momentum about G:

$$H_G = (2mv')l$$

Recalling Eq. (2):

$$\begin{aligned} 2mv'l &= 2lmv_0 \sin^2 \theta \\ v' &= v_0 \sin^2 \theta \end{aligned} \quad (4)$$



PROBLEM 14.49 (Continued)

(c) Energy lost:

Considering system of 3 spheres:

Initially,
$$v_C = \left(\frac{l'}{l}\right)v_A = (2 \cos \theta)v_0$$

Therefore,

$$T_0 = \frac{1}{2}mv_A^2 + \frac{1}{2}mv_B^2 + \frac{1}{2}mv_C^2 = mv_0^2(1 + 2\cos^2 \theta)$$

$$\begin{aligned} T_f &= \frac{1}{2}(2m)v_D^2 + 2\left(\frac{1}{2}mv'^2\right) + \frac{1}{2}mv_C^2 \\ &= m\left[v_0^2 \cos^2 \theta + m(v_0 \sin^2 \theta)^2 + 2v_0^2 \cos^2 \theta\right] \\ &= mv_0^2(3 \cos^2 \theta + \sin^4 \theta) \end{aligned}$$

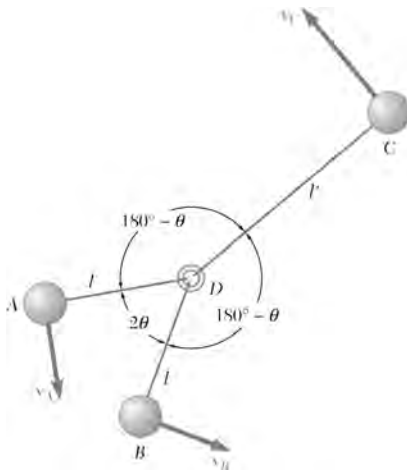
$$\begin{aligned} \% \text{ loss} &= 100 \frac{T_0 - T_f}{T_0} \\ &= 100 \frac{1 + 2\cos^2 \theta - 3\cos^2 \theta - \sin^4 \theta}{1 + 2\cos^2 \theta} \\ &= 100 \frac{\sin^2 \theta \cos^2 \theta}{1 + 2\cos^2 \theta} \end{aligned} \quad (5)$$

Making $\theta = 30^\circ$ in Eqs. (3), (4), and (5)

(a) $0.866v_0$, (b) $0.250 v_0$, (c) 7.50% ◀

PROBLEM 14.50

Solve Problem 14.49, assuming that $\theta = 45^\circ$.



PROBLEM 14.49 Three identical spheres A , B , and C , which can slide freely on a frictionless horizontal surface, are connected by means of inextensible, inelastic cords to a small ring D located at the mass center of the three spheres ($l' = 2l \cos \theta$). The spheres are rotating initially about ring D , which is at rest, at speeds proportional to their distances from D . We denote by v_0 the original speed of A and B and assume that $\theta = 30^\circ$. Suddenly, cord CD breaks, causing sphere C to slide away. Considering the motion of spheres A and B and of ring D after the other two cords have again become taut, determine (a) the speed of ring D , (b) the relative speed at which spheres A and B rotate about D , (c) the percent of energy of the original system which is dissipated when cords AD and BD again become taut.

SOLUTION

We consider the following two positions of the spheres A and B and the ring D .

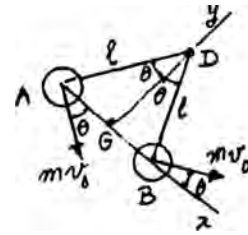
Position 1: Immediately after cord CD breaks linear momentum:

$$\mathbf{L} = (2mv_0 \cos \theta)\mathbf{i} \quad (1)$$

Angular momentum about G :

$$H_G = 2(l \sin \theta)(mv_0 \sin \theta)\mathbf{k}$$

$$H_G = (2lmv_0 \sin^2 \theta)\mathbf{k} \quad (2)$$



Position 2: After cords AD and BD become taut:

(a) Speed of mass center (now located at D).

Recalling Eq. (1):

$$\mathbf{L} = (2m)\bar{\mathbf{v}} = (2mv_0 \cos \theta)\mathbf{i} \quad \bar{\mathbf{v}} = (v_0 \cos \theta)\mathbf{i}$$

$$v_D = \bar{v} = v_0 \cos \theta \quad (3)$$

(b) Relative speed v' at which A and B rotate about D .

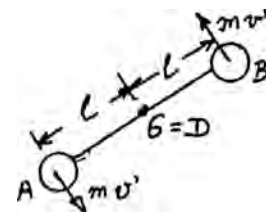
Angular momentum about G :

$$H_G = (2mv')l$$

Recalling Eq. (2):

$$2mv'l = 2lmv_0 \sin^2 \theta$$

$$v' = v_0 \sin^2 \theta \quad (4)$$



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PROBLEM 14.50 (Continued)

(c) Energy lost:

Considering system of 3 spheres:

Initially,

$$v_C = (l'/l)v_A = (2 \cos \theta)v_0$$

Therefore,

$$T_0 = \frac{1}{2}mv_A^2 + \frac{1}{2}mv_B^2 + \frac{1}{2}mv_C^2 = mv_0^2(1 + 2\cos^2 \theta)$$

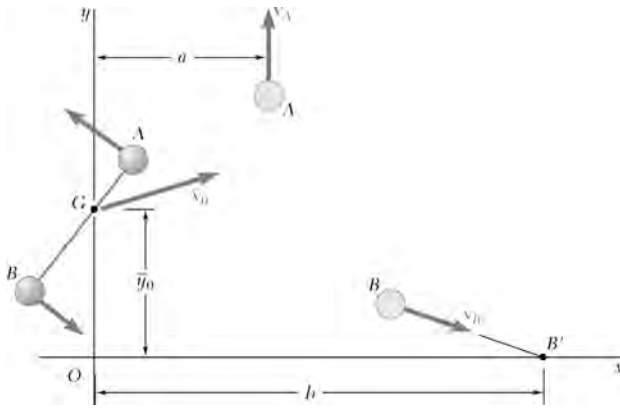
$$\begin{aligned} T_f &= \frac{1}{2}(2m)v_D^2 + 2\left(\frac{1}{2}mv'^2\right) + \frac{1}{2}mv_C^2 \\ &= m\left[v_0^2 \cos^2 \theta + m(v_0 \sin^2 \theta)^2 + 2v_0^2 \cos^2 \theta\right] \\ &= mv_0^2(3\cos^2 \theta + \sin^4 \theta) \end{aligned}$$

$$\begin{aligned} \% \text{ loss} &= 100 \frac{T_0 - T_f}{T_0} \\ &= 100 \frac{1 + 2\cos^2 \theta - 3\cos^2 \theta - \sin^4 \theta}{1 + 2\cos^2 \theta} \\ &= 100 \frac{\sin^2 \theta \cos^2 \theta}{1 + 2\cos^2 \theta} \end{aligned} \quad (5)$$

Making $\theta = 45^\circ$ in Eqs. (3), (4), and (5):

(a) $0.707v_0$, (b) $0.500 v_0$, (c) 12.50% ◀

PROBLEM 14.51



Two small disks A and B , of mass 3 kg and 1.5 kg , respectively, may slide on a horizontal, frictionless surface. They are connected by a cord 600 mm long and spin counterclockwise about their mass center G at the rate of 10 rad/s . At $t = 0$, the coordinates of G are $\bar{x}_0 = 0$, $\bar{y}_0 = 2\text{ m}$, and its velocity is $\bar{v}_0 = (1.2\text{ m/s})\mathbf{i} + (0.96\text{ m/s})\mathbf{j}$. Shortly thereafter, the cord breaks; disk A is then observed to move along a path parallel to the y axis and, disk B along a path which intersects the x axis at a distance $b = 7.5\text{ m}$ from O . Determine (a) the velocities of A and B after the cord breaks, (b) the distance a from the y -axis to the path of A .

SOLUTION

Initial conditions.

Location of G :

$$\frac{AG}{m_B} = \frac{BG}{m_A} = \frac{AG + GB}{m_B + m_A} = \frac{AB}{m} = \frac{0.6\text{ m}}{4.5\text{ kg}}$$

$$AG = 1.5 \left(\frac{0.6}{4.5} \right) = 0.2\text{ m}$$

$$BG = 0.4\text{ m}$$

Linear momentum:

$$\begin{aligned} \mathbf{L}_0 &= m\bar{\mathbf{v}}_0 = (4.5\text{ kg})(1.2\mathbf{i} + 0.96\mathbf{j}) \\ &= 5.4\mathbf{i} + 4.32\mathbf{j} \end{aligned}$$

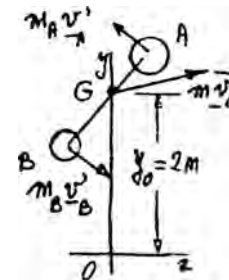
Angular momentum:

About G :

$$\begin{aligned} (\mathbf{H}_G)_0 &= \overline{GA} \times m_A \mathbf{v}'_A + \overline{GB} \times m_B \mathbf{v}'_B \\ &= (0.2\text{ m})(3\text{ kg})(0.2\text{ m} \times 10\text{ rad/s})\mathbf{k} \\ &\quad + (0.4\text{ m})(1.5\text{ kg})(0.4\text{ m} \times 10\text{ rad/s})\mathbf{k} \\ (\mathbf{H}_G)_0 &= (3.6\text{ kg} \cdot \text{m}^2/\text{s})\mathbf{k} \end{aligned}$$

About O : Using formula derived in Problem 14.27,

$$\begin{aligned} (\mathbf{H}_O)_0 &= \bar{\mathbf{r}} \times m\bar{\mathbf{v}}_0 + (\mathbf{H}_G)_0 \\ &= 2\mathbf{j} \times (5.4\mathbf{i} + 4.32\mathbf{j}) + 3.6\mathbf{k} \\ &= -10.8\mathbf{k} + 3.6\mathbf{k} = -(7.2\text{ kg} \cdot \text{m}^2/\text{s})\mathbf{k} \end{aligned}$$



PROBLEM 14.51 (Continued)

Kinetic energy: Using Eq. (14.29),

$$\begin{aligned} T_0 &= \frac{1}{2} m \bar{v}_0^2 + \frac{1}{2} \sum_i m v_i'^2 = \frac{1}{2} m \bar{v}_0^2 + \frac{1}{2} m_A v_A'^2 + \frac{1}{2} m_B v_B'^2 \\ &= \frac{1}{2} (4.5) [(1.2)^2 + (0.96)^2] + \frac{1}{2} (3) (0.2 \times 10)^2 + \frac{1}{2} (1.5) (0.4 \times 10)^2 \\ &= (5.3136 + 6 + 12) = 23.314 \text{ J} \end{aligned}$$

(a) Conservation of linear momentum:

$$\begin{aligned} \mathbf{L}_0 &= \mathbf{L} \\ 5.4\mathbf{i} + 4.32\mathbf{j} &= m_A \mathbf{v}_A + m_B \mathbf{v}_B \\ &= 3(v_A \mathbf{j}) + 1.5[(v_B)_x \mathbf{i} + (v_B)_y \mathbf{j}] \end{aligned}$$

Equating coefficients of $\mathbf{i}$: $5.4 = 1.5(v_B)_x$

$$(v_B)_x = 3.6 \text{ m/s} \quad (1)$$

Equating coefficients of $\mathbf{j}$: $4.32 = 3v_A + 1.5(v_B)_y$

$$(v_B)_y = 2.88 - 2v_A \quad (2)$$

Conservation of energy.

$$\begin{aligned} T_0 &= T: \quad T_0 = \frac{1}{2} m_A v_A^2 + \frac{1}{2} m_B v_B^2 \\ 23.314 \text{ J} &= \frac{1}{2} (3) (v_A^2) + \frac{1}{2} (1.5) [(v_B)_x^2 + (v_B)_y^2] \end{aligned}$$

Substituting from Eqs. (1) and (2):

$$23.314 = 1.5v_A^2 + 0.75(3.6)^2 + 0.75(2.88 - 2v_A)^2$$

$$4.5v_A^2 - 8.64v_A - 7.373 = 0$$

$$v_A^2 - 1.92v_A - 1.6389 = 0$$

$$v_A = 0.96 + 1.60 = 2.56 \text{ m/s} \quad \mathbf{v}_A = 2.56 \text{ m/s} \uparrow \blacktriangleleft$$

and $v_A = 0.96 - 1.60 = -0.64 \text{ m/s}$ (rejected, since $\mathbf{v}_A$ is shown directed up)

From Eqs. (1) and (2):

$$(v_B)_x = 3.6 \text{ m/s}$$

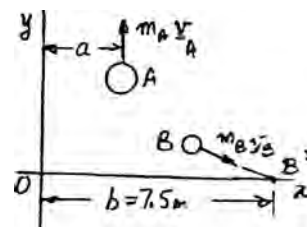
$$(v_B)_y = 2.88 - 2(2.56) = -2.24 \text{ m/s}$$

$$\mathbf{v}_B = 3.6\mathbf{i} - 2.24\mathbf{j} \quad \mathbf{v}_B = 4.24 \text{ m/s} \searrow 31.9^\circ \blacktriangleleft$$

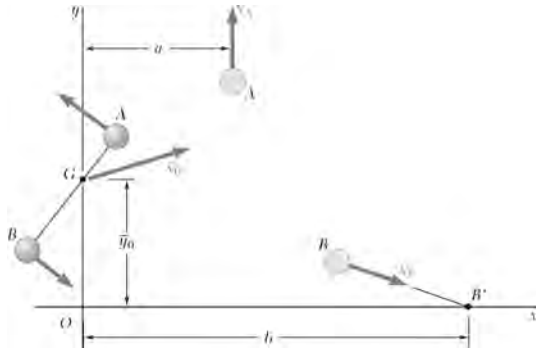
(b) Conservation of angular momentum about O:

$$\begin{aligned} (\mathbf{H}_O)_0 &= \mathbf{H}_O: \quad -7.2\mathbf{k} = a\mathbf{i} \times m_A \mathbf{v}_A + b\mathbf{i} \times m_B \mathbf{v}_B \\ -7.2\mathbf{k} &= a\mathbf{i} \times 3(2.56\mathbf{j}) + 7.5\mathbf{i} \times 1.5(3.6\mathbf{i} - 2.24\mathbf{j}) \\ -7.2\mathbf{k} &= 7.68a\mathbf{k} - 25.2\mathbf{k} \end{aligned}$$

$$a = 2.34 \text{ m} \blacktriangleleft$$



PROBLEM 14.52



Two small disks A and B , of mass 2 kg and 1 kg , respectively, may slide on a horizontal and frictionless surface. They are connected by a cord of negligible mass and spin about their mass center G . At $t = 0$, G is moving with the velocity $\bar{\mathbf{v}}_0$ and its coordinates are $\bar{x}_0 = 0$, $\bar{y}_0 = 1.89\text{ m}$. Shortly thereafter, the cord breaks and disk A is observed to move with a velocity $\mathbf{v}_A = (5\text{ m/s})\mathbf{j}$ in a straight line and at a distance $a = 2.56\text{ m}$ from the y axis, while B moves with a velocity $\mathbf{v}_B = (7.2\text{ m/s})\mathbf{i} - (4.6\text{ m/s})\mathbf{j}$ along a path intersecting the x axis at a distance $b = 7.48\text{ m}$ from the origin O . Determine (a) the initial velocity $\bar{\mathbf{v}}_0$ of the mass center G of the two disks, (b) the length of the cord initially connecting the two disks, (c) the rate in rad/s at which the disks were spinning about G .

SOLUTION

Initial conditions.

Location of G :

$$\frac{AG}{m_B} = \frac{BG}{m_A} = \frac{AG + GB}{m_B + m_A} = \frac{l}{m}$$

$$AG = \frac{m_B}{m} l = \frac{1}{3} l$$

$$BG = \frac{m_A}{m} l = \frac{2}{3} l$$

Linear momentum:

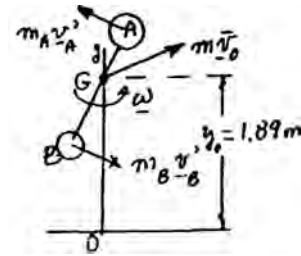
$$L_0 = m\bar{\mathbf{v}}_0 = 3\bar{\mathbf{v}}_0$$

Angular momentum about G :

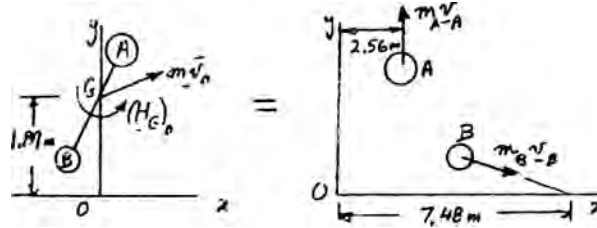
$$\begin{aligned} (H_G)_0 &= \bar{\mathbf{GA}} \times m_A \mathbf{v}'_A + \bar{\mathbf{GB}} \times m_B \mathbf{v}'_B \\ &= \left(\frac{1}{3}l\right)(2\text{ kg})\left(\frac{1}{3}lw\right)\mathbf{k} + \left(\frac{2}{3}l\right)(1\text{ kg})\left(\frac{2}{3}lw\right)\mathbf{k} \\ &= \frac{2}{3}l^2 w \mathbf{k}^2 \end{aligned}$$

Kinetic energy: Using Eq. (14.29),

$$\begin{aligned} T_0 &= \frac{1}{2} m \bar{v}_0^2 + \frac{1}{2} \sum m_i v_i'^2 = \frac{1}{2} m \bar{v}_0^2 + \frac{1}{2} m_A v_A'^2 + \frac{1}{2} m_B v_B'^2 \\ &= \frac{1}{2} (3) \bar{v}_0^2 + \frac{1}{2} (2) \left(\frac{1}{3} lw\right)^2 + \frac{1}{2} (1) \left(\frac{2}{3} lw\right)^2 \\ T_0 &= \frac{3}{2} \bar{v}_0^2 + \frac{1}{3} l^2 w^2 \end{aligned}$$



PROBLEM 14.52 (Continued)



Conservation of linear momentum:

$$m\bar{\mathbf{v}}_0 = m_A\mathbf{v}_A + m_B\mathbf{v}_B$$

$$3\bar{\mathbf{v}}_0 = (2)(5\mathbf{j}) + (1)(7.2\mathbf{i} - 4.6\mathbf{j}) = 7.2\mathbf{i} + 5.4\mathbf{j}$$

(a) $\bar{\mathbf{v}}_0 = (2.4 \text{ m/s})\mathbf{i} + (1.8 \text{ m/s})\mathbf{j} \blacktriangleleft$

Conservation of angular momentum about O:

$$\curvearrowright (1.89\mathbf{j}) \times m\bar{\mathbf{v}}_0 + (\mathbf{H}_G)_0 = (2.56\mathbf{i}) \times m_A\mathbf{v}_A + (7.48\mathbf{i}) \times m_B\mathbf{v}_B$$

Substituting for $\bar{\mathbf{v}}_0$, $(\mathbf{H}_G)_0$, $\mathbf{v}_A$, $\mathbf{v}_B$ and masses:

$$(1.89\mathbf{j}) \times 3(2.4\mathbf{i} + 1.8\mathbf{j}) + \frac{2}{3}l^2w\mathbf{k} = (2.56\mathbf{i}) \times 2(5\mathbf{j}) + (7.48\mathbf{i}) \times (7.2\mathbf{i} - 4.6\mathbf{j})$$

$$-13.608\mathbf{k} + \frac{2}{3}l^2w\mathbf{k} = 25.6\mathbf{k} - 34.408\mathbf{k}$$

$$\frac{2}{3}l^2w = 4.80 \quad l^2w = 7.20 \tag{1}$$

Conservation of energy:

$$T_0 = T: \quad \frac{3}{2}\bar{v}_0^2 + \frac{1}{3}l^2w^2 = \frac{1}{2}m_Av_A^2 + \frac{1}{2}m_Bv_B^2$$

$$\frac{3}{2}[(2.4)^2 + (1.8)^2] + \frac{1}{3}l^2w^2 = \frac{1}{2}(2)(5)^2 + \frac{1}{2}(1)[(7.2)^2 + (4.6)^2]$$

$$13.5 + \frac{1}{3}l^2w^2 = 25 + 36.5 \quad l^2w^2 = 144.0 \tag{2}$$

Dividing Eq. (2) by Eq. (1), member by member:

$$w = \frac{144.0}{7.20} = 20.0 \text{ rad/s}$$

(c) Original rate of spin = 20.0 rad/s $\blacktriangleleft$

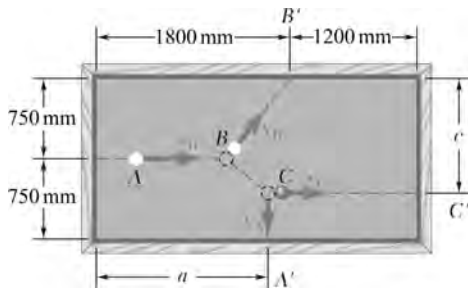
Substituting for w into Eq. (1):

$$l^2(20.0) = 7.20 \quad l^2 = 0.360 \quad l = 0.600 \text{ m}$$

(b) Length of cord = 600 mm $\blacktriangleleft$

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PROBLEM 14.53



In a game of billiards, ball A is given an initial velocity $\mathbf{v}_0$ along the longitudinal axis of the table. It hits ball B and then ball C , which are both at rest. Balls A and C are observed to hit the sides of the table squarely at A' and C' , respectively, and ball B is observed to hit the side obliquely at B' . Knowing that $v_0 = 4 \text{ m/s}$, $v_A = 2 \text{ m/s}$, and $a = 1600 \text{ mm}$, determine (a) the velocities $\mathbf{v}_B$ and $\mathbf{v}_C$ of balls B and C , (b) the Point C' where ball C hits the side of the table. Assume frictionless surfaces and perfectly elastic impacts (i.e., conservation of energy).

SOLUTION

Conservation of linear momentum.

$$mv_0 \mathbf{i} = -mv_A \mathbf{j} + m(v_B)_x \mathbf{i} + m(v_B)_y \mathbf{j} + mv_C \mathbf{i}$$

Equating coefficients of unit vectors:

$$\mathbf{i}: \quad mv_0 = m(v_B)_x + mv_C \quad (v_B)_x + v_C = v_0 = 4 \text{ m/s} \quad (1)$$

$$\mathbf{j}: \quad 0 = -mv_A + (v_B)_y \quad (v_B)_y = v_A = 2 \text{ m/s} \quad (2)$$

Conservation of angular momentum about corner D .

$$\curvearrowright (750 \text{ mm})v_0 = -(1600 \text{ mm})v_A + (1800 \text{ mm})(v_B)_y + cv_C$$

$$(750)(4) = (1600)(2) + (1800)(2) + cv_C$$

$$cv_C = 2600 \quad (3)$$

Conservation of energy.
$$\frac{1}{2}mv_0^2 = \frac{1}{2}mv_A^2 + \frac{1}{2}m[(v_B)_x^2 + (v_B)_y^2] + \frac{1}{2}mv_C^2$$

Dividing by m , multiplying by 2, and substituting for v_0 ; v_A , $(v_B)_y$, their values and $(v_B)_x = 4 - v_C$ from Eq. (1):

$$(4)^2 = (2)^2 + (4 - v_C)^2 + (2)^2 + v_C^2$$

$$v_C^2 - 4v_C + 4 = 0, \quad v_C = 2 \text{ m/s}$$

With $v_C = 2$, Eq. (3) yields $C = 1300$

(a) Therefore:

$$\mathbf{v}_C = 2 \text{ m/s} \rightarrow \blacktriangleleft$$

From Eq. (1): $(v_B)_x = 4 - 2 = 2 \text{ m/s}$

$$\mathbf{v}_B = (2 \text{ m/s})\mathbf{i} + (2 \text{ m/s})\mathbf{j}$$

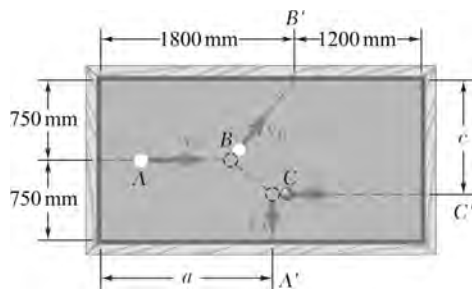
or

$$\mathbf{v}_B = 2.328 \text{ m/s} \nearrow 45^\circ \blacktriangleleft$$

(b) From Eq. (3): $C(7.68) = 322.56$

$$C = 1300 \text{ mm} \blacktriangleleft$$

PROBLEM 14.54



For the game of billiards of Problem 14.53, it is now assumed that $v_0 = 5 \text{ m/s}$, $v_C = 3 \text{ m/s}$, and $c = 1200 \text{ mm}$. Determine (a) the velocities $\mathbf{v}_A$ and $\mathbf{v}_B$ of balls A and B, (b) the point A' where ball A hits the side of the table.

PROBLEM 14.53 In a game of billiards, ball A is given an initial velocity $\mathbf{v}_0$ along the longitudinal axis of the table. It hits ball B and then ball C, which are both at rest. Balls A and C are observed to hit the sides of the table squarely at A' and C', respectively, and ball B is observed to hit the side obliquely at B'. Knowing that $v_0 = 4 \text{ m/s}$, $v_A = 2 \text{ m/s}$, and $a = 160 \text{ mm}$, determine (a) the velocities $\mathbf{v}_B$ and $\mathbf{v}_C$ of balls B and C, (b) the Point C' where ball C hits the side of the table. Assume frictionless surfaces and perfectly elastic impacts (i.e., conservation of energy).

SOLUTION

Conservation of linear momentum.

$$mv_0 \mathbf{i} = -mv_A \mathbf{j} + m(v_B)_x \mathbf{i} + m(v_B)_y \mathbf{j} + mv_C \mathbf{i}$$

$$\mathbf{i}: \quad mv_0 = m(v_B)_x + mv_C \quad (v_B)_x = 5 - 3 = 2 \text{ m/s} \quad (1)$$

$$\mathbf{j}: \quad 0 = -mv_A + (v_B)_y \quad (v_B)_y = v_A \quad (2)$$

Conservation of angular momentum about corner D.

$$\curvearrowright (750 \text{ mm})v_0 = -av_A + (1800 \text{ mm})(v_B)_y + cv_C$$

Substituting given data and using Eq. (2):

$$\begin{aligned} 750(5 \text{ m/s}) &= -av_A + 1800v_A + 1200(3) \\ 150 &= (1800 - a)v_A \end{aligned} \quad (3)$$

Conservation of energy.

$$\frac{1}{2}mv_0^2 = \frac{1}{2}mv_A^2 + \frac{1}{2}m[(v_B)_x^2 + (v_B)_y^2] + \frac{1}{2}mv_C^2$$

Dividing by m , multiplying by 2, and substituting

$$(5)^2 = v_A^2 + (2)^2 + v_A^2 + (3)^2$$

$$(a) \quad v_A^2 = 6 \quad v_A = 2.4495 \text{ m/s} \quad \mathbf{v}_A = 2.45 \text{ m/s} \downarrow \blacktriangleleft$$

$$\text{From Eqs. (1) and (2): } \mathbf{v}_B = (v_B)_x \mathbf{i} + (v_B)_y \mathbf{j} = (2 \text{ m/s})\mathbf{i} + (2.45 \text{ m/s})\mathbf{j}$$

$$\text{or} \quad \mathbf{v}_B = 3.16 \text{ m/s} \nearrow 50.8^\circ \blacktriangleleft$$

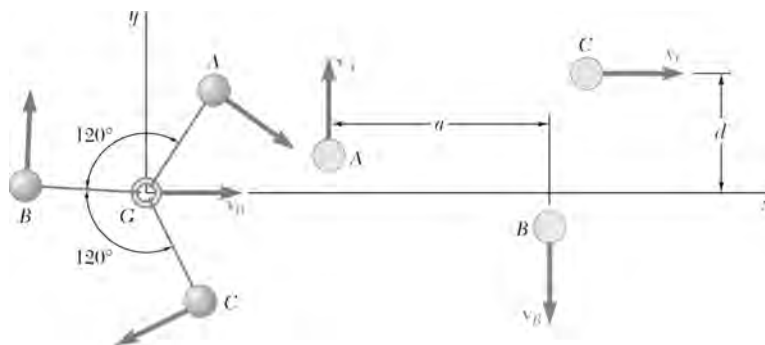
$$(b) \quad \text{From Eq. (3):} \quad (1800 - a)(2.4495) = 150$$

$$a = 1738.7 \text{ mm} \quad a = 1739 \text{ mm} \blacktriangleleft$$

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PROBLEM 14.55

Three small identical spheres A , B , and C , which can slide on a horizontal, frictionless surface, are attached to three 200-mm-long strings, which are tied to a ring G . Initially, the spheres rotate clockwise about the ring with a relative velocity of 0.8 m/s and the ring moves along the x -axis with a velocity $\mathbf{v}_0 = (0.4 \text{ m/s})\mathbf{i}$. Suddenly, the ring breaks and the three spheres move freely in the xy plane with A and B following paths parallel to the y -axis at a distance $a = 346 \text{ mm}$ from each other and C following a path parallel to the x axis. Determine (a) the velocity of each sphere, (b) the distance d .



SOLUTION

Conservation of linear momentum.

Before break: $\mathbf{L}_0 = (3m)\bar{\mathbf{v}} = 3m(0.4\mathbf{i}) = m(1.2 \text{ m/s})\mathbf{i}$

After break: $\mathbf{L} = mv_A\mathbf{j} - mv_B\mathbf{j} + mv_C\mathbf{i}$

$\mathbf{L} = \mathbf{L}_0$: $mv_C\mathbf{i} + m(v_A - v_B)\mathbf{j} = m(1.2 \text{ m/s})\mathbf{i}$

Therefore, $v_A = v_B$ (1)

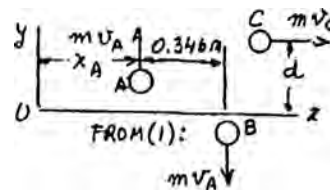
$v_C = 1.200 \text{ m/s} \quad v_C = 1.200 \text{ m/s} \rightarrow$ (2)

Conservation of angular momentum.

Before break: $\begin{aligned} \curvearrowright (H_O)_0 &= 3mlv' = 3m(0.2\text{ m})(0.8 \text{ m/s}) \\ &= 0.480 \text{ m} \end{aligned}$

After break: $\begin{aligned} \curvearrowright H_O &= m\cancel{v_A} x_A \\ &\quad + mv_A(\cancel{x_A} + 0.346) \\ &\quad + mv_C d \end{aligned}$

$H_O = (H_O)_0$: $0.346mv_A + mv_C d = 0.480 \text{ m}$



PROBLEM 14.55 (Continued)

Recalling Eq. (2):

$$\begin{aligned} 0.346v_A + 1.200d &= 0.480 \\ d &= 0.400 - 0.28833v_A \end{aligned} \quad (3)$$

Conservation of energy.

Before break:

$$\begin{aligned} T_0 &= \frac{1}{2}(3m)\bar{v}^2 + 3\left(\frac{1}{2}mv'^2\right) \\ &= \frac{3}{2}m(v_0^2 + v'^2) = \frac{3}{2}[(0.4)^2 + (0.8)^2]m = 1.200m \end{aligned}$$

After break:

$$T = \frac{1}{2}mv_A^2 + \frac{1}{2}mv_B^2 + \frac{1}{2}mv_C^2$$

$T = T_0$: Substituting for v_B from Eq. (1) and v_C from Eq. (2),

$$\begin{aligned} \frac{1}{2}[v_A^2 + v_A^2 + (1.200)^2] &= 1.200 \\ v_A^2 &= 0.480 \end{aligned}$$

$$v_A = v_B = 0.69282 \text{ m/s}$$

(a) Velocities:

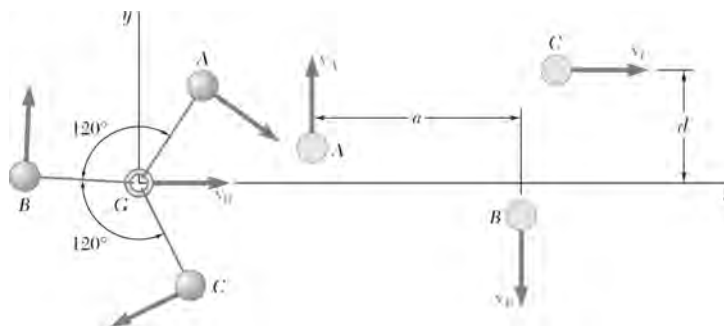
$$\mathbf{v}_A = 0.693 \text{ m/s } \uparrow; \quad \mathbf{v}_B = 0.693 \text{ m/s } \downarrow; \quad \mathbf{v}_C = 1.200 \text{ m/s } \rightarrow \quad \blacktriangleleft$$

(b) Distance d :

$$\text{From Eq. (3):} \quad d = 0.400 - 0.28833(0.69282) = 0.20024 \text{ m} \quad d = 200 \text{ mm } \blacktriangleleft$$

PROBLEM 14.56

Three small identical spheres A , B , and C , which can slide on a horizontal, frictionless surface, are attached to three strings of length l which are tied to a ring G . Initially, the spheres rotate clockwise about the ring which moves along the x axis with a velocity $\mathbf{v}_0$. Suddenly, the ring breaks and the three spheres move freely in the xy plane. Knowing that $\mathbf{v}_A = (1.039 \text{ m/s})\mathbf{j}$, $\mathbf{v}_C = (1.800 \text{ m/s})\mathbf{i}$, $a = 416 \text{ mm}$, and $d = 240 \text{ mm}$, determine (a) the initial velocity of the ring, (b) the length l of the strings, (c) the rate in rad/s at which the spheres were rotating about G .



SOLUTION

Conservation of linear momentum.

$$(3m)\bar{\mathbf{v}} = m\mathbf{v}_A + m\mathbf{v}_B + m\mathbf{v}_C$$

$$3m v_0 \mathbf{i} = m(1.039 \text{ m/s})\mathbf{j} - m v_B \mathbf{j} + m(1.800 \text{ m/s})\mathbf{i}$$

Equating coefficients of unit vectors:

$$\mathbf{i}: \quad 3v_0 = 1.800 \text{ m/s}$$

$$\mathbf{j}: \quad 0 = 1.039 \text{ m/s} - v_B \quad v_B = 1.039 \text{ m/s} \quad (1)$$

$$(a) \quad v_0 = 0.600 \text{ m/s} \rightarrow$$

Conservation of angular momentum.

$$\text{Before break:} \quad \curvearrowright (H_O)_0 = 3ml^2\dot{\theta}$$

$$\text{After break:} \quad \curvearrowright H_O = -m v_A x_A$$

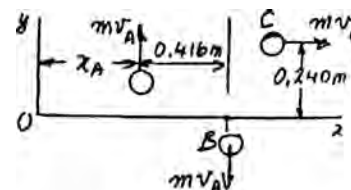
$$+m v_A (x_A + 0.416)$$

$$+m v_C (0.240)$$

$$= m(1.039)(0.416) + m(1.800)(0.240)$$

$$= m(0.864224)$$

$$(H_O)_0 = H_O: \quad 3ml^2\dot{\theta} = m(0.864224) \quad l^2\dot{\theta} = 0.28807 \quad (2)$$



PROBLEM 14.56 (Continued)

Conservation of energy.

Before break:

$$\begin{aligned} T_0 &= \frac{1}{2}(3m)\bar{v}^2 + 3\left(\frac{1}{2}mv'^2\right) = \frac{3}{2}mv_0^2 + \frac{3}{2}m(l\dot{\theta})^2 \\ &= \frac{3}{2}m(0.600)^2 + \frac{3}{2}ml^2\dot{\theta}^2 \end{aligned}$$

After break:

$$\begin{aligned} T &= \frac{1}{2}mv_A^2 + \frac{1}{2}mv_B^2 + \frac{1}{2}mv_C^2 \\ &= \frac{1}{2}m[(1.039)^2 + (1.039)^2 + (1.800)^2] = \frac{1}{2}m(5.399) \end{aligned}$$

$$T = T_0: \quad \frac{1}{2}m(5.399) = \frac{3}{2}m(0.600)^2 + \frac{3}{2}ml^2\dot{\theta}^2$$

$$l^2\dot{\theta}^2 = 1.4397 \quad (3)$$

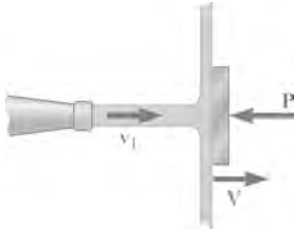
$$\text{Dividing Eq. (3) by Eq. (2):} \quad \ddot{\theta} = \frac{1.4397}{0.28807} = 4.9976$$

$$(b) \quad \text{From Eq. (2):} \quad l^2 = \frac{0.28807}{4.9976} \quad l = 0.2401 \text{ m}$$

$$l = 240 \text{ mm} \quad \blacktriangleleft$$

(c) Rate of rotation:

$$\dot{\theta} = 5.00 \text{ rad/s} \quad \curvearrowright \quad \blacktriangleleft$$



PROBLEM 14.57

A stream of water of cross-sectional area A and velocity v_1 strikes a plate which is held motionless by a force $\mathbf{P}$. Determine the magnitude of $\mathbf{P}$, knowing that $A = 500 \text{ mm}^2$, $v_1 = 25 \text{ m/s}$, and $V = 0$.

SOLUTION

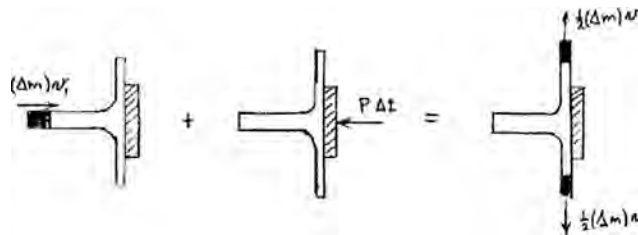
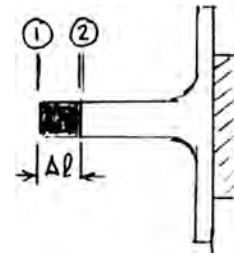
Mass flow rate. As the fluid moves from Section 1 to Section 2 in time Δt , the mass Δm moved is

$$\Delta m = \rho A(\Delta l)$$

Then
$$\frac{dm}{dt} = \frac{\Delta m}{\Delta t} = \frac{\rho A(\Delta l)}{\Delta t} = \rho A v_1$$

Data: $\rho = 1000 \text{ kg/m}^3$, $A = 500 \text{ mm}^2 = 500 \times 10^{-6} \text{ m}^2$, $v_1 = 25 \text{ m/s}$

$$\frac{dm}{dt} = (1000)(500 \times 10^{-6})(25) = 12.5 \text{ kg/s}$$



Principle of impulse and momentum.

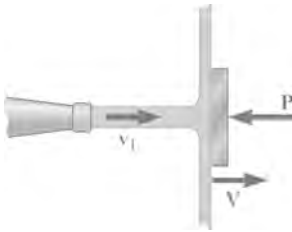
$$\rightarrow : (\Delta m)v_1 - P\Delta t = 0$$

$$P = \frac{\Delta m}{\Delta t} v = \frac{dm}{dt} v$$

$$P = (12.5)(25)$$

$$P = 312 \text{ N} \blacktriangleleft$$

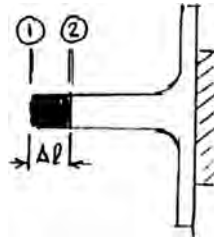




PROBLEM 14.58

A stream of water of cross-sectional area A and velocity v_1 strikes a plate which moves to the right with a velocity V . Determine the magnitude of V , knowing that $A = 600 \text{ mm}^2$, $v_1 = 30 \text{ m/s}$, and $P = 400 \text{ N}$.

SOLUTION



Consider velocities measured with respect to the plate, which is moving with velocity V . The velocity of the stream relative to the plate is

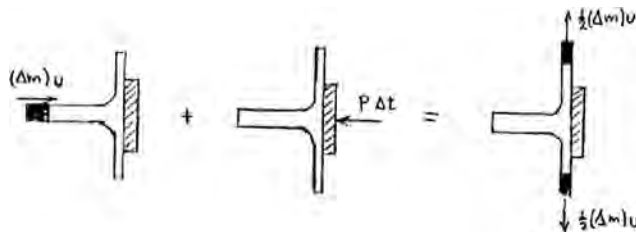
$$u = v_1 - V \quad (1)$$

Mass flow rate. As the fluid moves from Section 1 to Section 2 in time Δt , the mass Δm moved is

$$\Delta m = \rho A(\Delta l)$$

Then

$$\frac{dm}{dt} = \frac{\Delta m}{\Delta t} = \frac{\rho A(\Delta l)}{\Delta t} = \rho A u \quad (2)$$



Principle of impulse and momentum.

$$+ \rightarrow (\Delta m)u - P(\Delta t) = 0$$

$$P = \frac{\Delta m}{\Delta t} u = \frac{dm}{dt} u = \rho A u^2$$

$$u = \sqrt{\frac{P}{\rho A}}$$

PROBLEM 14.58 (Continued)

From Eq. (1),

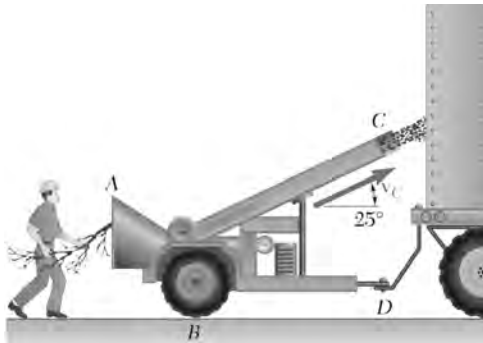
$$V = v_1 - u = v_1 - \sqrt{\frac{P}{\rho A}}$$

Data:

$$P = 400 \text{ N}, \quad A = 600 \text{ mm}^2 = 600 \times 10^{-6} \text{ m}^2$$

$$v_1 = 30 \text{ m/s}, \quad \rho = 1000 \text{ kg/m}^3$$

$$V = 30 - \sqrt{\frac{400}{(1000)(600 \times 10^{-6})}} \qquad V = 4.18 \text{ m/s} \blacktriangleleft$$



PROBLEM 14.59

Tree limbs and branches are being fed at A at the rate of 5 kg/s into a shredder which spews the resulting wood chips at C with a velocity of 20 m/s. Determine the horizontal component of the force exerted by the shredder on the truck hitch at D.

SOLUTION

Eq. (14.38):

$$(\Delta m) \mathbf{v}_A + \Sigma \mathbf{F} \Delta t = (\Delta m) \mathbf{v}_C$$

$$\Sigma \mathbf{F} = \frac{\Delta m}{\Delta t} \mathbf{v}_C = (5 \text{ kg/s})(20 \text{ m/s} \angle 25^\circ)$$

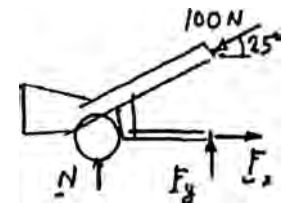
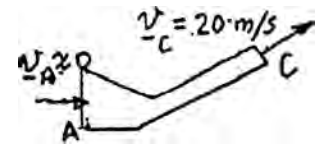
Force exerted on chips = $\Sigma \mathbf{F} = 100 \text{ N} \angle 25^\circ$

Free body: shredder.

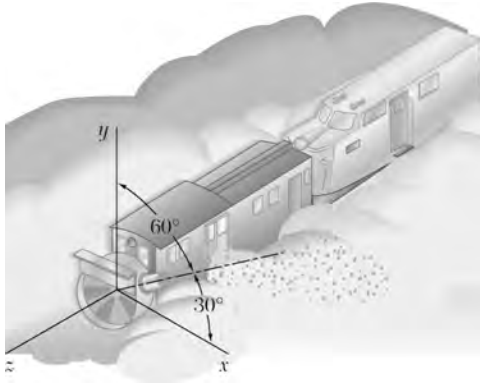
$$\rightarrow \Sigma F_x = 0: F_x - (100 \text{ N}) \cos 25^\circ = 0$$

$$F_x = 90.6 \text{ N} \rightarrow$$

On hitch:



$$F_x = 90.6 \text{ N} \leftarrow$$



PROBLEM 14.60

A rotary power plow is used to remove snow from a level section of railroad track. The plow car is placed ahead of an engine which propels it at a constant speed of 20 km/h. The plow car clears 18×10^4 kg of snow per minute, projecting it in the direction shown with a velocity of 12 m/s relative to the plow car. Neglecting friction, determine (a) the force exerted by the engine on the plow car, (b) the lateral force exerted by the track on the plow car.

SOLUTION

Velocity of the plow: $v_P = 20 \text{ km/h} = \frac{50}{9} \text{ m/s}$

Velocity of thrown snow:

$$\mathbf{v}_s = (12 \text{ m/s})(\cos 30^\circ \mathbf{i} + \sin 30^\circ \mathbf{j}) + \left(\frac{50}{9} \text{ m/s}\right) \mathbf{k}$$

Mass flow rate:

$$\frac{dm}{dt} = \frac{18 \times 10^4 \text{ kg}}{60 \text{ s/min}} = 3000 \text{ kg/s}$$

Let F be the force exerted on the plow and the snow.

Apply impulse-momentum, noting that the snow is initially at rest and that the velocity of the plow is constant. Neglect gravity.

$$\mathbf{F}(\Delta t) = (\Delta m)\mathbf{v}_s$$

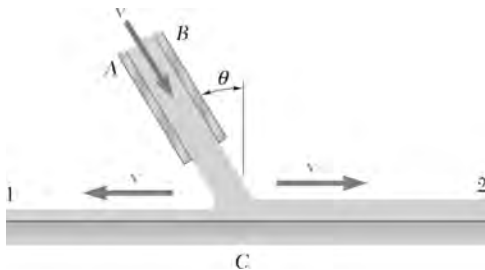
$$\begin{aligned} \mathbf{F} &= \left(\frac{dm}{dt}\right)\mathbf{v}_s = (3000) \left(12 \cos 30^\circ \mathbf{i} + 12 \sin 30^\circ \mathbf{j} + \frac{50}{9} \mathbf{k}\right) \\ &= (31177 \text{ N})\mathbf{i} + (18000 \text{ N})\mathbf{j} + (16666.7 \text{ N})\mathbf{k} \end{aligned}$$

(a) Force exerted by engine.

$$F_z = 16.67 \text{ kN} \quad \blacktriangleleft$$

(b) Lateral force exerted by track.

$$F_x = 31.2 \text{ kN} \quad \blacktriangleleft$$



PROBLEM 14.61

Water flows in a continuous sheet from between two plates A and B with a velocity $\mathbf{v}$ of magnitude 30 m/s. The stream is split into two parts by a smooth horizontal plate C. Knowing that the rates of flow in each of the two resulting streams are, respectively, $Q_1 = 100$ L/min and $Q_2 = 500$ L/min, determine (a) the angle θ , (b) the total force exerted by the stream on the horizontal plate.

SOLUTION

For steady flow

$$Q_1 + Q_2 = Q \quad (1)$$

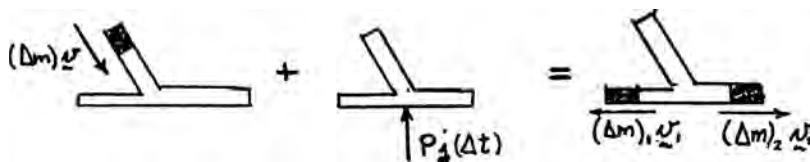
Assume that the fluid speed is constant.

Velocity vectors:

$$\mathbf{v} = v(\sin \theta \mathbf{i} - \cos \theta \mathbf{j}), \quad \mathbf{v}_1 = -v\mathbf{i}, \quad \mathbf{v}_2 = v\mathbf{i}$$

Let $P\mathbf{j}$ be the force that plate C exerts on the fluid.

Impulse-momentum principle:



$$(\Delta m)\mathbf{v} + P\mathbf{j}(\Delta t) = (\Delta m)_1\mathbf{v}_1 + (\Delta m)_2\mathbf{v}_2$$

$$\begin{aligned} P\mathbf{j} &= \frac{(\Delta m)_1}{\Delta t} \mathbf{v}_1 + \frac{(\Delta m)_2}{\Delta t} \mathbf{v}_2 - \frac{\Delta m}{\Delta t} \mathbf{v} \\ &= \left(\frac{dm}{dt} \right)_1 (-v\mathbf{i}) + \left(\frac{dm}{dt} \right)_2 (v\mathbf{i}) - \left(\frac{dm}{dt} \right) (v \sin \theta \mathbf{i} - v \cos \theta \mathbf{j}) \\ &= -\rho Q_1 v \mathbf{i} + \rho Q_2 v \mathbf{i} - \rho Q v (\sin \theta \mathbf{i} - \cos \theta \mathbf{j}) \end{aligned}$$

Resolve into components.

$$\mathbf{i}: \quad 0 = -\rho Q_1 v + \rho Q_2 v - \rho Q v \sin \theta \quad Q \sin \theta = Q_2 - Q_1 \quad (2)$$

$$\mathbf{j}: \quad P = \rho Q v \cos \theta \quad (3)$$

Data:

$$\rho = 1000 \text{ kg/m}^3, \quad v = 30 \text{ m/s}$$

$$Q_1 = 100 \text{ L/min} = 1.6667 \times 10^{-3} \text{ m}^3/\text{s}$$

$$Q_2 = 500 \text{ L/min} = 8.3333 \times 10^{-3} \text{ m}^3/\text{s}$$

From Eq. (1),

$$Q = 600 \text{ L/min} = 10 \times 10^{-3} \text{ m}^3/\text{s}$$

PROBLEM 14.61 (Continued)

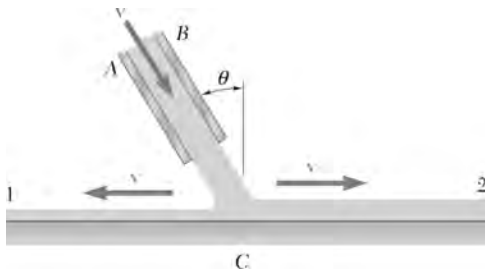
(a) From Eq. (2),
$$\sin \theta = \frac{Q_2 - Q_1}{Q} = 0.66667$$
$$\theta = 41.810^\circ$$

$$\theta = 41.8^\circ \blacktriangleleft$$

From Eq. (3),
$$P = (1000)(10 \times 10^{-3})(30) \cos 41.810^\circ$$
$$= 224 \text{ N}$$

(b) Force that stream exerts on plate C:

$$-P \mathbf{j} = 224 \text{ N} \downarrow \blacktriangleleft$$



PROBLEM 14.62

Water flows in a continuous sheet from between two plates A and B with a velocity $\mathbf{v}$ of magnitude 40 m/s. The stream is split into two parts by a smooth horizontal plate C. Determine the rates of flow Q_1 and Q_2 in each of the two resulting streams, knowing that $\theta = 30^\circ$ and that the total force exerted by the stream on the horizontal plate is a 500-N vertical force.

SOLUTION

For steady flow

$$Q_1 + Q_2 = Q \quad (1)$$

Assume that the velocity is constant.

Velocity vectors:

$$\mathbf{v} = v(\sin \theta \mathbf{i} - \cos \theta \mathbf{j}), \quad \mathbf{v}_1 = -v\mathbf{i}, \quad \mathbf{v}_2 = v\mathbf{i}$$

Let $P\mathbf{j}$ be the force that plate C exerts on the fluid.

Impulse-momentum principle:



$$(\Delta m)\mathbf{v} + P\mathbf{j}(\Delta t) = (\Delta m)_1\mathbf{v}_1 + (\Delta m)_2\mathbf{v}_2$$

$$\begin{aligned} P\mathbf{j} &= \frac{(\Delta m)_1}{\Delta t}\mathbf{v}_1 + \frac{(\Delta m)_2}{\Delta t}\mathbf{v}_2 - \frac{\Delta m}{\Delta t}\mathbf{v} \\ &= \left(\frac{dm}{dt}\right)_1(-v\mathbf{i}) + \left(\frac{dm}{dt}\right)_2(v\mathbf{i}) - \left(\frac{dm}{dt}\right)(v\sin \theta \mathbf{i} - v\cos \theta \mathbf{j}) \\ &= -\rho Q_1 v\mathbf{i} + \rho Q_2 v\mathbf{i} - \rho Q v(\sin \theta \mathbf{i} - \cos \theta \mathbf{j}) \end{aligned}$$

Resolve into components.

$$\mathbf{i}: 0 = -\rho Q_1 v + \rho Q_2 v - \rho Q v \sin \theta \quad Q \sin \theta = Q_2 - Q_1 \quad (2)$$

$$\mathbf{j}: P = \rho Q v \cos \theta \quad (3)$$

Data:

$$\rho = 1000 \text{ kg/m}^3, \quad v = 40 \text{ m/s}$$

From Eq. (3),

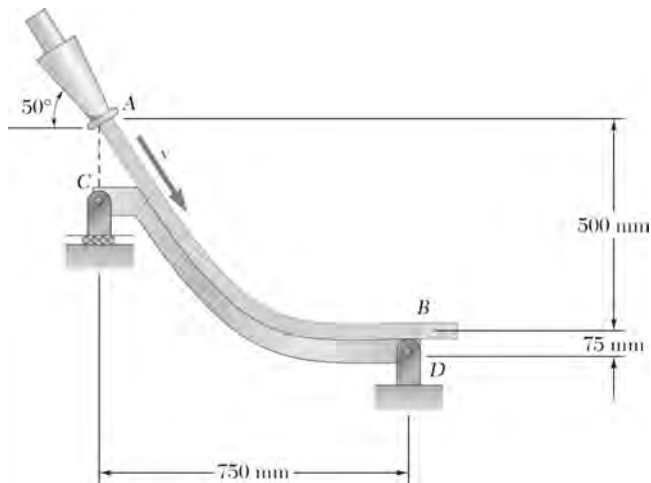
$$\begin{aligned} Q &= \frac{P}{\rho v \cos \theta} \\ &= \frac{500}{(1000)(40) \cos 30^\circ} \\ &= 14.4334 \times 10^{-3} \text{ m}^3/\text{s} \\ &= 866.03 \text{ L/min} \end{aligned}$$

PROBLEM 14.62 (Continued)

Solving Eqs. (1) and (2) simultaneously for Q_1 and Q_2 ,

$$\begin{aligned} Q_1 &= \frac{1}{2} Q(1 - \sin \theta) \\ &= \frac{1}{2} (866.03)(1 - \sin 30^\circ) \end{aligned} \quad Q_1 = 217 \text{ L/min} \blacktriangleleft$$

$$\begin{aligned} Q_2 &= \frac{1}{2} Q(1 + \sin \theta) \\ &= \frac{1}{2} (866.03)(1 + \sin 30^\circ), \end{aligned} \quad Q_2 = 650 \text{ L/min} \blacktriangleleft$$



PROBLEM 14.63

The nozzle discharges water at the rate of $1.3 \text{ m}^3/\text{min}$. Knowing the velocity of the water at both A and B has a magnitude of 20 m/s and neglecting the weight of the vane, determine the components of the reactions at C and D.

SOLUTION

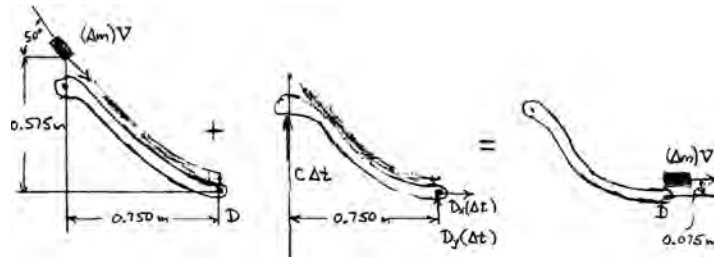
Volumetric flow rate: $Q = 1.3 \text{ m}^3/\text{min} = 21.667 \times 10^{-3} \text{ m}^3/\text{s}$

Mass density of water: $\rho = 1000 \text{ kg/m}^3$

Mass flow rate: $\frac{dm}{dt} = \rho Q = 21.667 \text{ kg/s}$

Assume that the flow speed remains constant.

Principle of impulse and momentum.



$\curvearrowright$ Moments about D:

$$[(\Delta m)V \cos 50^\circ](0.575) - [(\Delta m)V \sin 50^\circ](0.750) + C(\Delta t)(0.750) = (\Delta m)V(0.075)$$

$$C = \frac{(0.750 \sin 50^\circ - 0.575 \cos 50^\circ + 0.075) \frac{\Delta m}{\Delta t} V}{0.750}$$

$$= 0.37324 \frac{dm}{dt} V$$

$$= (0.37324)(21.667 \text{ kg/s})(20 \text{ m/s})$$

$$C = 161.7 \text{ N} \uparrow \blacktriangleleft$$

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PROBLEM 14.63 (Continued)

$\rightarrow$ Horizontal components:

$$(\Delta m)V \cos 50^\circ + D_x(\Delta t) = (\Delta m)V$$

$$D_x = (1 - \cos 50^\circ) \frac{\Delta m}{\Delta t} V$$

$$= 0.35721 \frac{dm}{dt} V$$

$$= (0.35721)(21.667 \text{ kg/s})(20 \text{ m/s})$$

$$\mathbf{D}_x = 154.8 \text{ N} \rightarrow \blacktriangleleft$$

$\uparrow$ Vertical components:

$$-(\Delta m)V \sin 50^\circ + C(\Delta t) + D_y(\Delta t) = 0$$

$$D_y = (\sin 50^\circ) \frac{\Delta m}{\Delta t} V - C$$

$$= 0.76604 \frac{dm}{dt} V - C$$

$$= (0.76604)(21.667 \text{ kg/s})(20 \text{ m/s}) - 161.7 \text{ N}$$

$$\mathbf{D}_y = 170.2 \text{ N} \uparrow \blacktriangleleft$$

PROBLEM 14.64

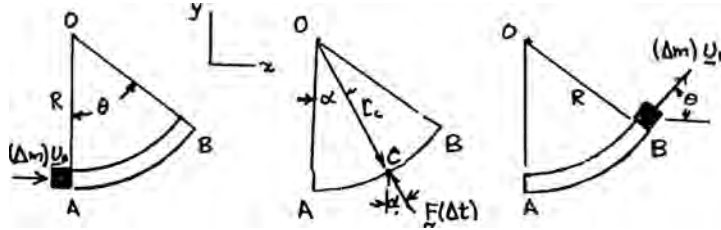
Knowing that the blade AB of Sample Problem 14.7 is in the shape of an arc of circle, show that the resultant force $\mathbf{F}$ exerted by the blade on the stream is applied at the midpoint C of the arc AB . (*Hint: First show that the line of action of $\mathbf{F}$ must pass through the center O of the circle.*)

SOLUTION

Assume

$$u_A = u_B = u$$

Principle of impulse and momentum.



Moments about $O \curvearrowright$:

$$\begin{aligned} R(\Delta m)u_A \mathbf{k} + r_c \times \mathbf{F}(\Delta t) &= R(\Delta m)u_B \mathbf{k} \\ r_c \times \mathbf{F}(\Delta t) &= R(\Delta m)(u_B - u_A) \mathbf{k} = 0 \end{aligned}$$

The line of action of $\mathbf{F}$ passes through Point O .

Components $\rightarrow$:

$$(\Delta m)u_A + F(\Delta t) \sin \alpha = (\Delta m)u_B \cos \theta$$

$$F \sin \alpha = \frac{\Delta m}{\Delta t} u(1 - \cos \theta) \quad (1)$$

Components $\uparrow$:

$$0 + F(\Delta t) \cos \alpha = (\Delta m)u \sin \theta$$

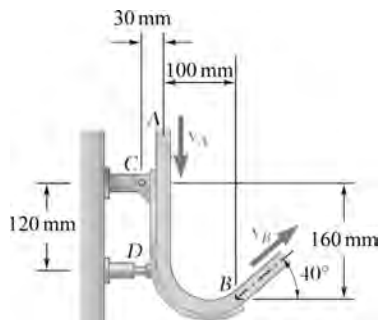
$$F \cos \alpha = \frac{\Delta m}{\Delta t} u \sin \theta \quad (2)$$

Dividing Eq. (2) by Eq. (1),

$$\tan \alpha = \frac{1 - \cos \theta}{\sin \theta} = \frac{2 \sin^2 \frac{\theta}{2}}{2 \sin \frac{\theta}{2} \cos \frac{\theta}{2}} = \tan \frac{\theta}{2}$$

$$\alpha = \frac{\theta}{2}$$

Thus, Point C lies at the midpoint of arc AB .



PROBLEM 14.65

The stream of water shown flows at a rate of 600 L/min and moves with a velocity of magnitude 20 m/s at both A and B. The vane is supported by a pin and bracket at C and by a load cell at D which can exert only a horizontal force. Neglecting the weight of the vane, determine the components of the reactions at C and D.

SOLUTION

Mass flow rate:

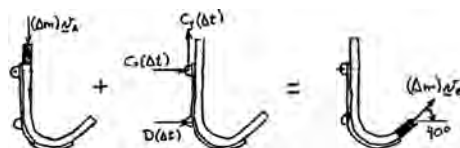
$$\frac{dm}{dt} = \rho Q = \left(1000 \frac{\text{kg}}{\text{m}^3} \right) (600 \times 10^{-3} \text{ m}^3) \left(\frac{1}{60 \text{ s}} \right)$$

$$\frac{dm}{dt} = 10 \text{ kg/s}$$

Velocity vectors:

$$\mathbf{v}_A = 20 \text{ m/s} \downarrow \quad \mathbf{v}_B = 20 \text{ m/s} \nearrow 40^\circ$$

Apply the impulse-momentum principle.



$\curvearrowright$ Moments about C :

$$-30(\Delta m)v_A + 120D(\Delta t) = 160(\Delta m)v_B \cos 40^\circ + 130v_B \sin 40^\circ$$

$$\begin{aligned} D &= \frac{1}{120} \left(\frac{\Delta m}{\Delta t} \right) [160v_B \cos 40^\circ + 130v_B \sin 40^\circ + 30v_A] \\ &= \frac{1}{120} (10) [(160)(20 \cos 40^\circ) + (130)(20 \sin 40^\circ) + (30)(20)] \\ &= 393.549 \text{ N} \end{aligned}$$

$$D_x = 394 \text{ N} \quad \blacktriangleleft$$

$$D_y = 0 \quad \blacktriangleleft$$

$\rightarrow$ x components:

$$C_x(\Delta t) + D(\Delta t) = (\Delta m)v_B \cos 40^\circ$$

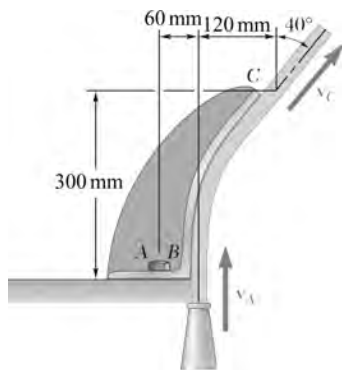
$$C_x = \left(\frac{\Delta m}{\Delta t} \right) v_B \cos 40^\circ - D = (10)(20 \cos 40^\circ) - 393.549 \quad C_x = -240 \text{ N} \quad \blacktriangleleft$$

$\uparrow$ y components:

$$-(\Delta m)v_A + C_y(\Delta t) = (\Delta m)v_B \sin 40^\circ$$

$$C_y = \left(\frac{\Delta m}{\Delta t} \right) v_A + \frac{\Delta m}{\Delta t} v_B \sin 40^\circ = (10)(20 + 20 \sin 40^\circ) \quad C_y = 329 \text{ N} \quad \blacktriangleleft$$

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PROBLEM 14.66

The nozzle shown discharges water at the rate of 800 L/min. Knowing that at both B and C the stream of water moves with a velocity of magnitude 30 m/s, and neglecting the weight of the vane, determine the force-couple system which must be applied at A to hold the vane in place.

SOLUTION

$$Q = \frac{800 \times 10^{-3}}{60}$$

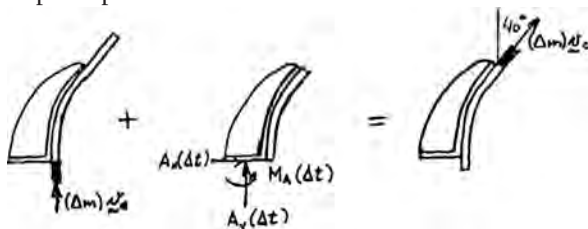
$$= \frac{1}{75} \text{ m}^3/\text{s}$$

$$\frac{dm}{dt} = \rho Q = 1000 \left(\frac{1}{75} \right) = \frac{40}{3} \text{ kg/s}$$

$$\mathbf{v}_B = (30 \text{ m/s})\mathbf{j}$$

$$\mathbf{v}_C = (30 \text{ m/s})(\sin 40^\circ \mathbf{i} + \cos 40^\circ \mathbf{j})$$

Apply the impulse-momentum principle.



$\rightarrow$ x components:

$$0 + A_x(\Delta t) = (\Delta m)(30 \sin 40^\circ)$$

$$A_x = \frac{\Delta m}{\Delta t}(30 \sin 40^\circ)$$

$$= \left(\frac{40}{3} \right)(30 \sin 40^\circ)$$

$$A_x = 257 \text{ N} \rightarrow$$

$\uparrow$ y components:

$$(\Delta m)(30) + A_y(\Delta t) = (\Delta m)(30 \cos 40^\circ)$$

$$A_y = \frac{\Delta m}{\Delta t}(30)(\cos 40^\circ - 1)$$

$$= \left(\frac{40}{3} \right)(30)(\cos 40^\circ - 1)$$

$$= -93.582 \text{ N}$$

$$A_y = 93.6 \text{ N} \downarrow$$

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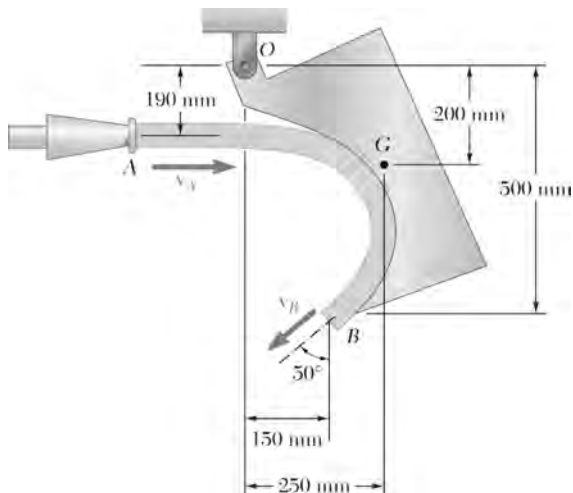
PROBLEM 14.66 (Continued)

$$\begin{aligned} \curvearrowright \text{ Moments about A: } \quad 60(\Delta m)(30) + M_A(\Delta t) &= (180)(\Delta m)(30 \cos 40^\circ) \\ &\quad - (300)(\Delta m)(30 \sin 40^\circ) \end{aligned}$$

$$\begin{aligned} M_A &= \left(\frac{\Delta m}{\Delta t} \right) (5400 \cos 40^\circ - 9000 \sin 40^\circ - 1800) \\ &= \left(\frac{40}{3} \right) (-3448.448) \\ &= -45979 \text{ N} \cdot \text{mm} \\ &= -45.979 \text{ N} \cdot \text{m} \end{aligned}$$

$$\mathbf{M}_A = 46.0 \text{ N} \cdot \text{m} \curvearrowright \blacktriangleleft$$

$$\mathbf{A} = 274 \text{ N} \nearrow 20.0^\circ \blacktriangleleft$$



PROBLEM 14.67

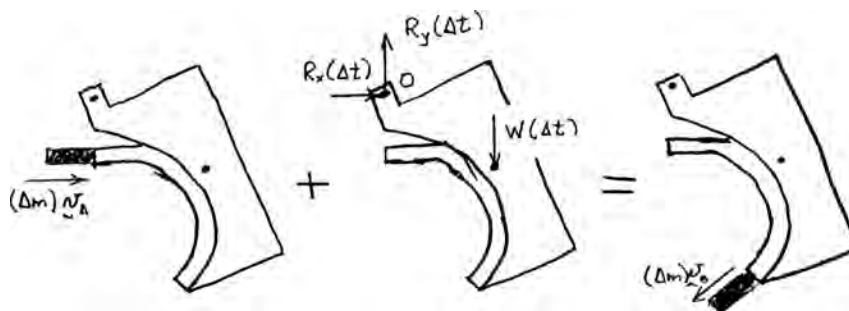
A high speed jet of air issues from the nozzle A with a velocity of $\mathbf{v}_A$ and mass flow rate of 0.36 kg/s. The air impinges on a vane causing it to rotate to the position shown. The vane has a mass of 6 kg. Knowing that the magnitude of the air velocity is equal at A and B, determine (a) the magnitude of the velocity at A, (b) the components of the reactions at O.

SOLUTION

Assume that the speed of the air jet is the same at A and B.

$$v_A = v_B = v$$

Apply the principle of impulse and momentum.



$$(a) \quad \curvearrowright \text{ Moments about } O: (0.190)(\Delta m)v - (0.250)W(\Delta t) = -(0.250)(\Delta m)v \cos 50^\circ - (0.500)(\Delta m)v \sin 50^\circ$$

$$\begin{aligned} v &= \frac{\Delta t}{\Delta m} \cdot \frac{0.250W}{0.150 \cos 50^\circ + 0.500 \sin 50^\circ + 0.190} \\ &= \frac{0.250W}{0.66944} \frac{dm}{dt} \\ &= \frac{(0.250)(6)(9.81)}{(0.66944)(0.36)} \\ &= 61.058 \text{ m/s} \end{aligned}$$

$$v_A = 61.1 \text{ m/s} \rightarrow \blacktriangleleft$$

PROBLEM 14.67 (Continued)

(b) $\rightarrow$ x components: $(\Delta m)v + R_x(\Delta t) = -(\Delta m)v \sin 50^\circ$

$$\begin{aligned} R_x &= \frac{\Delta m}{\Delta t} v(1 + \sin 50^\circ) \\ &= -(0.36)(61.058)(1 + \sin 50^\circ) \\ &= -38.82 \text{ N} \end{aligned}$$

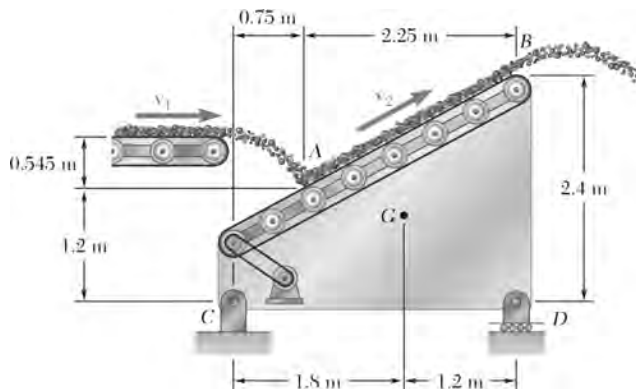
$\uparrow$ y components: $0 + R_y(\Delta t) - W(\Delta t) = -(\Delta m)v \cos 50^\circ$

$$\begin{aligned} R_y &= W + \frac{\Delta m}{\Delta t} v \cos 50^\circ \\ &= (6)(9.81) - (0.36)(61.058) \cos 50^\circ \\ &= 44.73 \text{ N} \end{aligned}$$

$$\begin{aligned} R &= \sqrt{(38.82)^2 + (44.73)^2} \\ &= 59.2 \text{ N} \end{aligned}$$

$$\tan \alpha = \frac{44.73}{38.82} \quad \alpha = 49.0^\circ$$

$$\mathbf{R} = 59.2 \text{ N} \quad \nwarrow 49.0^\circ \quad \blacktriangleleft$$



PROBLEM 14.68

Coal is being discharged from a first conveyor belt at the rate of 120 kg/s. It is received at A by a second belt which discharges it again at B. Knowing that $v_1 = 3$ m/s and $v_2 = 4.25$ m/s and that the second belt assembly and the coal it supports have a total mass of 472 kg, determine the components of the reactions at C and D.

SOLUTION

Velocity before impact at A:

$$(v_A)_x = v_1 = 3 \text{ m/s} \rightarrow$$

$$(v_A)_y^2 = 2g(\Delta y) = (2)(9.8)(0.545) = 10.685 \text{ m}^2/\text{s}^2 \quad (v_A)_y = 3.268 \text{ m/s} \downarrow$$

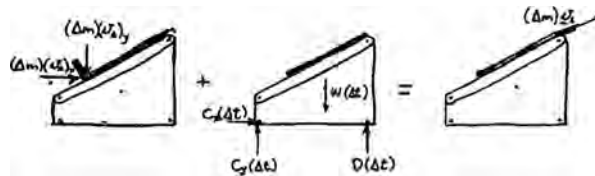
Slope of belt:

$$\tan \theta = \frac{2.4 - 1.2}{2.25}, \quad \theta = 28.07^\circ$$

Velocity of coal leaving at B:

$$v_2 = 4.25 (\cos \theta \mathbf{i} + \sin \theta \mathbf{j})$$

Apply the impulse-momentum principle.



$$\rightarrow x \text{ components:} \quad (\Delta m)(v_A)_x + C_x(\Delta t) = (\Delta m)v_2 \cos \theta$$

$$C_x = \frac{\Delta m}{\Delta t} [v_2 \cos \theta - (v_A)_x] = (120)(4.25 \cos 28.07^\circ - 3)$$

$$C_x = 90.0 \text{ N} \rightarrow$$

$$\curvearrowright \text{ moments about C: } (\Delta m)[-1.2(v_A)_x - 0.75(v_A)_y] + 3.00D(\Delta t) - 1.8W(\Delta t) = (\Delta m)[-2.4 v_2 \cos \theta + 3v_2 \sin \theta]$$

$$\frac{\Delta m}{\Delta t} [-(1.2)(3) - (0.75)(3.268)] + 3D - (1.8)(472)(9.8)$$

$$= \frac{\Delta m}{\Delta t} [-(2.4)(4.25 \cos \theta) + (3)(4.25 \sin \theta)]$$

$$D = 2775 + 1.0168 \frac{dm}{dt} = 2775 + (1.0168)(120) = 2897 \text{ N} \uparrow$$

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PROBLEM 14.68 (Continued)

+↑ y components: $(\Delta m)(-v_A)_y + (C_y + D - W)(\Delta t) = (\Delta m)v_2 \sin \theta$

$$C_y + D - W = \frac{\Delta m}{\Delta t} (3.268 + 4.25 \sin \theta)$$

$$= (120)(5.268) = 632.2 \text{ N}$$

$$C_y = 4625.6 - 2897 + 632.2$$

$$C_y = 2361 \text{ N} \uparrow$$

$$C_x = 90.0 \text{ N}, \quad C_y = 2360 \text{ N} \nwarrow$$

$$D_x = 0, \quad D_y = 2900 \text{ N} \nwarrow$$

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PROBLEM 14.69

While cruising in level flight at a speed of 900 km/h, a jet plane scoops in air at the rate of 90 kg/s and discharges it with a velocity of 660 m/s relative to the airplane. Determine the total drag due to air friction on the airplane.

SOLUTION

Flight speed: $v = 900 \text{ km/h} = 250 \text{ m/s}$

Mass flow rate:

$$\frac{dm}{dt} = 90 \text{ kg/s}$$

$$\Sigma \mathbf{F} = \mathbf{D} = \frac{dm}{dt}(\mathbf{u} - \mathbf{v}) \quad \text{or} \quad D = \frac{dm}{dt}(v - u)$$

where, for a frame of reference moving with the plane, v is the free stream velocity (equal to the air speed) and u is the relative exhaust velocity.

$$D = (90)(660 - 250) = 36,900 \text{ N}$$

$$D = 36.9 \text{ kN} \quad \blacktriangleleft$$

PROBLEM 14.70

The total drag due to air friction on a jet airplane cruising in level flight at a speed of 900 km/h is 40 kN. Knowing that the exhaust velocity is 550 m/s relative to the airplane, determine the rate in kg/s at which the air must pass through the engine.

SOLUTION

Flight speed:

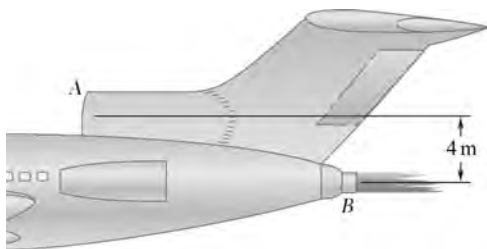
$$v = 900 \text{ km/h} = 250 \text{ m/s}$$

$$\Sigma \mathbf{F} = \mathbf{D} = \frac{dm}{dt}(\mathbf{u} - \mathbf{v}) \quad \text{or} \quad D = \frac{dm}{dt}(u - v)$$

where, for a frame of reference moving with the plane, v is the free stream velocity (equal to the air speed) and u is the relative exhaust velocity.

$$\frac{dm}{dt} = \frac{D}{u - v} = \frac{40000}{550 - 250} = \frac{400}{3} \text{ kg/s} = 133.3 \text{ kg/s}$$

$$\frac{dW}{dt} = 133.3 \text{ kg/s} \quad \blacktriangleleft$$

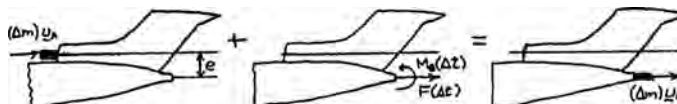


PROBLEM 14.71

The jet engine shown scoops in air at A at a rate of 100 kg/s and discharges it at B with a velocity of 600 m/s relative to the airplane. Determine the magnitude and line of action of the propulsive thrust developed by the engine when the speed of the airplane is (a) 500 km/h, (b) 1000 km/h.

SOLUTION

Use a frame of reference moving with the plane. Apply the impulse-momentum principle. Let F be the force that the plane exerts on the air.



$\rightarrow$ x components:

$$(\Delta m)u_A + F(\Delta t) = (\Delta m)u_B$$

$$F = \frac{\Delta m}{\Delta t}(u_B - u_A) = \frac{dm}{dt}(u_B - u_A) \quad (1)$$

$\curvearrowright$ moments about B:

$$-e(\Delta m)u_A + M_B(\Delta t) = 0$$

$$M_B = e \frac{dm}{dt} u_A \quad (2)$$

Let d be the distance that the line of action is below B.

$$Fd = M_B \quad d = \frac{M_B}{F} = \frac{eu_A}{u_B - u_A} \quad (3)$$

Data:

$$\frac{dm}{dt} = 100 \text{ kg/s}, \quad u_B = 600 \text{ m/s}, \quad e = 4 \text{ m}$$

(a)

$$u_A = 500 \text{ km/h} = 1250/9 \text{ m/s}$$

From Eq. (1),

$$F = 100 \left(600 - \frac{1250}{9} \right) = 46111.1 \text{ N} \quad F = 46100 \text{ N} \quad \blacktriangleleft$$

From Eq. (3),

$$d = \frac{(4) \left(\frac{1250}{9} \right)}{\left(600 - \frac{1250}{9} \right)} = 1.2048 \text{ m} \quad d = 1.205 \text{ m} \quad \blacktriangleleft$$

(b)

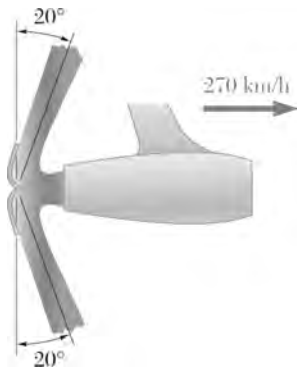
$$u_A = 1000 \text{ km/h} = \frac{2500}{9} \text{ m/s}$$

From Eq. (1),

$$F = 100 \left(600 - \frac{2500}{9} \right) \quad F = 32200 \text{ N} \quad \blacktriangleleft$$

From Eq. (3),

$$d = \frac{(4) \left(\frac{2500}{9} \right)}{\left(600 - \frac{2500}{9} \right)} \quad d = 3.45 \text{ m} \quad \blacktriangleleft$$



PROBLEM 14.72

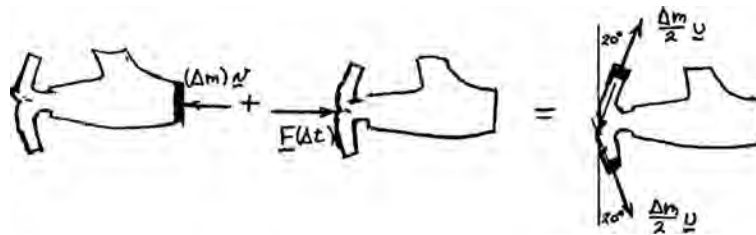
In order to shorten the distance required for landing, a jet airplane is equipped with movable vanes, which partially reverse the direction of the air discharged by each of its engines. Each engine scoops in the air at a rate of 120 kg/s and discharges it with a velocity of 600 m/s relative to the engine. At an instant when the speed of the airplane is 270 km/h, determine the reverse thrust provided by each of the engines.

SOLUTION

Apply the impulse-momentum principle to the moving air. Use a frame of reference that is moving with the airplane. Let $\mathbf{F}$ be the force on the air.

$$v = 270 \text{ km/h} = 75 \text{ m/s}$$

$$u = 600 \text{ m/s}$$



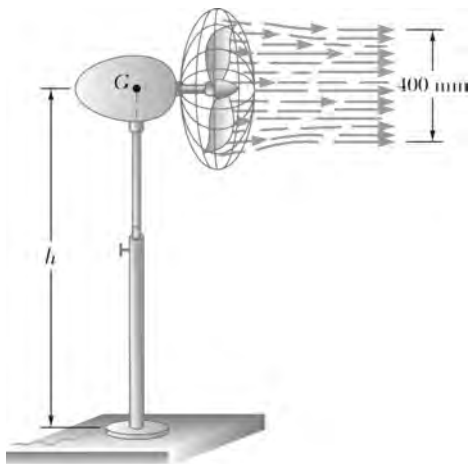
$$-(\Delta m)v + F(\Delta t) = 2 \frac{(\Delta m)}{2} u \sin 20^\circ$$

$$F = \frac{\Delta m}{\Delta t} (v + u \sin 20^\circ) = \frac{dm}{dt} (v + u \sin 20^\circ)$$

$$F = (120)(75 + 600 \sin 20^\circ) = 33.6 \times 10^3 \text{ N}$$

Force on airplane is $-\mathbf{F}$.

$$\mathbf{F} = 33.6 \text{ kN} \leftarrow \blacktriangleleft$$



PROBLEM 14.73

A floor fan designed to deliver air at a maximum velocity of 6 m/s in a 400-mm-diameter slipstream is supported by a 200-mm-diameter circular base plate. Knowing that the total weight of the assembly is 60 N and that its center of gravity is located directly above the center of the base plate, determine the maximum height h at which the fan may be operated if it is not to tip over. Assume $\rho = 1.21 \text{ kg/m}^3$ for air and neglect the approach velocity of the air.

SOLUTION

Calculation of $\frac{dm}{dt}$ at a section in the airstream:

mass = density $\times$ volume = density $\times$ area $\times$ length

$$\Delta m = \rho A(\Delta l) = \rho A v(\Delta t)$$

$$\frac{\Delta m}{\Delta t} = \frac{dm}{dt} = \rho A v$$

Thrust on the airstream:

$$\mathbf{F} = \frac{dm}{dt}(\mathbf{v}_B - \mathbf{v}_A)$$

where $\mathbf{v}_B$ is the velocity just downstream of the fan and $\mathbf{v}_A$ is the velocity for upstream. Assume that $\mathbf{v}_A$ is negligible.

$$F = (\rho A v)v = \rho \left(\frac{\pi}{4} D^2 \right) v^2$$

$$F = (1.21 \text{ kg/m}^3) \left(\frac{\pi}{4} \right) (0.400 \text{ m})^2 (6 \text{ m/s})^2 = 5.474 \text{ N}$$

$$\mathbf{F} = 5.474 \text{ N} \rightarrow$$

Force on fan:

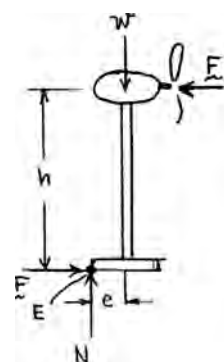
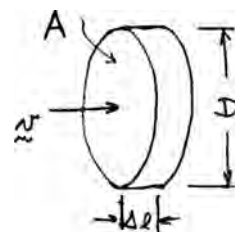
$$\mathbf{F}' = -\mathbf{F} = 5.474 \text{ N} \leftarrow$$

Maximum height h . $\left(e = \frac{1}{2} d = 100 \text{ mm} = 0.1 \text{ m} \right)$

$$\sum M_E = 0$$

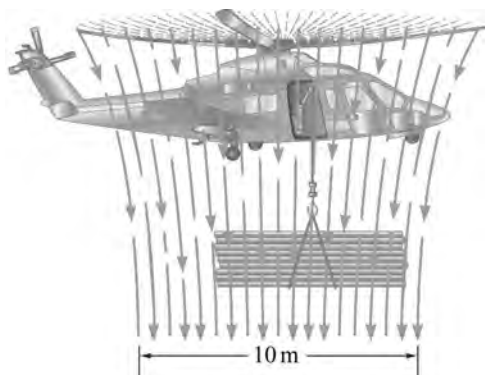
$$F'h - We = 0$$

$$h = \frac{We}{F'} = \frac{(60)(0.1)}{5.474}$$



$$h = 1.096 \text{ m} \quad \blacktriangleleft$$

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PROBLEM 14.74

The helicopter shown can produce a maximum downward air speed of 25 m/s in a 10-m-diameter slipstream. Knowing that the weight of the helicopter and its crew is 18 kN and assuming $\rho = 1.21 \text{ kg/m}^3$ for air, determine the maximum load that the helicopter can lift while hovering in midair.

SOLUTION

The thrust is

$$F = \frac{dm}{dt}(v_B - v_A)$$

Calculation of $\frac{dm}{dt}$.

mass = density $\times$ volume = density $\times$ area $\times$ length

$$\Delta m = \rho A_B (\Delta l) = \rho A_B v_B (\Delta t)$$

$$\frac{\Delta m}{\Delta t} = \rho A_B v_B = \frac{dm}{dt}$$



where A_B is the area of the slipstream well below the helicopter and v_B is the corresponding velocity in the slipstream. Well above the blade, $v_A \approx 0$.

Hence,

$$\begin{aligned} F &= \rho A_B v_B^2 \\ &= (1.21 \text{ kg/m}^3) \left(\frac{\pi}{4} \right) (10 \text{ m})^2 (25 \text{ m/s})^2 \\ &= 59,384.5 \text{ N} \\ F &= 59,384.5 \text{ N} \downarrow \end{aligned}$$

The force on the helicopter is 59.385 kN $\uparrow$

Weight of helicopter: $\mathbf{W}_H = 18 \text{ kN} \downarrow$

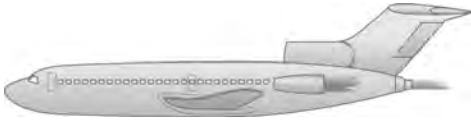
Weight of payload: $\mathbf{W}_P = W_P \downarrow$

Statics: $+\uparrow \sum F_y = F - W_H - W_P = 0$

$$W_P = F - W_H = 59.385 - 18 = 41.385 \text{ kN}$$

$$W = 41.4 \text{ kN} \blacktriangleleft$$

PROBLEM 14.75



A jet airliner is cruising at a speed of 1000 km/h with each of its three engines discharging air with a velocity of 600 m/s relative to the plane. Determine the speed of the airliner after it has lost the use of (a) one of its engines, (b) two of its engines. Assume that the drag due to air friction is proportional to the square of the speed and that the remaining engines keep operating at the same rate.

SOLUTION

Let v be the air liner speed and u be the discharge relative velocity.

$$u = 600 \text{ m/s.}$$

Thrust formula for one engine:

$$F = \frac{dm}{dt}(u - v)$$

Drag formula:

$$D = kv^2$$

Three engines working. Cruising speed = $v_0 = 1000 \text{ km/h} = \frac{2500}{9} \text{ m/s}$

$$3F - D = 3 \frac{dm}{dt}(u - v_0) - kv_0^2 = 0$$

$$\frac{dm}{dt} = \frac{kv_0^2}{3(u - v_0)} = \frac{k \left(\frac{2500}{9} \right)^2}{3 \left(600 - \frac{2500}{9} \right)} = 79.8212k$$

(a) *One engine fails.* Two engines working. Cruising speed = v_1

$$2F - D = 2 \frac{dm}{dt}(u - v_1) - kv_1^2 = 0$$

$$(2)(79.8212k)(600 - v_1) - kv_1^2 = 0$$

$$v_1^2 + 159.642v_1 - 95.785 \times 10^3 = 0$$

$$v_1 = 239.798 \text{ m/s}$$

$$v_1 = 863 \text{ km/h} \quad \blacktriangleleft$$

(b) *Two engines fail.* One engine working. Cruising speed = v_2

$$F - D = \frac{dm}{dt}(u - v_2) - kv_2^2 = 0$$

$$(79.8212k)(600 - v_2) - kv_2^2 = 0$$

$$v_2^2 + 79.8212v_2 - 47.893 \times 10^3 = 0$$

$$v_2 = 182.543 \text{ m/s}$$

$$v_2 = 657 \text{ km/h} \quad \blacktriangleleft$$

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PROBLEM 14.76



A 16-Mg jet airplane maintains a constant speed of 774 km/h while climbing at an angle $\alpha = 18^\circ$. The airplane scoops in air at a rate of 300 kg/s and discharges it with a velocity of 665 m/s relative to the airplane. If the pilot changes to a horizontal flight while maintaining the same engine setting, determine (a) the initial acceleration of the plane, (b) the maximum horizontal speed that will be attained. Assume that the drag due to air friction is proportional to the square of the speed.

SOLUTION

Calculate the propulsive force using velocities relative to the airplane.

$$F = \frac{dm}{dt}(v_B - v_A)$$

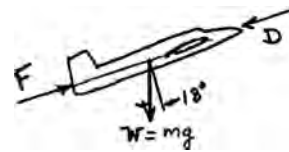
Data:

$$\frac{dm}{dt} = 300 \text{ kg/s}$$

$$v_A = 774 \text{ km/h} \\ = 215 \text{ m/s}$$

$$v_B = 665 \text{ m/s}$$

$$F = (300)(665 - 215) \\ = 135,000 \text{ N}$$



Since there is no acceleration while the airplane is climbing, the forces are in equilibrium.

$$+\nearrow 18^\circ \Sigma F = 0: \quad F - D - mg \sin \alpha = 0$$

$$F - D = mg \sin \alpha = (16,000)(9.8) \sin 18^\circ = 48,454 \text{ N}$$

(a) Initial acceleration of airplane in horizontal flight

$$ma = F - D: \quad 16,000a = 48,454 \times 10^3 \quad a = 3.03 \text{ m/s}^2 \nearrow 18^\circ \nwarrow$$

Corresponding drag force: $D = 135,000 - 48,454 \\ = 86,546 \text{ N}$

Drag force factor: $D = kv_A^2$

or $k = \frac{D}{v_A^2} \\ = \frac{86,546}{(215)^2} \\ = 1.87228 \text{ N} \cdot \text{s}^2/\text{m}^2$

PROBLEM 14.76 (Continued)

(b) *Maximum speed in horizontal flight*

Since the acceleration is zero, the forces are in equilibrium.

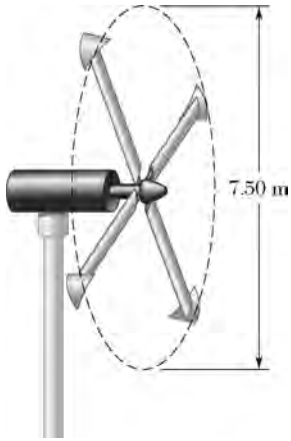
$$F - D = 0$$

$$\frac{dm}{dt}(v_B - v_A) - kv_A^2 = 0 \quad kv_A^2 + \frac{dm}{dt}v_A - \frac{dm}{dt}v_B = 0$$

$$1.87228v_A^2 + 300v_A - (300)(665) = 0$$

$$v_A = 256.0 \text{ m/s}$$

$$v_A = 922 \text{ km/h} \quad \blacktriangleleft$$



PROBLEM 14.77

The wind turbine-generator shown has an output-power rating of 5 kW for a wind speed of 30 km/h. For the given wind speed, determine (a) the kinetic energy of the air particles entering the 7.50-m-diameter circle per second, (b) the efficiency of this energy-conversion system. Assume $\rho = 1.21 \text{ kg/m}^3$ for air.

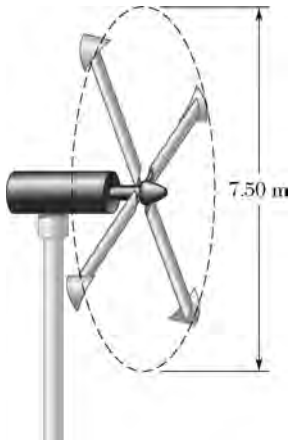
SOLUTION

(a) Kinetic energy per second.

$$\begin{aligned}
 &= \frac{1}{2} \frac{\Delta m}{\Delta t} v^2 = \frac{1}{2} \rho Q v^2 = \frac{1}{2} \rho (Av) v^2 = \frac{1}{2} \rho A v^3 \\
 &= \frac{1}{2} (1.21 \text{ kg/m}^3) \frac{\pi}{4} (7.50 \text{ m})^2 \left(\frac{30 \times 10^3 \text{ m}}{3.6 \times 10^3 \text{ s}} \right)^3 \quad \frac{dT}{dt} = 15.47 \text{ kJ/s} \blacktriangleleft
 \end{aligned}$$

(b) Efficiency.

$$\eta = \frac{5 \text{ kW}}{15.47 \text{ kJ/s}} \quad \eta = 0.323 \blacktriangleleft$$



PROBLEM 14.78

For a certain wind speed, the wind turbine-generator shown produces 28 kW of electric power and has an efficiency of 0.35 as an energy-conversion system. Assuming $\rho = 1.21 \text{ kg/m}^3$ for air determine (a) the kinetic energy of the air particles entering the 7.50-m-diameter circle per second, (b) the wind speed.

SOLUTION

(a) Kinetic energy per second.

$$= \text{input power} = \frac{\text{output power}}{\text{efficiency}} = \frac{28 \text{ kW}}{0.35} \qquad \frac{dT}{dt} = 80 \text{ kJ/s} \blacktriangleleft$$

(b) Wind speed.

$$\begin{aligned} \text{K.E. per second} &= \frac{1}{2} \frac{\Delta m}{\Delta t} v^2 \\ &= \frac{1}{2} \rho Q v^2 \\ &= \frac{1}{2} \rho (A v) v^2 \\ &= \frac{1}{2} \rho A v^3 \end{aligned}$$

Therefore,

$$80 \text{ kJ/s} = \frac{1}{2} (1.21 \text{ kg/m}^3) \frac{\pi}{4} (7.50 \text{ m})^2 v^3$$

$$v^3 = 2993.1$$

$$v = 14.411 \text{ m/s}$$

$$v = 51.9 \text{ km/h} \blacktriangleleft$$

PROBLEM 14.79

While cruising in level flight at a speed of 900 km/h, a jet airplane scoops in air at a rate of 120 kg/s and discharges it with a velocity of 650 m/s relative to the airplane. Determine (a) the power actually used to propel the airplane, (b) the total power developed by the engine, (c) the mechanical efficiency of the airplane.

SOLUTION

Data:

$$\frac{dm}{dt} = 120 \text{ kg/s}$$

$$u = 650 \text{ m/s}$$

$$v = 900 \text{ km/h} = 250 \text{ m/s}$$

$$\begin{aligned} F &= \frac{dm}{dt}(u - v) \\ &= 120(650 - 250) \\ &= 48,000 \text{ N} \end{aligned}$$

(a) *Power used to propel airplane:*

$$\begin{aligned} P_1 &= Fv \\ &= (48,000)(250) \\ &= 12 \times 10^6 \text{ W} = 12 \text{ MW} \end{aligned}$$

Propulsion power = 12 MW ◀

Power of kinetic energy of exhaust:

$$\begin{aligned} P_2(\Delta t) &= \frac{1}{2}(\Delta m)(u - v)^2 \\ P_2 &= \frac{1}{2} \frac{dm}{dt}(u - v)^2 \\ &= \frac{1}{2}(120)(650 - 250)^2 \\ &= 9.6 \times 10^6 \text{ W} = 9.6 \text{ MW} \end{aligned}$$

(b) *Total power:*

$$\begin{aligned} P &= P_1 + P_2 \\ &= 21.6 \text{ MW} \end{aligned}$$

Total power = 21.6 MW ◀

(c) *Mechanical efficiency:*

$$\begin{aligned} \frac{P_1}{P} &= \frac{12}{21.6} \\ &= \frac{5}{9} = 0.556 \end{aligned}$$

Mechanical efficiency = 0.556 ◀

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PROBLEM 14.80

The propeller of a small airplane has a 2-m-diameter slipstream and produces a thrust of 4000 N when the airplane is at fix on the ground. Assuming $\rho = 1.21 \text{ kg/m}^3$ for air, determine (a) the speed of the air in the slipstream, (b) the volume of air passing through the propeller per second, (c) the kinetic energy imparted per second to the air in the slipstream.

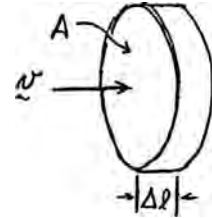
SOLUTION

Calculation of $\frac{dm}{dt}$ at a section in the airstream:

$$\begin{aligned}\text{mass} &= \text{density} \times \text{volume} \\ &= \text{density} \times \text{area} \times \text{length}\end{aligned}$$

$$\Delta m = \rho A(\Delta l) = \rho A v \Delta t$$

$$\frac{\Delta m}{\Delta t} = \frac{dm}{dt} = \rho A v$$



- (a) Thrust $= \frac{dm}{dt}(\mathbf{v}_B - \mathbf{v}_A)$ where $\mathbf{v}_B$ is the velocity just downstream of propeller and $\mathbf{v}_A$ is the velocity far upstream. Assume $\mathbf{v}_A$ is negligible.

$$\text{Thrust} = (\rho A v)v = \rho \left(\frac{\pi}{4} D^2 \right) v^2$$

$$4000 = (1.21) \left(\frac{\pi}{4} \right) (2)^2 v^2$$

$$= 3.80133 v^2$$

$$v = 32.4386 \text{ m/s}$$

$$v = 32.4 \text{ m/s} \quad \blacktriangleleft$$

- (b) $Q = \frac{1}{\rho} \frac{dm}{dt} = A v = \left(\frac{\pi}{4} D^2 \right) v = \frac{\pi}{4} (2)^2 (32.4386)$

$$Q = 101.9 \text{ m}^3/\text{s} \quad \blacktriangleleft$$

- (c) Kinetic energy of mass Δm :

$$\Delta T = \frac{1}{2} (\Delta m) v^2 = \frac{1}{2} \rho A (\Delta l) v^2 = \frac{1}{2} \rho A v (\Delta t) v^3$$

$$\frac{\Delta T}{\Delta t} = \frac{dT}{dt}$$

$$= \frac{1}{2} \rho A v^3$$

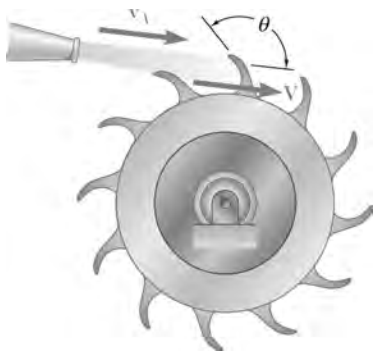
$$= \frac{1}{2} \rho \left(\frac{\pi D^2}{4} \right) v^3$$

$$= \frac{1}{2} (1.21) \left(\frac{\pi}{4} \right) (2)^2 (32.4386)^3$$

$$= 64,877.1 \text{ W}$$

$$\frac{dT}{dt} = 64,900 \text{ W} \quad \blacktriangleleft$$

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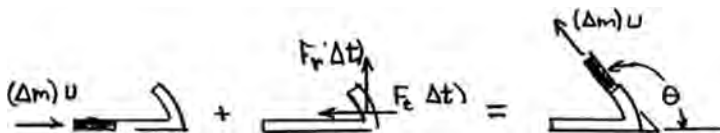
PROBLEM 14.81

In a Pelton-wheel turbine, a stream of water is deflected by a series of blades so that the rate at which water is deflected by the blades is equal to the rate at which water issues from the nozzle ($\Delta m/\Delta t = \rho A v_A$). Using the same notation as in Sample Problem 14.7, (a) determine the velocity $\mathbf{V}$ of the blades for which maximum power is developed, (b) derive an expression for the maximum power, (c) derive an expression for the mechanical efficiency.

SOLUTION

Let $\mathbf{u}$ be the velocity of the stream relative to the velocity of the blade.

$$u = (v - V)$$



Mass flow rate:

$$\frac{\Delta m}{\Delta t} = \rho A v_A$$

Principle of impulse and momentum.

$$\rightarrow (\Delta m)u - F_t(\Delta t) = (\Delta m)u \cos \theta$$

$$\begin{aligned} F_t &= \frac{\Delta m}{\Delta t} u(1 - \cos \theta) \\ &= \rho A v_A (v_A - V)(1 - \cos \theta) \end{aligned}$$

where F_t is the tangential force on the fluid.

The force F_t on the fluid is directed to the left as shown. By Newton's law of action and reaction, the tangential force on the blade is F_t to the right.

Output power:

$$\begin{aligned} P_{\text{out}} &= F_t V \\ &= \rho A v_A (v_A - V)V(1 - \cos \theta) \end{aligned}$$

(a) V for maximum power output.

$$\frac{dP_{\text{out}}}{dV} = \rho A (v_A - 2V)(1 - \cos \theta) = 0 \qquad v_A = \frac{1}{2}V \quad \blacktriangleleft$$

PROBLEM 14.81 (Continued)

(b) *Maximum power.*

$$(P_{\text{out}})_{\text{max}} = \rho A v_A \left(v_A - \frac{1}{2} v_A \right) \left(\frac{1}{2} v_A \right) (1 - \cos \theta)$$

$$(P_{\text{out}})_{\text{max}} = \frac{1}{4} \rho A v_A^3 (1 - \cos \theta) \quad \blacktriangleleft$$

Input power = rate of supply of kinetic energy of the stream

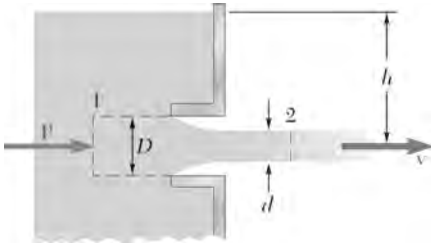
$$\begin{aligned} P_{\text{in}} &= \frac{1}{\Delta t} \left[\frac{1}{2} (\Delta m) v_A^2 \right] \\ &= \frac{1}{2} \frac{\Delta m}{\Delta t} v_A^2 \\ &= \frac{1}{2} \rho A v_A^3 \end{aligned}$$

(c) *Efficiency.*

$$\begin{aligned} \eta &= \frac{P_{\text{out}}}{P_{\text{in}}} \\ \eta &= \frac{\rho A v_A (v_A - V) V (1 - \cos \theta)}{\frac{1}{2} \rho A v_A^3} \end{aligned}$$

$$\eta = 2 \left(1 - \frac{V}{v_A} \right) \frac{V}{v_A} (1 - \cos \theta) \quad \blacktriangleleft$$

PROBLEM 14.82



A circular reentrant orifice (also called Borda's mouthpiece) of diameter D is placed at a depth h below the surface of a tank. Knowing that the speed of the issuing stream is $v = \sqrt{2gh}$ and assuming that the speed of approach v_1 is zero, show that the diameter of the stream is $d = D/\sqrt{2}$. (Hint: Consider the section of water indicated, and note that P is equal to the pressure at a depth h multiplied by the area of the orifice).

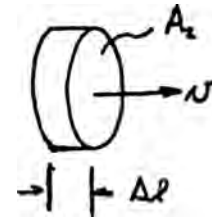
SOLUTION

From hydrostatics, the pressure at Section 1 is $p_1 = rh = \rho gh$.

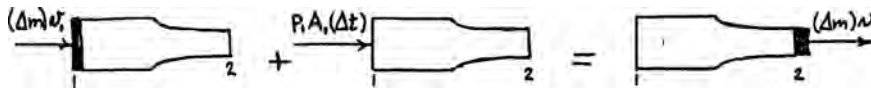
The pressure at Section 2 is $p_2 = 0$.

Calculate the mass flow rate using Section 2.

$$\begin{aligned} \text{mass} &= \text{density} \times \text{volume} \\ &= \text{density} \times \text{area} \times \text{length} \\ \Delta m &= \rho A_2 (\Delta l) = \rho A_2 v (\Delta t) \\ \frac{dm}{dt} &= \frac{\Delta m}{\Delta t} = \rho A_2 v \end{aligned}$$



Apply the impulse-momentum principle to fluid between Sections 1 and 2.



$$\begin{aligned} (\Delta m)v_1 + p_1 A_1 (\Delta t) &= (\Delta m)v \\ \frac{dm}{dt} v_1 + p_1 A_1 &= \frac{dm}{dt} v \\ p_1 A_1 &= \frac{dm}{dt} (v - v_1) \\ &= \rho A_2 v (v - v_1) \end{aligned}$$

But v_1 is negligible,

$$\begin{aligned} p_1 &= \rho gh \quad \text{and} \quad v = \sqrt{2gh} \\ \rho gh A_1 &= \rho A_2 (2gh) \quad \text{or} \quad A_1 = 2A_2 \end{aligned}$$

$$\frac{\pi}{4} D^2 = 2 \left(\frac{\pi}{4} d^2 \right) \quad d = \frac{D}{\sqrt{2}} \quad \blacktriangleleft$$





PROBLEM 14.83*

The depth of water flowing in a rectangular channel of width b at a speed v_1 and a depth d_1 increases to a depth d_2 at a *hydraulic jump*. Express the rate of flow Q in terms of b , d_1 , and d_2 .

SOLUTION

Mass flow rate:

$$\begin{aligned}\text{mass} &= \text{density} \times \text{volume} \\ &= \text{density} \times \text{area} \times \text{length}\end{aligned}$$

$$\Delta m = \rho b d (\Delta l) = \rho b d v (\Delta t)$$

$$\frac{dm}{dt} = \frac{\Delta m}{\Delta t} = \rho b d v$$

$$Q = \frac{1}{\rho} \frac{dm}{dt} = b d v$$

Continuity of flow:

$$Q_1 = Q_2 = Q$$

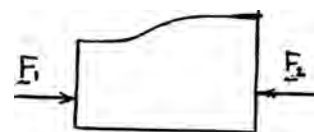
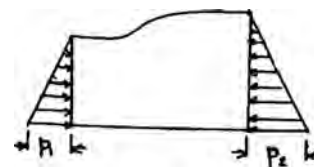
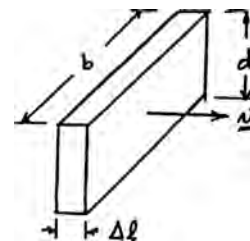
$$v_1 = \frac{Q}{b d_1} \quad v_2 = \frac{Q}{b d_2}$$

Resultant pressure forces:

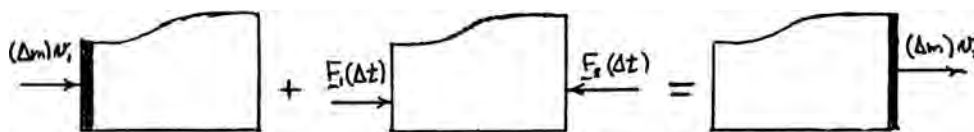
$$p_1 = \gamma d_1 \quad p_2 = \gamma d_2$$

$$F_1 = \frac{1}{2} p_1 b d_1 = \frac{1}{2} \gamma b d_1^2$$

$$F_2 = \frac{1}{2} p_2 b d_2 = \frac{1}{2} \gamma b d_2^2$$



Apply impulse-momentum principle to water between Sections 1 and 2.



$$(\Delta m)v_1 + F_1(\Delta t) - F_2(\Delta t) = (\Delta m)v_2$$

$$\frac{\Delta m}{\Delta t}(v_1 - v_2) = F_2 - F_1 \quad \rho Q \cdot \left(\frac{Q}{b d_1} - \frac{Q}{b d_2} \right) = \frac{1}{2} \gamma b (d_2^2 - d_1^2)$$

$$\frac{\rho Q^2 (d_2 - d_1)}{b d_1 d_2} = \frac{1}{2} \gamma b (d_1 + d_2)(d_2 - d_1)$$

Noting that $\gamma = \rho g$,

$$Q = b \sqrt{\frac{1}{2} g d_1 d_2 (d_1 + d_2)} \quad \blacktriangleleft$$

PROBLEM 14.84*



Determine the rate of flow in the channel of Problem 14.83, knowing that $b = 3.6$ m, $d_1 = 1.2$ m and $d_2 = 1.5$ m.

PROBLEM 14.83 The depth of water flowing in a rectangular channel of width b at a speed v_1 and a depth d_1 increases to a depth d_2 at a *hydraulic jump*. Express the rate of flow Q in terms of b , d_1 , and d_2 .

SOLUTION

Mass flow rate:

$$\begin{aligned}\text{mass} &= \text{density} \times \text{volume} \\ &= \text{density} \times \text{area} \times \text{length}\end{aligned}$$

$$\Delta m = \rho b d (\Delta l) = \rho b d v (\Delta t)$$

$$\frac{dm}{dt} = \frac{\Delta m}{\Delta t} = \rho b d v$$

$$Q = \frac{1}{\rho} \frac{dm}{dt} = b d v$$

Continuity of flow:

$$Q_1 = Q_2 = Q$$

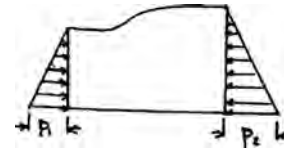
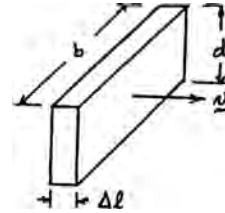
$$v_1 = \frac{Q}{b d_1} \quad v_2 = \frac{Q}{b d_2}$$

Resultant pressure forces:

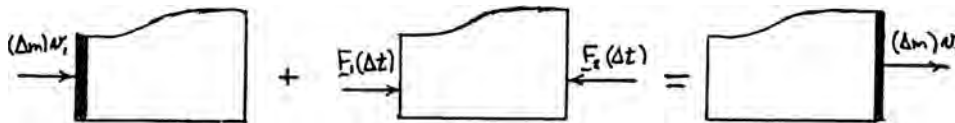
$$p_1 = \gamma d_1 \quad p_2 = \gamma d_2$$

$$F_1 = \frac{1}{2} p_1 b d_1 = \frac{1}{2} \gamma b d_1^2$$

$$F_2 = \frac{1}{2} p_2 b d_2 = \frac{1}{2} \gamma b d_2^2$$



Apply impulse-momentum principle to water between Sections 1 and 2.



$$(\Delta m)v_1 + F_1(\Delta t) - F_2(\Delta t) = (\Delta m)v_2$$

$$\frac{\Delta m}{\Delta t}(v_1 - v_2) = F_2 - F_1 \quad \rho Q \cdot \left(\frac{Q}{b d_1} - \frac{Q}{b d_2} \right) = \frac{1}{2} \gamma b (d_2^2 - d_1^2)$$

$$\frac{\rho Q^2 (d_2 - d_1)}{b d_1 d_2} = \frac{1}{2} \gamma b (d_1 + d_2)(d_2 - d_1)$$

PROBLEM 14.84* (Continued)

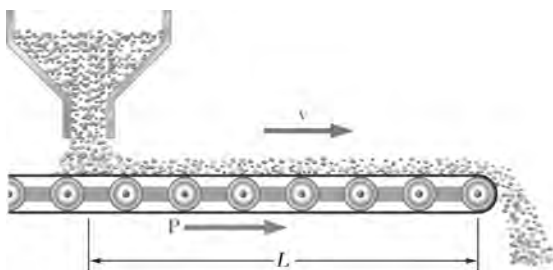
Noting that $\gamma = \rho g$,

$$Q = b \sqrt{\frac{1}{2} g d_1 d_2 (d_1 + d_2)}$$

Data:

$$g = 9.81 \text{ m/s}^2 \quad b = 3.6 \text{ m}, \quad d_1 = 1.2 \text{ m}, \quad d_2 = 1.5 \text{ m}$$

$$Q = 3.6 \sqrt{\frac{1}{2} (9.81)(1.2)(1.5)(2.7)} \qquad Q = 17.58 \text{ m}^3/\text{s} \blacktriangleleft$$



PROBLEM 14.85

Gravel falls with practically zero velocity onto a conveyor belt at the constant rate $q = dm/dt$. (a) Determine the magnitude of the force $\mathbf{P}$ required to maintain a constant belt speed v . (b) Show that the kinetic energy acquired by the gravel in a given time interval is equal to half the work done in that interval by the force $\mathbf{P}$. Explain what happens to the other half of the work done by $\mathbf{P}$.

SOLUTION

- (a) We apply the impulse-momentum principle to the gravel on the belt and to the mass Δm of gravel hitting and leaving belt in interval Δt .



$$\rightarrow x \text{ comp.: } mv + P\Delta t = mv + (\Delta m)v$$

$$P = \frac{\Delta m}{\Delta t} v = qv$$

$$P = qv \quad \blacktriangleleft$$

- (b) Kinetic energy acquired for unit time:

$$\begin{aligned} \Delta T &= \frac{1}{2}(\Delta m)v^2 \\ \frac{\Delta T}{\Delta t} &= \frac{1}{2} \frac{\Delta m}{\Delta t} v^2 = \frac{1}{2} qv^2 \end{aligned} \quad (1)$$

Work done per unit time:

$$\frac{\Delta U}{\Delta t} = \frac{P\Delta x}{\Delta t} = Pv$$

Recalling the result of Part a:

$$\begin{aligned} \Delta U &= P(\Delta x) \\ \frac{\Delta U}{\Delta t} &= (qv)v = qv^2 \end{aligned} \quad (2)$$

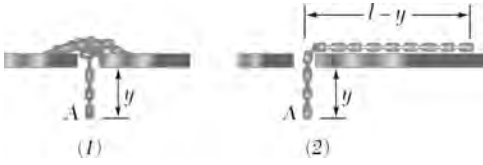
Comparing Eqs. (1) and (2), we conclude that

$$\frac{\Delta T}{\Delta t} = \frac{1}{2} \frac{\Delta U}{\Delta t} \quad \text{Q.E.D.} \quad \blacktriangleleft$$

The other half of the work of $\mathbf{P}$ is dissipated into heat by friction as the gravel slips on the belt before reaching the speed v . $\blacktriangleleft$



PROBLEM 14.86

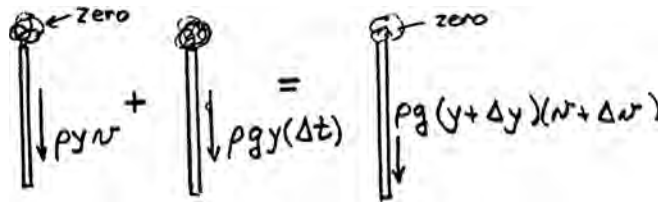


A chain of length l and mass m falls through a small hole in a plate. Initially, when y is very small, the chain is at rest. In each case shown, determine (a) the acceleration of the first link A as a function of y , (b) the velocity of the chain as the last link passes through the hole. In Case 1, assume that the individual links are at rest until they fall through the hole; in Case 2, assume that at any instant all links have the same speed. Ignore the effect of friction.

SOLUTION

Let ρ be the mass per unit length of chain. Assume that the weight of any chain above the hole is supported by the floor. It and the corresponding upward reaction of the floor are not shown in the diagrams.

Case 1. Apply the impulse-momentum principle to the entire chain.



$$\begin{aligned}\rho y v + \rho g y \Delta t &= \rho (y + \Delta y) (v + \Delta v) \\ &= \rho y v + \rho (\Delta y) v + \rho y (\Delta v) + \rho (\Delta y) (\Delta v) \\ \rho g y &= \rho \frac{\Delta y}{\Delta t} v + \rho y \frac{\Delta v}{\Delta t} + \rho \frac{(\Delta y)(\Delta v)}{\Delta t}\end{aligned}$$

Let $\Delta t \rightarrow 0$.

$$\begin{aligned}\rho g y &= \rho \frac{dy}{dt} v + \rho y \frac{dv}{dt} \\ &= \rho \frac{d}{dt} (y v)\end{aligned}$$

Multiply both sides by yv .

$$\rho g y^2 v = \rho y v \frac{d}{dt} (y v)$$

Let $v = \frac{dy}{dt}$ on left hand side.

$$\rho g y^2 \frac{dy}{dt} = \rho y v \frac{d}{dt} (y v)$$

Integrate with respect to time.

$$\begin{aligned}\rho g \int y^2 dy &= \rho \int (y v) d(y v) \\ \frac{1}{3} \rho g y^3 &= \frac{1}{2} \rho (y v)^2 \quad \text{or} \quad v^2 = \frac{2}{3} g y\end{aligned}\tag{1}$$

Differentiate with respect to time.

$$2v \frac{dv}{dt} = \frac{2}{3} g \frac{dy}{dt} = \frac{2}{3} g v$$

PROBLEM 14.86 (Continued)

$$(a) \quad a = \frac{dv}{dt} = \frac{1}{3}g \quad \mathbf{a} = 0.333g \downarrow \blacktriangleleft$$

$$(b) \quad \text{Set } y = l \text{ in Eq. (1).} \quad v^2 = \frac{2}{3}gl \quad \mathbf{v} = 0.817\sqrt{gl} \downarrow \blacktriangleleft$$

Case 2. Apply conservation of energy using the floor as the level from which the potential energy is measured.

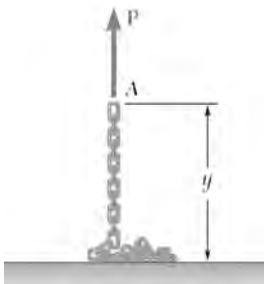
$$\begin{aligned} T_1 &= 0 & V_1 &= 0 \\ T_2 &= \frac{1}{2}mv^2 & V_2 &= -\rho gy \frac{y}{2} \\ T_1 + V_1 &= T_2 + V_2 \\ 0 &= \frac{1}{2}mv^2 - \frac{1}{2}\rho gy^2 & v^2 &= \frac{\rho gy^2}{m} = \frac{gy^2}{l} \end{aligned} \quad (2)$$

$$\text{Differentiating with respect to } y, \quad 2v \frac{dv}{dy} = \frac{2gy}{l}$$

$$(a) \quad \text{Acceleration.} \quad a = v \frac{dv}{dy} = \frac{gy}{l} \quad \mathbf{a} = \frac{gy}{l} \downarrow \blacktriangleleft$$

$$(b) \quad \text{Setting } y = l \text{ in Eq. (2),} \quad v^2 = gl \quad \mathbf{v} = \sqrt{gl} \downarrow \blacktriangleleft$$

Note: The impulse-momentum principle may be used to obtain the force that the edge of the hole exerts on the chain.



PROBLEM 14.87

A chain of length l and mass m lies in a pile on the floor. If its end A is raised vertically at a constant speed v , express in terms of the length y of chain which is off the floor at any given instant (a) the magnitude of the force $\mathbf{P}$ applied at A , (b) the reaction of the floor.

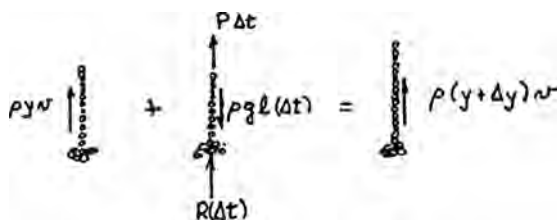
SOLUTION

Let ρ be the mass per unit length of chain. Apply the impulse-momentum to the entire chain. Assume that the reaction from the floor is equal to the weight of chain still in contact with the floor.

Calculate the floor reaction.

$$R = \rho g(l - y)$$

$$R = mg \left(1 - \frac{y}{l} \right)$$



Apply the impulse-momentum principle.

$$\rho y v + P(\Delta t) + R(\Delta t) - \rho g L(\Delta t) = \rho(y + \Delta y)v$$

$$P \Delta t = \rho(\Delta y)v + \rho g L(\Delta t) - R(\Delta t)$$

(a)

$$P = \rho \frac{\Delta y}{\Delta t} v + \rho g L - \rho(L - y)g$$

$$= \rho v^2 + \rho g y$$

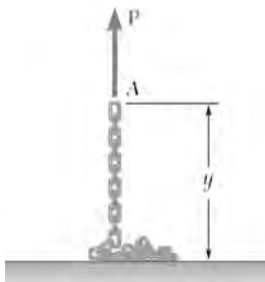
$$P = \frac{m}{l}(v^2 + gy) \quad \blacktriangleleft$$

Let

$$\frac{\Delta y}{\Delta t} = \frac{dy}{dt} = v$$

(b) From above,

$$\mathbf{R} = mg \left(1 - \frac{y}{l} \right) \uparrow \quad \blacktriangleleft$$



PROBLEM 14.88

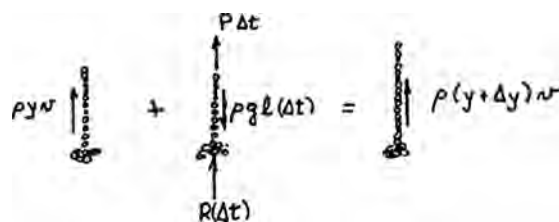
Solve Problem 14.87, assuming that the chain is being *lowered* to the floor at a constant speed v .

SOLUTION

- (a) Let ρ be the mass per unit length of chain. The force P supports the weight of chain still off the floor.

$$P = \rho gy \qquad P = \frac{mgy}{l} \quad \blacktriangleleft$$

- (b) Apply the impulse-momentum principle to the entire chain.



$$\begin{aligned} -\rho yv + P(\Delta t) + R(\Delta t) - \rho gL(\Delta t) &= -\rho g(y + \Delta y)v \\ R(\Delta t) &= \rho gL(\Delta t) - P(\Delta t) - \rho g(\Delta y)v \\ R &= \rho gL - \rho gy - \rho \frac{\Delta y}{\Delta t}v \end{aligned}$$

Let $\Delta t \rightarrow 0$. Then

$$\frac{\Delta y}{\Delta t} = \frac{dy}{dt} = -v$$

$$R = \rho g(L - y) + \rho v^2 \qquad \mathbf{R} = \frac{m}{l}[g(L - y) + v^2] \uparrow \quad \blacktriangleleft$$

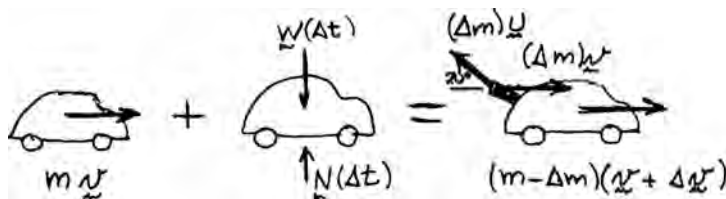


PROBLEM 14.89

A toy car is propelled by water that squirts from an internal tank at a constant 2 m/s relative to the car. The weight of the empty car is 200 g and it holds 1 kg of water. Neglecting other tangential forces, determine the top speed of the car.

SOLUTION

Consider a time interval Δt . Let m be the mass of the car plus the water in the tank at the beginning of the interval and $(m - \Delta m)$ the corresponding mass at the end of the interval. m_0 is the initial value of m . Let v be the velocity of the car. Apply the impulse and momentum principle over the time interval.



Horizontal components $\rightarrow$:

$$mv + 0 = (\Delta m)(v - U \cos 20^\circ) + (m - \Delta m)(v + \Delta v)$$

$$\Delta v = U \cos 20^\circ \frac{\Delta m}{m - \Delta m} \quad (1)$$

Let Δv be replaced by differential dv and Δm be replaced by the small differential $-dm$, the minus sign meaning that dm is the infinitesimal increase in m , and neglecting the dm in denominator of (1)

$$dv = -U \cos 20^\circ \frac{dm}{m}$$

Integrating,

$$v = v_0 - U \cos 20^\circ \ln \frac{m}{m_0}$$

Since $v_0 = 0$,

$$v = U \cos 20^\circ \ln \frac{m_0}{m}$$

The velocity is maximum when $m = m_f$, the value of m when all of the water is expelled.

$$v_{\max} = U \cos 20^\circ \ln \frac{m_0}{m_f}$$

$$v_{\max} = (2 \text{ m/s}) \cos 20^\circ \ln \frac{1 + 0.2}{0.2} \quad v_{\max} = 3.37 \text{ m/s} \quad \blacktriangleleft$$

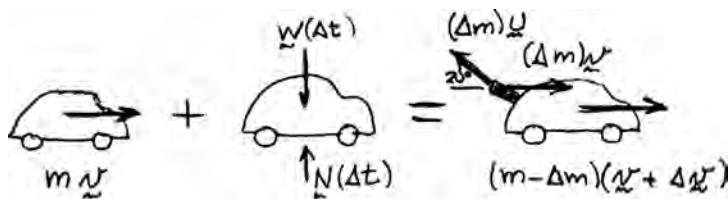


PROBLEM 14.90

A toy car is propelled by water that squirts from an internal tank. The weight of the empty car is 200 g and it holds 1 kg of water. Knowing the top speed of the car is 2.5 m/s, determine the relative velocity of the water that is being ejected.

SOLUTION

Consider a time interval Δt . Let m be the mass of the car plus the water in the tank at the beginning of the interval and $(m - \Delta m)$ the corresponding mass at the end of the interval. m_0 is the initial value of m . Let v be the velocity of the car. Apply the impulse and momentum principle over the time interval.



Horizontal components $\rightarrow$:

$$mv + 0 = (\Delta m)(v - U \cos 20^\circ) + (m - \Delta m)(v + \Delta v)$$

$$\Delta v = U \cos 20^\circ \frac{\Delta m}{m - \Delta m} \quad (1)$$

Let Δv be replaced by differential dv and Δm be replaced by the small differential $-dm$, the minus sign meaning that dm is the infinitesimal increase in m and neglecting dm in denominator of (1)

$$dv = -U \cos 20^\circ \frac{dm}{m}$$

Integrating,

$$v = v_0 - U \cos 20^\circ \ln \frac{m}{m_0}$$

Since $v_0 = 0$,

$$v = U \cos 20^\circ \ln \frac{m_0}{m}$$

The velocity is maximum when $m = m_f$, the value of m when all of the water is expelled.

$$v_{\max} = U \cos 20^\circ \ln \frac{m_0}{m_f}$$

$$2.5 \text{ m/s} = U \cos 20^\circ \ln \frac{1 + 0.2}{0.2} \quad U = 1.485 \text{ m/s} \quad \blacktriangleleft$$



PROBLEM 14.91

The main propulsion system of a space shuttle consists of three identical rocket engines, each of which burns the hydrogen-oxygen propellant at the rate of 340 kg/s and ejects it with a relative velocity of 3750 m/s. Determine the total thrust provided by the three engines.

SOLUTION

From Eq. (14.44) for each engine:

$$\begin{aligned} P &= \frac{dm}{dt} u \\ &= (340 \text{ kg/s})(3750 \text{ m/s}) \\ &= 1.275 \times 10^6 \text{ N} \end{aligned}$$

For the 3 engines:

$$\text{Total thrust} = 3(1.275 \times 10^6 \text{ N})$$

$$\text{Total thrust} = 3.03 \text{ MN} \quad \blacktriangleleft$$



PROBLEM 14.92

The main propulsion system of a space shuttle consists of three identical rocket engines which provide a total thrust of 6 MN. Determine the rate at which the hydrogen-oxygen propellant is burned by each of the three engines, knowing that it is ejected with a relative velocity of 3750 m/s.

SOLUTION

Thrust of each engine: $P = \frac{1}{3}(6 \text{ MN}) = 2 \times 10^6 \text{ N}$

Eq. (14.44): $P = \frac{dm}{dt}u$
 $2 \times 10^6 \text{ N} = \frac{dm}{dt}(3750 \text{ m/s})$

$$\frac{dm}{dt} = \frac{2 \times 10^6 \text{ N}}{3750 \text{ m/s}}$$

$$\frac{dm}{dt} = 533 \text{ kg/s} \quad \blacktriangleleft$$

PROBLEM 14.93



A space vehicle describing a circular orbit at a speed of 24×10^3 km/h releases its front end a capsule which has a gross mass of 600 kg, including 400 kg of fuel. If the fuel is consumed at the constant rate of 18 kg/s and ejected with a relative velocity of 3000 m/s, determine
(a) the tangential acceleration of the capsule as the engine is fired,
(b) the maximum speed attained by the capsule.

SOLUTION

Thrust:

$$\begin{aligned} P &= \left| \frac{dm}{dt} \right| u \\ &= (18 \text{ kg/s})(3000 \text{ m/s}) \\ &= 54 \times 10^3 \text{ N} \end{aligned}$$

$$(a) \quad (a_t)_0 = \frac{P}{m_0} = \frac{54 \times 10^3}{600} = 90 \text{ m/s}^2 \quad (a_t)_0 = 90.0 \text{ m/s}^2 \blacktriangleleft$$

(b) Maximum speed is attained when all the fuel is used up.

$$\begin{aligned} v_1 &= v_0 + \int_0^{t_1} a_t dt = v_0 + \int_0^{t_1} \frac{P}{m} dt \\ &= v_0 + \int_0^{t_1} \frac{u \left(\frac{dm}{dt} \right)}{m} dt = v_0 + u \int_{m_0}^{m_1} \left(-\frac{dm}{m} \right) \\ v_1 &= v_0 + u \left(-\ln \frac{m_1}{m_0} \right) = v_0 + u \ln \frac{m_0}{m_1} \end{aligned}$$

Data:

$$\begin{aligned} v_0 &= 24 \times 10^3 \text{ km/h} \\ &= 6.6667 \times 10^3 \text{ m/s} \end{aligned}$$

$$u = 3000 \text{ m/s}$$

$$m_0 = 600 \text{ kg}$$

$$m_1 = 600 - 400 = 200 \text{ kg}$$

$$\begin{aligned} v_1 &= 6.6667 \times 10^3 + 3000 \ln \frac{600}{200} \\ &= 9.9625 \times 10^3 \text{ m/s} \end{aligned}$$

$$v_1 = 35.9 \times 10^3 \text{ km/h} \blacktriangleleft$$



PROBLEM 14.94

A rocket has a mass of 1200 kg, including 1000 kg of fuel, which is consumed at the rate of 12.5 kg/s and ejected with a relative velocity of 4000 m/s. Knowing that the rocket is fired vertically from the ground, determine its acceleration (*a*) as it is fired, (*b*) as the last particle of fuel is being consumed.

SOLUTION

From Eq. (14.44) of the text book, the thrust is

$$\begin{aligned}P &= \frac{dm}{dt}u \\&= (12.5 \text{ kg/s})(4000 \text{ m/s}) \\&= 50 \times 10^3 \text{ kg} \cdot \text{m/s}^2 \\&= 50 \times 10^3 \text{ N}\end{aligned}$$

$$\Sigma F = ma$$

$$P - mg = ma \quad a = \frac{P}{m} - g \quad (1)$$

(a) *At the start of firing,*

$$m = m_0 = 960 \text{ kg}$$

$$g = 9.81 \text{ m/s}^2$$

From Eq. (1),

$$a = \frac{50 \times 10^3}{1200} - 9.81 = 31.86 \text{ m/s}^2 \quad \mathbf{a = 31.9 \text{ m/s}^2 \uparrow \blacktriangleleft}$$

(b) *As the last particle of fuel is consumed,*

$$m = 1200 - 1000 = 200 \text{ kg} \quad g = 9.81 \text{ m/s}^2 \text{ (assumed)}$$

From Eq. (1),

$$a = \frac{54 \times 10^3}{200} - 9.81 = 240.2 \text{ m/s}^2 \quad \mathbf{a = 240 \text{ m/s}^2 \uparrow \blacktriangleleft}$$

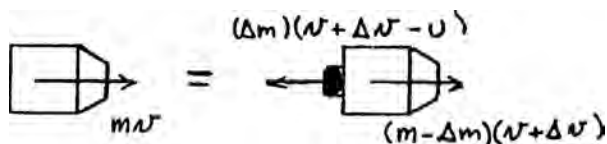


PROBLEM 14.95

A communication satellite weighing 50 kN, including fuel, has been ejected from a space shuttle describing a low circular orbit around the earth. After the satellite has slowly drifted to a safe distance from the shuttle, its engine is fired to increase its velocity by 2500 m/s as a first step to its transfer to a geosynchronous orbit. Knowing that the fuel is ejected with a relative velocity of 4000 m/s, determine the weight of fuel consumed in this maneuver.

SOLUTION

Apply the *principle of impulse and momentum* to the satellite plus the fuel expelled in time Δt .



$$\begin{aligned} \xrightarrow{+}: mv &= (m - \Delta m)(v + \Delta v) + (\Delta m)(v + \Delta v - u) \\ &= mv + m(\Delta v) - (\Delta m)v - (\Delta m)(\Delta v) + (\Delta m)v + (\Delta m)(\Delta v) - (\Delta m)u \\ m(\Delta v) - u(\Delta m) &= 0 \end{aligned}$$

$$\Delta m = -\frac{dm}{dt}(\Delta t)$$

$$\frac{\Delta v}{\Delta t} = \frac{dv}{dt} = -\frac{u}{m} \frac{dm}{dt}$$

$$\int_{v_0}^{v_1} dv = -\int_0^{t_1} \frac{u}{m} \frac{dm}{dt} dt = -\int_{m_0}^{m_1} \frac{dm}{m}$$

$$v_1 - v_0 = -u \ln \frac{m_1}{m_0} = u \ln \frac{m_0}{m_1}$$

$$\frac{m_0}{m_1} = \exp\left(\frac{v_1 - v_0}{u}\right)$$

PROBLEM 14.95 (Continued)

Data:

$$v_1 - v_0 = 2500 \text{ m/s}$$

$$u = 4000 \text{ m/s}$$

$$m_0 = \frac{50000}{9.81} \text{ kg}$$

$$\frac{50000}{9.81 m_1} = \exp \frac{2500}{4000}$$

$$= 1.8682$$

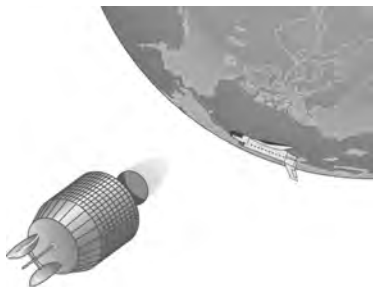
$$m_1 = 2728.14 \text{ kg}$$

$$m_{\text{fuel}} = m_0 - m_1 = 2368.7 \text{ kg}$$

$$\text{weight}_{\text{fuel}} = 23236.9 \text{ N}$$

$$w_{\text{fuel}} = 23200 \text{ N} \quad \blacktriangleleft$$

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PROBLEM 14.96

Determine the increase in velocity of the communication satellite of Problem 14.95 after 12.5 kN of fuel has been consumed.

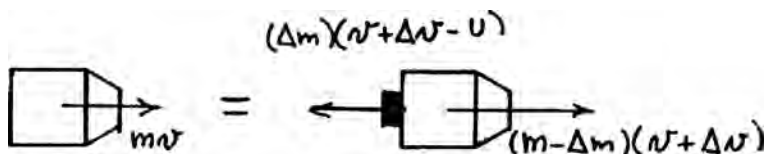
SOLUTION

Data from Problem 14.95:

$$m_0 = \frac{50000 \text{ kg}}{9.81} \quad u = 4000 \text{ m/s}$$

$$m_1 = m_0 - m_{\text{fuel}} = \frac{(50000 - 12500)}{9.81} = 3822.63 \text{ kg.}$$

Apply the *principle of impulse and momentum* to the satellite plus the fuel expelled in time Δt .



$$\begin{aligned} \xrightarrow{+}: \quad mv &= (m - \Delta m)(v + \Delta v) + (\Delta m)(v + \Delta v - u) \\ &= mv + m(\Delta v) - (\Delta m)v - (\Delta m)(\Delta v) \\ &\quad + (\Delta m)v + (\Delta m)(\Delta v) - (\Delta m)u \\ m(\Delta v) - u(\Delta m) &= 0 \end{aligned}$$

$$\Delta m = -\frac{dm}{dt}(\Delta t)$$

$$\frac{\Delta v}{\Delta t} = \frac{dv}{dt} = -\frac{u}{m} \frac{dm}{dt}$$

$$\int_{v_0}^{v_1} dv = -\int_0^{t_1} \frac{u}{m} \frac{dm}{dt} dt = -\int_{m_0}^{m_1} u \frac{dm}{m}$$

$$v_1 - v_0 = -u \ln \frac{m_1}{m_0} = u \ln \frac{m_0}{m_1}$$

$$\Delta v = v_1 - v_0 = 4000 \ln \frac{50000}{37500} \quad \Delta v = 1151 \text{ m/s} \quad \blacktriangleleft$$





PROBLEM 14.97

A 540-kg spacecraft is mounted on top of a rocket with a mass of 19 Mg, including 17.8 Mg of fuel. Knowing that the fuel is consumed at a rate of 225 kg/s and ejected with a relative velocity of 3600 m/s, determine the maximum speed imparted to the spacecraft if the rocket is fired vertically from the ground.

SOLUTION

See sample Problem 14.8 for derivation of

$$v = u \ln \frac{m_0}{m_0 - qt} - gt \quad (1)$$

Data:

$$u = 3600 \text{ m/s} \quad q = 225 \text{ kg/s}, \quad m_{\text{fuel}} = 17,800 \text{ kg}$$

$$m_0 = 19,000 \text{ kg} + 540 \text{ kg} = 19,540 \text{ kg}$$

We have

$$m_{\text{fuel}} = qt, \quad 17,800 \text{ kg} = (225 \text{ kg/s})t$$

$$t = \frac{17,800 \text{ kg}}{225 \text{ kg/s}} = 79.111 \text{ s}$$

Maximum velocity is reached when all fuel has been consumed, that is, when $qt = m_{\text{fuel}}$. Eq. (1) yields

$$\begin{aligned} v_m &= u \ln \frac{m_0}{m_0 - m_{\text{fuel}}} - gt \\ &= (3600 \text{ m/s}) \ln \frac{19,540}{19,540 - 17,800} - (9.81 \text{ m/s}^2)(79.111 \text{ s}) \\ &= (3600 \text{ m/s}) \ln 11.230 - 776.1 \text{ m/s} \\ &= 7930.8 \text{ m/s} \end{aligned} \quad v_m = 7930 \text{ m/s} \blacktriangleleft$$



PROBLEM 14.98

The rocket used to launch the 540-kg spacecraft of Problem 14.97 is redesigned to include two stages A and B, each of mass 9.5 Mg, including 8.9 Mg of fuel. The fuel is again consumed at a rate of 225 kg/s and ejected with a relative velocity of 3600 m/s. Knowing that when stage A expels its last particle of fuel its casing is released and jettisoned, determine (a) the speed of the rocket at that instant, (b) the maximum speed imparted to the spacecraft.

SOLUTION

Thrust force.

$$P = u \frac{dm}{dt} = uq$$

Mass of rocket + unspent fuel:

$$m = m_0 - qt$$

Corresponding weight force:

$$W = mg$$

Acceleration:

$$a = \frac{F}{m} = \frac{P - W}{m} = \frac{P}{m} - g = \frac{uq}{m_0 - qt} - g$$

Integrating with respect to time to obtain the velocity,

$$\begin{aligned} v &= v_0 + \int_0^t a dt = v_0 + u \int_0^t \frac{q dt}{m_0 - qt} - gt \\ &= v_0 - u \ln \frac{m_0 - qt}{m_0} - gt \end{aligned} \quad (1)$$

For each stage,

$$m_{\text{fuel}} = 8900 \text{ kg} \quad u = 3600 \text{ m/s}$$

$$q = 225 \text{ kg/s} \quad t = \frac{m_{\text{fuel}}}{q} = \frac{8900}{225} = 39.556 \text{ s}$$

For the first stage,

$$v_0 = 0 \quad m_0 = 540 + (2)(9500) = 19,540 \text{ kg}$$

$$(a) \quad v_1 = 0 - 3600 \ln \frac{19,540 - 8900}{19,540} - (9.81)(39.556) = 1800.1 \text{ m/s}$$

$$v_1 = 1800 \text{ m/s} \quad \blacktriangleleft$$

For the second stage,

$$v_0 = 1800.1 \text{ m/s}, \quad m_0 = 540 + 9500 = 10,040 \text{ kg}$$

$$(b) \quad v_2 = 1800.1 - 3600 \ln \frac{10,040 - 8900}{10,040} - (9.81)(39.556) = 9244 \text{ m/s}$$

$$v_2 = 9240 \text{ m/s} \quad \blacktriangleleft$$

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PROBLEM 14.99

Determine the altitude reached by the spacecraft of Problem 14.97 when all the fuel of its launching rocket has been consumed.

SOLUTION

See Sample Problem 14.8 for derivation of

$$v = u \ln \frac{m_0}{m_0 - qt} - gt = -u \ln \frac{m_0 - qt}{m_0} - gt \quad (1)$$

Note that g is assumed to be constant.

Set $\frac{dy}{dt} = v$ in Eq. (1) and integrate with respect to time.

$$\begin{aligned} h &= \int_0^h dy = \int_0^t v dt = \int_0^t \left(u \ln \frac{m_0}{m_0 - qt} - gt \right) dt \\ &= -u \int_0^t \ln \frac{m_0 - qt}{m_0} dt - \frac{1}{2} gt^2 \end{aligned}$$

Let

$$z = \frac{m_0 - qt}{m_0} \quad dz = -\frac{q}{m_0} dt \quad \text{or} \quad dt = -\frac{m_0}{q} dz$$

$$\begin{aligned} h &= \frac{m_0 u}{q} \int_{z_0}^z \ln z \, dz - \frac{1}{2} gt^2 = \frac{m_0 u}{q} \left[(z \ln z + z) \right]_{z_0}^z - \frac{1}{2} gt^2 \\ &= \frac{m_0 u}{q} \left[\frac{m_0 - qt}{m_0} \left(\ln \frac{m_0 - qt}{m_0} - 1 \right) - \frac{m_0}{m_0} \left(\ln \frac{m_0}{m_0} - 1 \right) \right] - \frac{1}{2} gt^2 \\ &= \frac{m_0 u}{q} \left[\left(1 - \frac{qt}{m_0} \right) \left(\ln \frac{m_0 - qt}{m_0} - 1 \right) + 1 \right] - \frac{1}{2} gt^2 \\ &= \frac{m_0 u}{q} \left[\ln \frac{m_0 - qt}{m_0} - 1 + 1 \right] - ut \left[\ln \frac{m_0 - qt}{m_0} - 1 \right] - \frac{1}{2} gt^2 \\ &= ut + \left(\frac{m_0 u}{q} - ut \right) \ln \frac{m_0 - qt}{m_0} - \frac{1}{2} gt^2 \\ h &= u \left[t - \left(\frac{m_0}{q} - t \right) \ln \frac{m_0}{m_0 - qt} \right] - \frac{1}{2} gt^2 \quad (2) \end{aligned}$$

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PROBLEM 14.99 (Continued)

Data:

$$u = 3600 \text{ m/s} \quad m_0 = 19,000 + 540 = 19,540 \text{ kg}$$

$$q = 225 \text{ kg/s} \quad m_{\text{fuel}} = 17,800 \text{ kg}$$

$$t = \frac{m_{\text{fuel}}}{q} = \frac{17,800}{225} = 79.111 \text{ s} \quad g = 9.81 \text{ m/s}^2$$

$$m_0 - qt = 1740 \text{ kg}$$

From Eq. (2),

$$\begin{aligned} h &= (3600) \left[79.111 - \left(\frac{19,540}{225} - 79.111 \right) \ln \frac{19,540}{1740} \right] - \frac{1}{2} (9.81) (79.111)^2 \\ &= 186,766 \text{ m} \end{aligned}$$

$$h = 186.8 \text{ km} \quad \blacktriangleleft$$



PROBLEM 14.100

For the spacecraft and the two-stage launching rocket of Problem 14.98, determine the altitude at which (a) stage A of the rocket is released, (b) the fuel of both stages has been consumed.

SOLUTION

Thrust force.

$$P = u \frac{dm}{dt} = uq$$

Mass of rocket + unspent fuel:

$$m = m_0 - qt$$

Corresponding weight force:

$$W = mg$$

Acceleration:

$$a = \frac{F}{m} = \frac{P - W}{m} = \frac{P}{m} - g = \frac{uq}{m_0 - qt} - g$$

Integrating with respect to time to obtain velocity,

$$\begin{aligned} v &= v_0 + \int_0^t a dt = v_0 + u \int_0^t \frac{q dt}{m_0 - qt} - gt \\ &= v_0 - u \ln \frac{m_0 - qt}{m_0} - gt \end{aligned} \quad (1)$$

Integrating again to obtain the displacement,

$$s = s_0 + v_0 t - u \int_0^t \ln \frac{m_0 - qt}{m_0} dt - \frac{1}{2} gt^2$$

Let

$$z = \frac{m_0 - qt}{m_0} \quad dz = -\frac{q}{m_0} dt \quad dt = -\frac{m_0}{q} dz$$

Then

$$\begin{aligned} s &= s_0 + v_0 t + \frac{m_0 u}{q} \int_{z_0}^z \ln z \, dz - \frac{1}{2} gt^2 \\ &= s_0 + v_0 t + \frac{m_0 u}{q} (z \ln z + z) \Big|_{z_0}^z - \frac{1}{2} gt^2 \\ &= s_0 + v_0 t + \frac{m_0 u}{q} \left[\frac{m_0 - qt}{m_0} \ln \frac{m_0 - qt}{m_0} + \frac{m_0 - qt}{m_0} - \frac{m_0}{m_0} \ln \frac{m_0}{m_0} + \frac{m_0}{m_0} \right] - \frac{1}{2} gt^2 \\ &= s_0 + v_0 t + u \left[t + \left(\frac{m_0}{q} - t \right) \ln \frac{m_0 - qt}{m_0} \right] - \frac{1}{2} gt^2 \end{aligned} \quad (2)$$

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PROBLEM 14.100 (Continued)

For each stage,

$$m_{\text{fuel}} = 8900 \text{ kg} \quad u = 3600 \text{ m/s}$$

$$q = 225 \text{ kg/s} \quad t = \frac{m_{\text{fuel}}}{q} = \frac{8900}{225} = 39.556 \text{ s}$$

For the first stage,

$$v_0 = 0 \quad s_0 = 0$$

$$m_0 = 540 + (2)(9500) = 19,540 \text{ kg}$$

From Eq. (1),

$$v_1 = 0 - 3600 \ln \frac{19,540 - 8900}{19,540} - (9.81)(39.556)$$

$$= 1800.1 \text{ m/s}$$

From Eq. (2),

$$(a) \quad s_1 = 0 + 0 + 3600 \left[39.556 + \left(\frac{19,540}{225} - 39.556 \right) \ln \frac{19,540 - 8900}{19,540} \right] - \frac{1}{2} (9.81)(39.556)^2$$

$$= 31,249 \text{ m} \quad h_1 = 31.2 \text{ km} \blacktriangleleft$$

For the second stage,

$$v_0 = 1800.1 \text{ m/s} \quad s_0 = 31,249 \text{ m}$$

$$m_0 = 540 + 9500 = 10,040 \text{ kg}$$

From Eq. (2),

$$(b) \quad s_2 = 31,249 + (1800.1)(39.556) + 3600 \left[39.556 + \left(\frac{10,040}{225} - 39.556 \right) \ln \frac{10,040 - 8900}{10,040} \right]$$

$$- \frac{1}{2} (9.81)(39.556)^2$$

$$= 197,502 \text{ m} \quad h_2 = 197.5 \text{ km} \blacktriangleleft$$



PROBLEM 14.101

Determine the distance separating the communication satellite of Problem 14.95 from the space shuttle 60 s after its engine has been fired, knowing that the fuel is consumed at a rate of 20 kg/s.

PROBLEM 14.95 A communication satellite weighing 50 kN, including fuel, has been ejected from a space shuttle describing a low circular orbit around the earth. After the satellite has slowly drifted to a safe distance from the shuttle, its engine is fired to increase its velocity by 2500 m/s as a first step to its transfer to a geosynchronous orbit. Knowing that the fuel is ejected with a relative velocity of 4000 m/s, determine the weight of fuel consumed in this maneuver.

SOLUTION

Apply the *principle of impulse and momentum* to the satellite plus the fuel expelled in time Δt .

$$\begin{aligned}
 & \text{Diagram showing a satellite of mass } m \text{ moving with velocity } v. \text{ After time } \Delta t, \text{ the satellite has mass } (m - \Delta m) \text{ and velocity } (v + \Delta v). \text{ The expelled fuel has mass } \Delta m \text{ and velocity } (v + \Delta v - u). \\
 & \xrightarrow{+}: \quad mv = (m - \Delta m)(v + \Delta v) + (\Delta m)(v + \Delta v - u) \\
 & \quad = mv + m(\Delta v) - (\Delta m)v - (\Delta m)(\Delta v) \\
 & \quad \quad + (\Delta m)v + (\Delta m)(\Delta v) - (\Delta m)u \\
 & m(\Delta v) - u(\Delta m) = 0 \\
 & \Delta m = -\frac{dm}{dt}(\Delta t) \\
 & \frac{\Delta v}{\Delta t} = \frac{dv}{dt} = -\frac{u}{m} \frac{dm}{dt} = -\frac{uq}{m} = -\frac{uq}{m_0 - qt} \\
 & v = v_0 + \int_0^t \frac{uq}{m_0 - qt} dt = v_0 - u \ln(m_0 - qt) \Big|_0^t \\
 & \quad = v_0 + u \ln(m_0 - qt) + u \ln m_0 \\
 & \quad = v_0 - u \ln \left(\frac{m_0 - qt}{m_0} \right) \tag{1}
 \end{aligned}$$

Set $\frac{dx}{dt} = v$ in Eq. (1) and integrate with respect to time.

$$x = x_0 + v_0 t + u \int_0^t \ln \left(\frac{m_0 - qt}{m_0} \right) dt$$



PROBLEM 14.101 (Continued)

Call the last term x' and let

$$z = \frac{m_0 - qt}{m_0} \quad dz = -\frac{q}{m_0} dt \quad \text{or} \quad dt = -\frac{m_0}{q} dz$$

$$\begin{aligned} x' &= \frac{m_0 u}{q} \int_{z_0}^z \ln z \, dz = \frac{m_0 u}{q} [(z \ln z + z)]_{z_0}^z \\ &= \frac{m_0 u}{q} \left[\frac{m_0 - qt}{m_0} \left(\ln \frac{m_0 - qt}{m_0} - 1 \right) - \frac{m_0}{m_0} \left(\ln \frac{m_0}{m_0} - 1 \right) \right] \\ &= \frac{m_0 u}{q} \left[\left(1 - \frac{qt}{m_0} \right) \left(\ln \frac{m_0 - qt}{m_0} - 1 \right) + 1 \right] \\ &= \frac{m_0 u}{q} \left[\ln \frac{m_0 - qt}{m_0} - 1 + 1 \right] - ut \left[\ln \frac{m_0 - qt}{m_0} - 1 \right] \\ &= ut + \left(\frac{m_0 u}{q} - ut \right) \ln \frac{m_0 - qt}{m_0} \\ &= u \left[t - \left(\frac{m_0}{q} - t \right) \ln \frac{m_0}{m_0 - qt} \right] \\ x &= x_0 + v_0 t + u \left[t - \left(\frac{m_0}{q} - t \right) \ln \frac{m_0}{m_0 - qt} \right] \end{aligned} \quad (2)$$

Data:

$$x_0 = 0 \quad v_0 = 0 \quad q = 20 \text{ kg/s}$$

$$m_0 = \frac{50000}{9.81} \text{ kg}, \quad t = 60 \text{ sec} \quad u = 4000 \text{ m/s}$$

$$\begin{aligned} x &= 0 + 0 + (4000) \left[60 - \left(\frac{50000}{(9.81)(20)} - 60 \right) \ln \frac{50000}{(9.81) \left(\frac{50000}{9.81} - (20)(60) \right)} \right] \\ &= 30,774.9 \text{ m} \end{aligned} \quad x = 30.8 \text{ km} \blacktriangleleft$$

PROBLEM 14.102

For the rocket of Problem 14.94, determine (a) the altitude at which all the fuel has been consumed, (b) the velocity of the rocket at that time.

PROBLEM 14.94 A rocket has a mass of 1200 kg, including 1000 kg of fuel, which is consumed at the rate of 12.5 kg/s and ejected with a relative velocity of 4000 m/s. Knowing that the rocket is fired vertically from the ground, determine its acceleration (a) as it is fired, (b) as the last particle of fuel is being consumed.

SOLUTION

See Sample Problem 14.8 for derivation of

$$v = u \ln \frac{m_0}{m_0 - qt} - gt = -u \ln \frac{m_0 - qt}{m_0} - gt \quad (1)$$

Note that g is assumed to be constant.

Set $\frac{dy}{dt} = v$ in Eq. (1) and integrate with respect to time.

$$\begin{aligned} h &= \int_0^h dy = \int_0^t v dt = \int_0^t \left(u \ln \frac{m_0}{m_0 - qt} - gt \right) dt \\ &= -u \int_0^t \ln \frac{m_0 - qt}{m_0} dt - \frac{1}{2} gt^2 \end{aligned}$$

Let

$$z = \frac{m_0 - qt}{m_0} \quad dz = -\frac{q}{m_0} dt \quad \text{or} \quad dt = -\frac{m_0}{q} dz$$

$$\begin{aligned} h &= \frac{m_0 u}{q} \int_{z_0}^z \ln z \, dz - \frac{1}{2} gt^2 = \frac{m_0 u}{q} [(z \ln z + z)]_{z_0}^z - \frac{1}{2} gt^2 \\ &= \frac{m_0 u}{q} \left[\frac{m_0 - qt}{m_0} \left(\ln \frac{m_0 - qt}{m_0} - 1 \right) - \frac{m_0}{m_0} \left(\ln \frac{m_0}{m_0} - 1 \right) \right] - \frac{1}{2} gt^2 \\ &= \frac{m_0 u}{q} \left[\left(1 - \frac{qt}{m_0} \right) \left(\ln \frac{m_0 - qt}{m_0} - 1 \right) + 1 \right] - \frac{1}{2} gt^2 \\ &= \frac{m_0 u}{q} \left[\ln \frac{m_0 - qt}{m_0} - 1 + 1 \right] - ut \left[\ln \frac{m_0 - qt}{m_0} - 1 \right] - \frac{1}{2} gt^2 \\ &= ut + \left(\frac{m_0 u}{q} - ut \right) \ln \frac{m_0 - qt}{m_0} - \frac{1}{2} gt^2 \\ h &= u \left[t + \left(\frac{m_0}{q} - t \right) \ln \frac{m_0}{m_0 - qt} \right] - \frac{1}{2} gt^2 \quad (2) \end{aligned}$$

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PROBLEM 14.102 (Continued)

Data:

$$m_0 = 1200 \text{ kg} \quad qt = m_{\text{fuel}} = 1000 \text{ kg} \quad q = 12.5 \text{ kg/s}$$

$$u = 4000 \text{ m/s} \quad g = 9.81 \text{ m/s}^2$$

$$t = \frac{m_{\text{fuel}}}{q} = \frac{1000}{12.5} = 80 \text{ s}$$

(a) From Eq. (2),

$$\begin{aligned} h &= (4000) \left[80 - \left(\frac{1200}{12.5} - 80 \right) \ln \frac{1200}{1200 - 1000} \right] - \frac{1}{2} (9.81) (80)^2 \\ &= (4000) (80 - 16 \ln 6) - 31,392 \\ &= 173,935 \text{ m} \end{aligned}$$

$$h = 173.9 \text{ km} \quad \blacktriangleleft$$

(b) From Eq. (1),

$$\begin{aligned} v &= -4000 \ln \frac{1200 - 1000}{1200} - (9.81) (80) \\ &= 4000 \ln 6 - 784.8 \\ &= 6382 \text{ m/s} \end{aligned}$$

$$v = 6.38 \text{ km/s} \quad \blacktriangleleft$$

PROBLEM 14.103

In a jet airplane, the kinetic energy imparted to the exhaust gases is wasted as far as propelling the airplane is concerned. The useful power is equal to the product of the force available to propel the airplane and the speed of the airplane. If v is the speed of the airplane and u is the relative speed of the expelled gases, show that the mechanical efficiency of the airplane is $\eta = 2v/(u + v)$. Explain why $\eta = 1$ when $u = v$.

SOLUTION

Let F be the thrust force, and $\frac{dm}{dt}$ be the mass flow rate.

Absolute velocity of exhaust: $v_e = u - v$

Thrust force: $F = \frac{dm}{dt}(u - v)$

Power of thrust force: $P_1 = Fv = \frac{dm}{dt}(u - v)v$

Power associated with exhaust: $P_2(\Delta t) = \frac{1}{2}(\Delta m)v_e^2 = \frac{1}{2}(\Delta m)(u - v)^2$
 $P_2 = \frac{1}{2}\frac{dm}{dt}(u - v)^2$

Total power supplied by engine: $P = P_1 + P_2$
 $P = \frac{dm}{dt}\left[(u - v)v - \frac{1}{2}(u - v)^2\right]$
 $= \frac{1}{2}\frac{dm}{dt}(u^2 - v^2)$

Mechanical efficiency: $\eta = \frac{\text{useful power}}{\text{total power}} = \frac{P_1}{P}$
 $\eta = \frac{2(u - v)v}{u^2 - v^2}$

$$\eta = \frac{2uv}{(u^2 + v)} \quad \blacktriangleleft$$

$\eta = 1$ when $u = v$. The exhaust, having zero velocity, carries no power away.

PROBLEM 14.104

In a rocket, the kinetic energy imparted to the consumed and ejected fuel is wasted as far as propelling the rocket is concerned. The useful power is equal to the product of the force available to propel the rocket and the speed of the rocket. If v is the speed of the rocket and u is the relative speed of the expelled fuel, show that the mechanical efficiency of the rocket is $\eta = 2uv/(u^2 + v^2)$. Explain why $\eta = 1$ when $u = v$.

SOLUTION

Let F be the thrust force, and $\frac{dm}{dt}$ be the mass flow rate.

Absolute velocity of exhaust: $v_e = u - v$

Thrust force: $F = \frac{dm}{dt}u - v$

Power of thrust force: $P_1 = Fv = \frac{dm}{dt}(u - v)v$

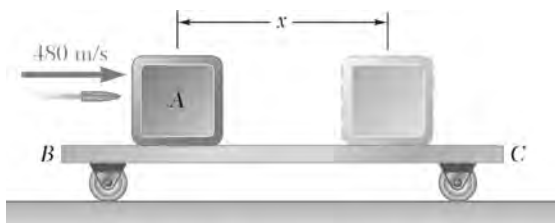
Power associated with exhaust: $P_2(\Delta t) = \frac{1}{2}(\Delta m)v_e^2 = \frac{1}{2}(\Delta m)(u - v)^2$
 $P_2 = \frac{1}{2} \frac{dm}{dt}(u - v)^2$

Total power supplied by engine: $P = P_1 + P_2$
 $P = \frac{dm}{dt} \left[uv - \frac{1}{2}(u - v)^2 \right] = \frac{1}{2} \frac{dm}{dt}(u^2 - v^2)$

Mechanical efficiency: $\eta = \frac{\text{useful power}}{\text{total power}} = \frac{P_1}{P}$

$$\eta = \frac{2uv}{(u^2 + v^2)} \quad \blacktriangleleft$$

$\eta = 1$ when $u = v$. The exhaust, having zero velocity, carries no power away.

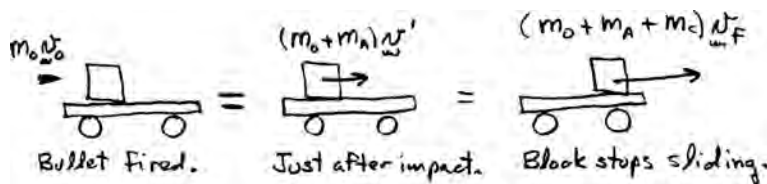


PROBLEM 14.105

A 30-g bullet is fired with a velocity of 480 m/s into block A, which has a mass of 5 kg. The coefficient of kinetic friction between block A and cart BC is 0.50. Knowing that the cart has a mass of 4 kg and can roll freely, determine (a) the final velocity of the cart and block, (b) the final position of the block on the cart.

SOLUTION

(a) Conservation of linear momentum.



$$m_0 v_0 = (m_0 + m_A) v' = (m_0 + m_A + m_C) v_f$$

$$(0.030 \text{ kg})(480 \text{ m/s}) = (5.030 \text{ kg}) v' = (9.030 \text{ kg}) v_f$$

$$v' = \frac{0.030}{5.030} (480 \text{ m/s}) = 2.863 \text{ m/s}$$

$$v_f = \frac{0.030}{9.030} (480 \text{ m/s}) = 1.5947 \text{ m/s} \quad v_f = 1.595 \text{ m/s} \blacktriangleleft$$

(b) Work-energy principle.

Just after impact:

$$T' = \frac{1}{2} (m_0 + m_A) v'^2$$

$$= \frac{1}{2} (5.030 \text{ kg}) (2.863 \text{ m/s})^2$$

$$= 20.615 \text{ J}$$

Final kinetic energy:

$$T_f = \frac{1}{2} (m_0 + m_A + m_C) v_f^2$$

$$= \frac{1}{2} (9.030 \text{ kg}) (1.5947 \text{ m/s})^2$$

$$= 11.482 \text{ J}$$

Work of friction force:

$$F = \mu_k N$$

$$= \mu_k (m_0 + m_A) g$$

$$= 0.50 (5.030) (9.81)$$

$$= 24.672 \text{ N}$$

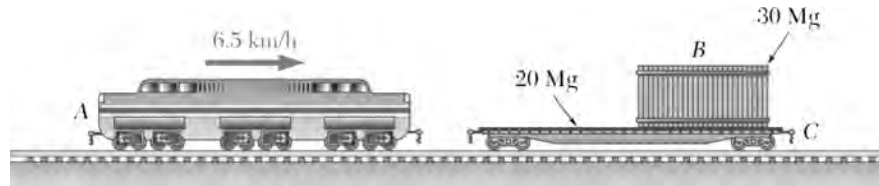
$$\text{Work} = U = -Fx = -24.672x$$

$$T' + U = T_f: \quad 20.615 - 24.672x = 11.482 \quad x = 0.370 \text{ m} \blacktriangleleft$$

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PROBLEM 14.106

An 80-Mg railroad engine *A* coasting at 6.5 km/h strikes a 20-Mg flatcar *C* carrying a 30-Mg load *B* which can slide along the floor of the car ($\mu_k = 0.25$). Knowing that the car was at rest with its brakes released and that it automatically coupled with the engine upon impact, determine the velocity of the car (a) immediately after impact, (b) after the load has slid to a stop relative to the car.



SOLUTION

The masses are the engine ($m_A = 80 \times 10^3$ kg), the load ($m_B = 30 \times 10^3$ kg), and the flat car ($m_C = 20 \times 10^3$ kg).

Initial velocities:

$$(v_A)_0 = 6.5 \text{ km/h} \\ = 1.80556 \text{ m/s} \\ (v_B)_0 = (v_C)_0 = 0.$$

No horizontal external forces act on the system during the impact and while the load is sliding relative to the flat car. Momentum is conserved.

Initial momentum:

$$m_A(v_A)_0 + m_B(0) + m_C(0) = m_A(v_A)_0 \quad (1)$$

- (a) Let v' be the common velocity of the engine and flat car immediately after impact. Assume that the impact takes place before the load has time to acquire velocity.

Momentum immediately after impact:

$$m_A v' + m_B(0) + m_C v' = (m_A + m_C) v' \quad (2)$$

Equating (1) and (2) and solving for v' ,

$$\begin{aligned} v' &= \frac{m_A(v_A)_0}{m_A + m_C} \\ &= \frac{(80 \times 10^3)(1.80556)}{(100 \times 10^3)} \\ &= 1.44444 \text{ m/s} \end{aligned}$$

$$v' = 5.20 \text{ km/h} \rightarrow \blacktriangleleft$$

- (b) Let v_f be the common velocity of all three masses after the load has slid to a stop relative to the car.

Corresponding momentum:

$$m_A v_f + m_B v_f + m_C v_f = (m_A + m_B + m_C) v_f \quad (3)$$

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PROBLEM 14.106 (Continued)

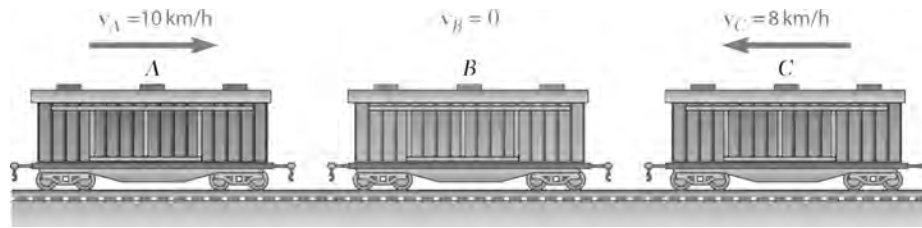
Equating (1) and (3) and solving for v_f ,

$$\begin{aligned}v_f &= \frac{m_A(v_A)_0}{m_A + m_B + m_C} \\&= \frac{(80 \times 10^3)(1.80556)}{(130 \times 10^3)} \\&= 1.11111 \text{ m/s}\end{aligned}$$

$$\mathbf{v_f = 4.00 \text{ km/h} \rightarrow \blacktriangleleft}$$

PROBLEM 14.107

Three identical freight cars have the velocities indicated. Assuming that car B is first hit by car A , determine the velocity of each car after all the collisions have taken place if (a) all three cars get automatically coupled, (b) cars A and B get automatically coupled while cars B and C bounce off each other with a coefficient of restitution $e = 0.8$.



SOLUTION

$$\mathbf{v}_A = 10 \text{ km/h} \rightarrow \quad \mathbf{v}_B = 0 \quad \mathbf{v}_C = 8 \text{ km/h} \leftarrow$$

(a) All cars are coupled.

Conservation of linear momentum. Components $\rightarrow$

$$m_A v_A + 0 - m_C v_C = (m_A + m_B + m_C) v_F$$

$$v_F = \frac{m_A v_A - m_C v_C}{m_A + m_B + m_C} = \frac{1}{3} v_A - \frac{1}{3} v_C = \frac{1}{3} (10 - 8)$$

$$v_F = \frac{2}{3} \text{ km/h}$$

$$\mathbf{v}_A = \mathbf{v}_B = \mathbf{v}_C = 0.667 \text{ km/h} \rightarrow \blacktriangleleft$$

(b) Cars A and B hit and couple.

Conservation of linear momentum for cars A and B .

$$m_A v_A + 0 = (m_A + m_B) v_{AB}$$

$$v_{AB} = \frac{m_A}{m_A + m_B} v_A = \left(\frac{1}{2} \right) (10) = 5 \text{ km/h}$$

Cars A and B are impacted by car C .

Conservation of linear momentum for unit AB and car C .

$$(m_A + m_B) v_{AB} - m_C v_C = (m_A + m_B) v'_{AB} + m_C v'_C$$

$$v_{AB} - \frac{1}{2} v_C = v'_{AB} + \frac{1}{2} v'_C$$

$$5 - \frac{1}{2} (8) = v'_{AB} + 0.5 v'_C \quad (1)$$

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PROBLEM 14.107 (Continued)

Coefficient of restitution.

$$\begin{aligned}v'_C - v'_{AB} &= e(v_{AB} + v_C) \\ &= (0.8)(5 + 8)\end{aligned}\tag{2}$$

Solving Eqs. (1) and (2) simultaneously,

$$v'_{AB} = -2.8 \text{ km/h}$$

$$v_C = 7.6 \text{ km/h}$$

$$\mathbf{v}_A = \mathbf{v}_B = 2.8 \text{ km/h} \leftarrow \qquad \mathbf{v}_C = 7.6 \text{ km/h} \rightarrow \blacktriangleleft$$

PROBLEM 14.108

A 4500 kg helicopter *A* was traveling due east in level flight at a speed of 120 km/h and at an altitude of 750 m when it was hit by a 6000 kg helicopter *B*. As a result of the collision, both helicopters lost their lift, and their entangled wreckage fell to the ground in 12 s at a point located 450 m east and 115.2 m south of the point of impact. Neglecting air resistance, determine the velocity components of helicopter *B* just before the collision.

SOLUTION

Let the *x* axis point east, the *y* axis point up, and the *z* axis point south. The origin of the *xyz* coordinate system is the point of the collision.

At the time of the collision, $t = 0 \quad x = 0 \quad y = 0 \quad z = 0$

At the time of impact with the ground, $t = 12 \text{ s} \quad x = 450 \text{ m} \quad y = -750 \text{ m} \quad z = 115.2 \text{ m}$

Let $\mathbf{v}'$ with components v'_x, v'_y, v'_z be the velocity of the entangled wreckage just after the collision. The entangled wreckage falls as a projectile.

$$x = 0 + v'_x t: \quad 450 = (v'_x)(12) \\ v'_x = 37.5 \text{ m/s}$$

$$y = 0 + v'_y t - \frac{1}{2} g t^2: \quad -750 = (v'_y)(12) - \frac{1}{2} (9.81)(12)^2 \\ v'_y = -3.64 \text{ m/s}$$

$$z = 0 + v'_z t: \quad 115.2 = (v'_z)(12) \\ v'_z = 9.6 \text{ m/s}$$

$$\mathbf{v}' = (37.5 \text{ m/s})\mathbf{i} - (3.64 \text{ m/s})\mathbf{j} + (9.6 \text{ m/s})\mathbf{k}$$

Let $\mathbf{v}_A$ and $\mathbf{v}_B$ be the velocities of helicopters *A* and *B*, respectively, before the collision.

$$\mathbf{v}_A = (120 \text{ km/h})\mathbf{i} = \left(\frac{100}{3} \text{ m/s}\right)\mathbf{i} \quad \mathbf{v}_B = ?$$

Conservation of linear momentum before and after the collision.

$$m_A \mathbf{v}_A + m_B \mathbf{v}_B = (m_A + m_B) \mathbf{v}' \\ \mathbf{v}_B = \frac{m_A + m_B}{m_B} \mathbf{v}' - \frac{m_A}{m_B} \mathbf{v}_A$$

Data:

$$\frac{m_A + m_B}{m_B} = \frac{4500 + 6000}{6000} = 1.75 \quad \frac{m_A}{m_B} = \frac{4500}{6000} = 0.75$$

$$\begin{aligned} \mathbf{v}_B &= (1.75)(37.5\mathbf{i} - 3.64\mathbf{j} + 9.6\mathbf{k}) - 0.75\left(\frac{100}{3}\right)\mathbf{i} \\ &= (40.625 \text{ m/s})\mathbf{i} - (6.37 \text{ m/s})\mathbf{j} + (16.8 \text{ m/s})\mathbf{k} \\ &= (146.25 \text{ km/h})\mathbf{i} - (22.932 \text{ km/h})\mathbf{j} + (60.48 \text{ km/h})\mathbf{k} \end{aligned}$$

Components of $\mathbf{v}_B$: $146.3 \text{ km/h} \quad 60.5 \text{ km/h} \quad 22.9 \text{ km/h}$ ◀

PROBLEM 14.109

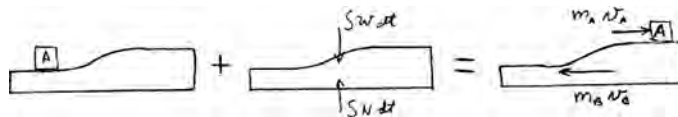


A 7.5 kg block B is at rest and a spring of constant $k = 15000 \text{ N/m}$ is held compressed 75 mm by a cord. After 2.5-kg block A is placed against the end of the spring, the cord is cut, causing A and B to move. Neglecting friction, determine the velocities of blocks A and B immediately after A leaves B .

SOLUTION

$$\begin{aligned} m_A &= 2.5 \text{ kg} \\ m_B &= 7.5 \text{ kg} \\ k &= 15000 \text{ N/m} \\ e &= 75 \text{ mm} = 0.075 \text{ m} \\ h &= 150 \text{ mm} = 0.15 \text{ m} \end{aligned}$$

Conservation of linear momentum.

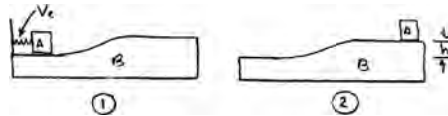


Horizontal components $\rightarrow$:

$$0 + 0 = m_A v_A - m_B v_B$$

$$v_B = \frac{m_A}{m_B} v_A = \frac{1}{3} v_A$$

Conservation of energy.



State 1.

$$V_{1e} = \frac{1}{2} k e^2 = \frac{1}{2} (15000)(0.075)^2 = 42.1875 \text{ J}$$

$$V_{1g} = 0$$

$$T_1 = 0$$

State 2.

$$V_{2e} = 0$$

$$V_{2g} = W_A h = (2.5)(9.81)(0.15) = 3.67875 \text{ J}$$

$$T_2 = \frac{1}{2} m_A v_A^2 + \frac{1}{2} m_B v_B^2$$

$$= \frac{1}{2} (2.5) v_A^2 + \frac{1}{2} (7.5) \left(\frac{v_A}{3} \right)^2 = \frac{5}{3} v_A^2$$

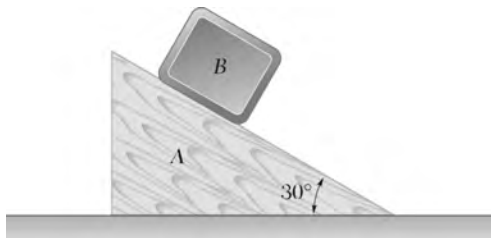
PROBLEM 14.109 (Continued)

$$T_1 + V_1 = T_2 + V_2: \quad 0 + 42.1875 = \frac{5}{3}v_A^2 + 3.67875$$

$$v_A^2 = 23.10525 \text{ m}^2$$

$$\mathbf{v}_A = 4.81 \text{ m/s} \rightarrow \blacktriangleleft$$

$$\mathbf{v}_B = 1.602 \text{ m/s} \leftarrow \blacktriangleleft$$



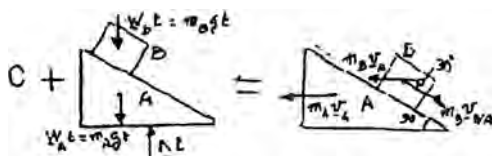
PROBLEM 14.110

A 9-kg block B starts from rest and slides down the inclined surface of a 15-kg wedge A , which is supported by a horizontal surface. Neglecting friction, determine (a) the velocity of B relative to A after it has slid 0.6 m down the surface of the wedge, (b) the corresponding velocity of the wedge.

SOLUTION

We resolve $\mathbf{v}_B$ into its components: $\mathbf{v}_A \leftarrow$ and $\mathbf{v}_{B/A} \searrow 30^\circ$

Impulse-momentum principle.



$$\Sigma m \mathbf{v}_0 + \Sigma \mathbf{F} t = \Sigma m \mathbf{v}$$

$\rightarrow$ x component:

$$C + 0 = m_B v_{B/A} \cos 30^\circ - m_A v_A - m_B v_A$$

$$v_A = \frac{m_B \cos 30^\circ}{m_A + m_B} v_{B/A} = \frac{(9 \text{ kg}) \cos 30^\circ}{15 \text{ kg} + 9 \text{ kg}} v_{B/A}$$

$$v_A = 0.32476 v_{B/A} \quad (1)$$

Conservation of energy.

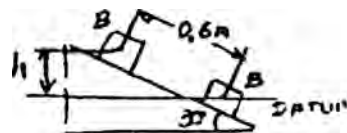
$$T_0 = 0$$

$$V_0 = m_B gh$$

$$= (9 \text{ kg})(9.81 \text{ m/s}^2)(0.6 \text{ m}) \sin 30^\circ$$

$$= 26.487 \text{ J}$$

$$T = \frac{1}{2} m_A v_A^2 + \frac{1}{2} m_B v_B^2 \quad V = 0$$



Referring to figures shown above and using the law of cosines:

$$T = \frac{1}{2} m_A v_A^2 + \frac{1}{2} m_B (v_A^2 + v_{B/A}^2 - 2 v_A v_{B/A} \cos 30^\circ)$$

Recalling Eq. (1) and substituting the given data:

$$T = \frac{1}{2} (15)(0.32476)^2 v_{B/A}^2 + \frac{1}{2} (9)[(0.32476)^2 + 1 - 2(0.32476) \cos 30^\circ] v_{B/A}^2$$

$$= 0.79102 v_{B/A}^2 + 2.04336 v_{B/A}^2 = 3.2344 v_{B/A}^2$$

$$T + V = T_0 + V_0: \quad 3.2344 v_{B/A}^2 = 26.487 \text{ J}$$

$$v_{B/A} = 2.8617 \text{ m/s}$$

PROBLEM 14.110 (Continued)

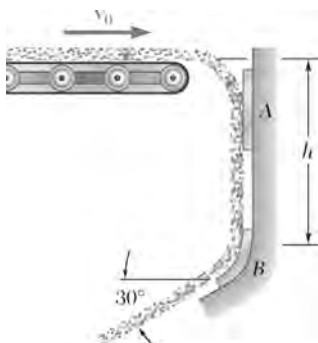
(a)

$$\mathbf{v}_{B/A} = 2.86 \text{ m/s} \swarrow 30^\circ \blacktriangleleft$$

(b) From Eq. (1):

$$\begin{aligned} v_A &= 0.32476(2.8617 \text{ m/s}) \\ &= 0.92936 \text{ m/s} \end{aligned}$$

$$\mathbf{v}_A = 0.929 \text{ m/s} \leftarrow \blacktriangleleft$$

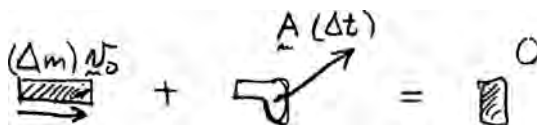


PROBLEM 14.111

A mass q of sand is discharged per unit time from a conveyor belt moving with a velocity v_0 . The sand is deflected by a plate at A so that it falls in a vertical stream. After falling a distance h , the sand is again deflected by a curved plate at B . Neglecting the friction between the sand and the plates, determine the force required to hold in the position shown (a) plate A , (b) plate B .

SOLUTION

- (a) When the sand impacts on plate A , it is momentarily brought to rest. Apply the principle of impulse and momentum to find the force on the sand.



$\rightarrow$ x component:

$$(\Delta m)v_0 + A_x(\Delta t) = 0$$

$$A_x = -\frac{\Delta m}{\Delta t}v_0 = -qv_0$$

$\uparrow$ y component:

$$0 + A_y(\Delta t) = 0 \quad A_y = 0$$

$$\mathbf{A} = qv_0 \leftarrow$$

The sand falls vertically. Use conservation of energy for mass element Δm . Let v be the speed at the curved portion of plate B .

$$T_1 + V_1 = T_2 + V_2: \quad 0 + (\Delta m)gh = \frac{1}{2}(\Delta m)v^2 + 0$$

$$v^2 = 2gh$$

$$v = \sqrt{2gh}$$

Over the curved portion of plate B , there is negligible change of elevation. Hence, by conservation of energy, v is both the entrance speed and exit speed of the curved portion of plate B .

- (b) Force exerted through plate B .

Entrance velocity:

$$\mathbf{v} = -\sqrt{2gh} \mathbf{j}$$

Exit velocity:

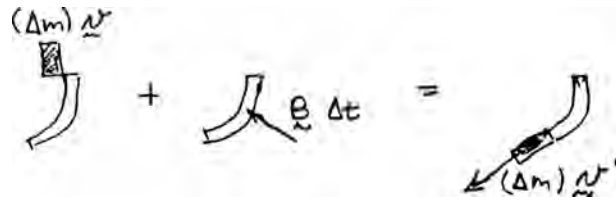
$$\mathbf{v}' = \sqrt{2gh}(-\cos 30^\circ \mathbf{i} - \sin 30^\circ \mathbf{j})$$

Mass flow rate:

$$\frac{dm}{dt} = \frac{\Delta m}{\Delta t} = q$$

PROBLEM 14.111 (Continued)

Principle of impulse and momentum.



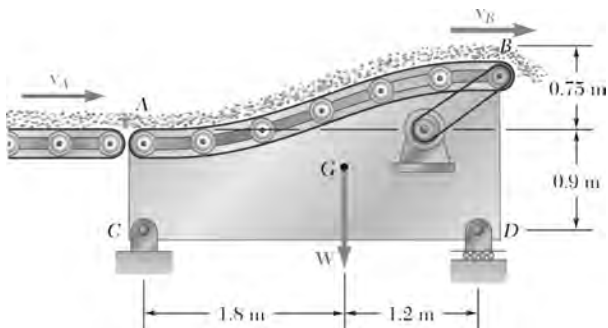
$$(\Delta m)\mathbf{v} + \mathbf{B}(\Delta t) = (\Delta m)\mathbf{v}'$$

$$\mathbf{B} = \left(\frac{\Delta m}{\Delta t} \right) (\mathbf{v}' - \mathbf{v}) = q\sqrt{2gh}(-\cos 30^\circ \mathbf{i} - \sin 30^\circ \mathbf{j} + \mathbf{j})$$

$$B_x = -\sqrt{2gh} \cos 30^\circ = -\frac{\sqrt{3}}{2} \sqrt{2gh}$$

$$B_y = \sqrt{2gh}(1 - \sin 30^\circ) = \frac{1}{2} \sqrt{2gh}$$

$$\mathbf{B} = \sqrt{2gh} \nearrow 30^\circ \blacktriangleleft$$

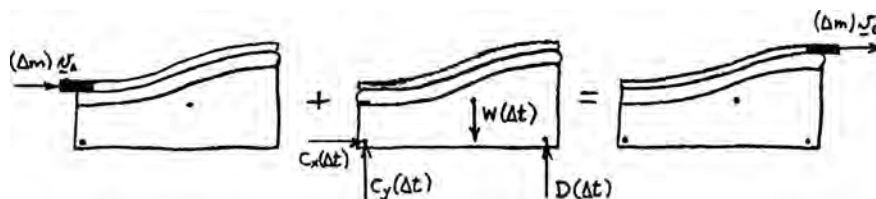


PROBLEM 14.112

The final component of a conveyor system receives sand at a rate of 100 kg/s at A and discharges it at B. The sand is moving horizontally at A and B with a velocity of magnitude $v_A = v_B = 4.5$ m/s. Knowing that the combined weight of the component and of the sand it supports is $W = 4$ kN, determine the reactions at C and D.

SOLUTION

Apply the impulse-momentum principle.



⤴ Moments about C: $-(0.9)(\Delta m)v_A + 3D(\Delta t) - 1.8W(\Delta t) = -1.65(\Delta m)v_B$

$$D = \frac{1.8}{3}W + \frac{1}{3} \frac{\Delta m}{\Delta t} (0.9v_A - 1.65v_B)$$

$$= \frac{(1.8)(4000)}{3} + \frac{1}{3}(100)[(0.9)(4.5) - (1.65)(4.5)] = 2287.5 \text{ N}$$

D = 2.29 kN ↑ ◀

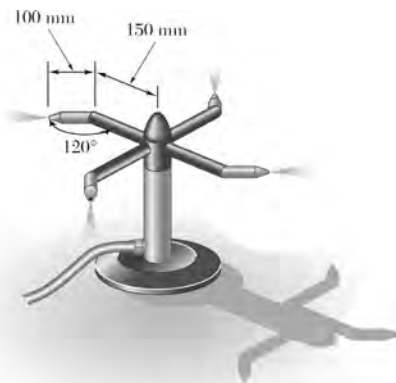
→ x components: $(\Delta m)v_A + C_x(\Delta t) = (\Delta m)v_B$

$$C_x = \left(\frac{\Delta m}{\Delta t} \right) (v_B - v_A) = (100)(4.5 - 4.5) = 0$$

↑ y components: $0 + C_y(\Delta t) + D(\Delta t) - W(\Delta t) = 0$

$$C_y = W - D = 4000 - 2287.5 = 1712.5 \text{ N}$$

C = 1.712 kN ↑ ◀



PROBLEM 14.113

A garden sprinkler has four rotating arms, each of which consists of two horizontal straight sections of pipe forming an angle of 120° with each other. Each arm discharges water at a rate of 20 L/min with a velocity of 18 m/s relative to the arm. Knowing that the friction between the moving and stationary parts of the sprinkler is equivalent to a couple of magnitude $M = 0.375 \text{ N} \cdot \text{m}$, determine the constant rate at which the sprinkler rotates.

SOLUTION

The flow through each arm is 20 L/min.

$$Q = \frac{20 \text{ L/min}}{1000 \text{ L/m}^3} \times \frac{1 \text{ min}}{60 \text{ s}} = 333.33 \times 10^{-6} \text{ m}^3/\text{s}$$

$$\begin{aligned} \frac{dm}{dt} &= \rho Q = (1000 \text{ kg/m}^3)(333.33 \times 10^{-6}) \\ &= 0.33333 \text{ kg/s} \end{aligned}$$

Consider the moment about O exerted on the fluid stream of one arm. Apply the impulse-momentum principle. Compute moments about O . First, consider the geometry of triangle OAB . Using first the law of cosines,

$$(OA)^2 = 150^2 + 100^2 - (2)(150)(100)\cos 120^\circ$$

$$OA = 217.95 \text{ mm} = 0.21795 \text{ m}$$

Law of sines.

$$\frac{\sin \beta}{100} = \frac{\sin 120^\circ}{217.95}$$

$$\beta = 23.413^\circ, \quad \alpha = 60^\circ - \beta = 36.587^\circ$$

⤵ Moments about O :

$$(\Delta m)(v_O)(0) + M_O(\Delta t) = (OA)(\Delta m)v_s \sin \alpha - (OA)(\Delta m)(OA)\omega$$

$$\begin{aligned} M_O &= \frac{\Delta m}{\Delta t} [(OA)v_s \sin \alpha - (OA)^2 \omega] \\ &= (0.33333)[(0.21795)(18) \sin 36.587^\circ - (0.21795)^2 \omega] \\ &= 0.77945 - 0.015834\omega \end{aligned}$$

Moment that the stream exerts on the arm is $-M_O$.

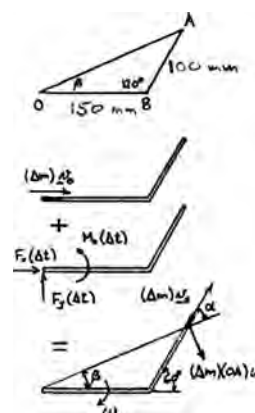
Balance of the friction couple and the four streams

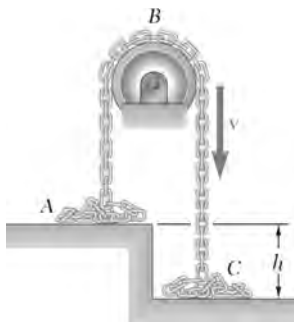
$$M_F - 4M_O = 0$$

$$0.375 - 4(0.77945 - 0.015834\omega) = 0$$

$$\omega = 43.305 \text{ rad/s}$$

$$\omega = 414 \text{ rpm} \quad \blacktriangleleft$$





PROBLEM 14.114

The ends of a chain lie in piles at A and C. When given an initial speed v , the chain keeps moving freely at that speed over the pulley at B. Neglecting friction, determine the required value of h .

SOLUTION

Let ρ be the mass per unit length of chain. Consider the impulse-momentum applied to the link being brought to rest at Point C.

Calculation of Δm .

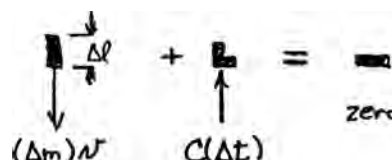
$$\Delta m = \rho(\Delta l) = \rho v(\Delta t)$$

Impulse-momentum principle:

$$-(\Delta m)v + C\Delta t = 0$$

$$-\rho v(\Delta t)v + C\Delta t = 0$$

$$C = \rho v^2$$



Impulse-momentum applied to the moving portion of the chain. Consider only the changes in momentum and forces contributing to moments about O in the diagram.

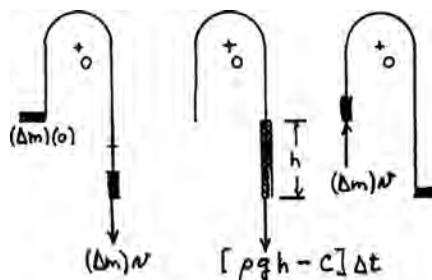
$\curvearrowright$ Moments about O:

$$(\Delta m)v + [\rho gh - C]\Delta t = (\Delta m)v$$

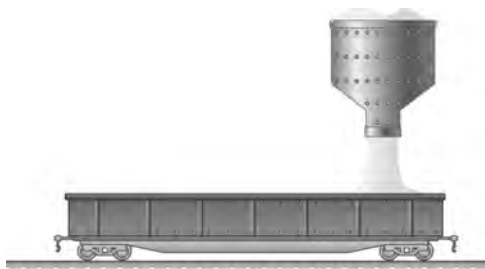
$$C = \rho gh$$

Equating the two expressions for C,

$$\rho v^2 = \rho gh$$



$$h = \frac{v^2}{g} \quad \blacktriangleleft$$

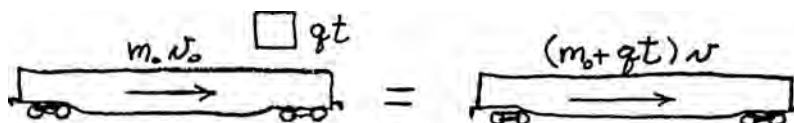


PROBLEM 14.115

A railroad car of length L and mass m_0 when empty is moving freely on a horizontal track while being loaded with sand from a stationary chute at a rate $dm/dt = q$. Knowing that the car was approaching the chute at a speed v_0 , determine (a) the mass of the car and its load after the car has cleared the chute, (b) the speed of the car at that time.

SOLUTION

Consider the conservation of the horizontal component of momentum of the railroad car of mass m_0 and the sand mass qt .



$$m_0 v_0 = (m_0 + qt)v \quad v = \frac{m_0 v_0}{m_0 + qt} \quad (1)$$

$$\frac{dx}{dt} = v = \frac{m_0 v_0}{m_0 + qt}$$

Integrating, using

$$x_0 = 0 \quad \text{and} \quad x = L \quad \text{when} \quad t = t_L,$$

$$\begin{aligned} L &= \int_0^{t_L} v dt = \int_0^{t_L} \frac{m_0 v_0}{m_0 + qt} dt = \frac{m_0 v_0}{q} [\ln(m_0 + qt_L) - \ln m_0] \\ &= \frac{m_0 v_0}{q} \ln \frac{m_0 + qt_L}{m_0} \end{aligned}$$

$$\ln \frac{m_0 + qt_L}{m_0} = \frac{qL}{m_0 v_0} \quad \frac{m_0 + qt_L}{m_0} = e^{qL/m_0 v_0}$$

(a) Final mass of railroad car and sand

$$m_0 + qt_L = m_0 e^{qL/m_0 v_0} \quad \blacktriangleleft$$

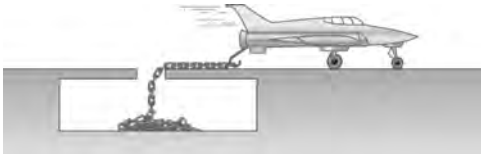
(b) Using Eq. (1),

$$v_L = \frac{m_0 v_0}{m_0 + qt_L} = \frac{m_0 v_0}{m_0} e^{-qL/m_0 v_0}$$

$$v_L = v_0 e^{-qL/m_0 v_0} \quad \blacktriangleleft$$

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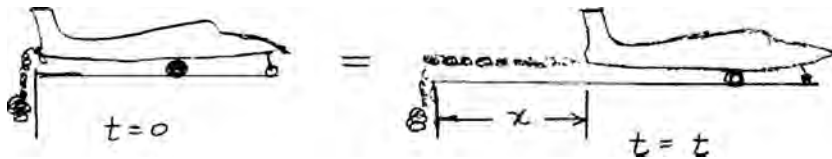
PROBLEM 14.116



A possible method for reducing the speed of a training plane as it lands on an aircraft carrier consists in having the tail of the plane hook into the end of a heavy chain of length l which lies in a pile below deck. Denoting by m the mass of the plane and by v_0 its speed at touchdown, and assuming no other retarding force, determine (a) the required mass of the chain if the speed of the plane is to be reduced to βv_0 , where $\beta < 1$, (b) the maximum value of the force exerted by the chain on the plane.

SOLUTION

Let m' be the mass of chain per unit length and x be the distance traveled in time t . Let v be the velocity of the plane at time t . Apply the principle of conservation of linear momentum.



$\rightarrow$ x-component:

$$mv_0 = (m + m'x)v$$

$$v = \frac{mv_0}{m + m'x} \quad (1)$$

(a) Speed is reduced to βv_0 when $x = l$.

$$\beta v_0 = \frac{mv_0}{m + m'l}$$

Mass of chain:

$$m'l = \frac{1 - \beta}{\beta} m \quad \blacktriangleleft$$

(b) Maximum force on the plane.

Acceleration of plane:

$$a = \frac{dv}{dt} = v \frac{dv}{dx} = \frac{mv_0}{m + m'x} \cdot \frac{(-m'v_0)}{(m + m'x)^2} = -\frac{m'm^2v_0^2}{(m + m'x)^2}$$

Force acting on the plane:

$$F = ma = -\frac{m'm^3v_0^2}{(m + m'x)^3}$$

$|F_{\max}|$ occurs when $x = 0$.

$$|F_{\max}| = m'v_0^2 = \frac{1 - \beta}{\beta} \frac{mv_0^2}{l} \quad |F_{\max}| = \frac{(1 - \beta)mv_0^2}{\beta l} \quad \blacktriangleleft$$

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